

Evolution of Multivariant Microstructures with Anisotropic Misfit

A Phase Field Study

A Thesis

Submitted for the Degree of

Doctor of Philosophy

in the Faculty of Engineering

By

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November 2007

To

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Synopsis

Many technologically important alloys such as Ni base superalloys and Ti-Al base alloys benefit from the precipitation of an ordered β phase from a disordered α matrix. When the crystallographic symmetry of the β phase is a subgroup of that of the disordered α phase, the microstructure may contain multiple orientational variants of the β phase, each with its own (anisotropic, crystallographically equivalent) misfit (lattice parameter mismatch) with the matrix phase. Examples include orthorhombic precipitates in a hexagonal matrix in Ti-Al-Nb alloys, and tetragonal precipitates in a cubic matrix in $\text{ZrO}_2\text{-Y}_2\text{O}_3$.

We have studied two-phase microstructures containing multiple variants of the precipitate phase. In particular, we have used phase field simulations to study the effect of elastic stresses in a two dimensional system containing a disordered matrix and three different orientational variants of the precipitate phase, with a view to elucidate the effect of different levels of anisotropy in misfit.

We consider a two dimensional, elastically homogeneous and isotropic model system in which the matrix (α) and precipitate (β) phases have hexagonal and rectangular symmetries, respectively, giving rise to three orientational variants of the β phase. Therefore, our phase field model has composition (c) and three order parameters (η_1, η_2, η_3) as the field variables. Due to the difference in crystallographic symmetry, the precipitate-matrix misfit strain tensor, ϵ^* , can be anisotropic. ϵ^* may be represented in its principal form as

$$\epsilon^* = \begin{pmatrix} \epsilon_{xx} & 0 \\ 0 & \epsilon_{yy} \end{pmatrix},$$

where ϵ_{xx} and ϵ_{yy} are the principal components of the misfit tensor.

We define $t = \epsilon_{yy}/\epsilon_{xx}$ as the parameter representing anisotropy in the misfit. In this thesis, we report the results of our systematic study of microstructural evolution in systems with different values of t , representing different levels of anisotropy in misfit:

- Case A: $t = 1$ (dilatational or isotropic misfit)

- Case B: $0 < t < 1$ (principal misfit components are unequal but have the same sign)
- Case C: $t = 0$ (the principal misfit along the y direction is zero)
- Case D: $-1 < t < 0$ (principal misfit components have opposite signs and unequal magnitudes)
- Case E: $t = -1$ (principal misfit components are equal, but with opposite signs; pure shear)

In Cases D and E, there is an invariant line along which the normal misfit is zero. In Case D, this invariant line is at $\pm 54.72^\circ$, and in Case E, it is at $\pm 45^\circ$, with respect to the x -axis.

Our simulations of microstructural evolution in this system are based on numerical integration of the Cahn-Hilliard and Cahn-Allen equations which govern the evolution of composition and order parameter fields, respectively. In each case, we have studied two different situations: isolated particle (single variant) and many interacting particles (multivariant).

Dynamical growth shape of an isolated precipitate

In systems with an isotropic misfit (Case A), the precipitate shape remains circular at all sizes. In Cases B and C, the precipitate shape is elongated along the y -axis, which is also the direction in which the magnitude of the misfit strain is lower. In all these cases, the symmetry of the particle shape remains unaltered at all sizes.

In contrast, in Cases D and E, the particle shape exhibits a symmetry-breaking transition. In Case D, the precipitate elongates initially along the y direction (i.e. the direction of lower absolute misfit), before undergoing a transition in which the mirror symmetry normal to x and y axes is lost. In Case E, the particle has an initial square-like shape (with its sides normal to the $\langle 11 \rangle$ directions) before losing its four-fold rotation axis to become rectangle-like with its long axis along one of the the $\langle 11 \rangle$ directions.

The critical precipitate size at which the symmetry-breaking shape transition occurs is obtained using bifurcation diagrams. In both Cases D and E, the critical size for the dynamical growth shapes is larger than those for equilibrium shapes [1]. This critical size is larger when the matrix supersaturation is higher or shear modulus is lower.

Microstructural Evolution

In all the five cases, the elastic stresses have a common effect: they lead to microstructures in which the precipitate volume fraction is lower than that in a system with no misfit. This observation is consistent with the results from the thermodynamics of stressed solids that show that a precipitate-matrix misfit increases the interfacial composition in both the matrix and the precipitate phase.

In systems with isotropic misfit (Case A), the microstructure consists of isolated circular domains of the precipitate phase that retain their circular shape during growth and subsequent coarsening. In Cases B and C with anisotropic misfit with $t \geq 0$, the three orientational variants of the precipitate phase are elongated along the directions of lower misfit (y -axis and $\pm 120^\circ$ to y -axis). At a given size, particles in Case C (in which one of the principal misfits is zero) are more elongated than those in Case B. Systems with a higher shear modulus enhance the effect of misfit stresses, and therefore, lead to thinner and longer precipitates. When the precipitate volume fraction is increased, these elongated precipitates interact with (and impinge against) one another to a greater extent, and acquire a more jagged appearance.

For Cases D and E, each orientation domain is associated with an invariant line along which the normal misfit is zero. Thus, in Case D, early stage microstructures show particles elongated along directions of lower absolute misfit (y -axis and $\pm 120^\circ$ to y -axis). At the later stages, the domains of the precipitate phase tend to orient along the invariant lines; this leads some of the particles to acquire a ‘Z’ shape before they completely re-orient themselves along the invariant line.

In Case E, each variant grows as a thin plate elongating along the invariant line. The growth and impingement of these thin plates leads to a microstructure exhibiting complex multi-domain patterns such as stars, wedges, triangles, and checkerboard. These patterns have been compared (and are in good agreement) with experimental observations in Ti-Al-Nb alloys containing the precipitate (O) and matrix (α_2) phases [2].

Since in Case E the sum of misfit strains of the three variants is zero, elastic energy considerations point to the possibility of compact, self-accommodating clusters of the three variants, sharing antiphase boundaries (APBs). Thus, if the APB energy is sufficiently low, the microstructure may consist of such compact clusters. Our simulations with such low APB energy do show triangle shaped clusters with six separate particles (two of each variant) in a self-accommodating pattern.

Acknowledgments

I express my heartfelt thanks to my research adviser Prof. T. A. Abinandanan for his guidance, encouragement, patience and friendliness. His insightful discussions has helped me a lot to resolve several critical issues.

I thank Prof. K. A. Natarajan and Prof. K. Chattopadhyay, Chairmen of the Department, for allowing me to use the facilities and resources of the department. I thank Chairman, SERC, for allowing me to access the computational facilities. I thank all the faculty and the staff of the Department for their help and support.

I thank all the members of the $\Sigma\tau$ group for many useful discussions. I would specially like to thank Prof. S. Ranganathan, Prof. K. Chattopadhyay and Prof. V. Jayaram for taking a keen interest in my work and giving me their invaluable suggestions and comments. I would also like to thank Dr. Karthikeyan for his suggestions and discussions.

I thank Prof. Dr. Ferdinand Haider at the University of Augsburg who helped me learn Monte Carlo simulation techniques. I had many a wonderful discussions with him and his group during my stay in Germany.

I thank Dr. Muraleedharan and Dr. Balamuralikrishna of DMRL, Hyderabad for giving me useful suggestions.

I thank all the past and present members of the Computational Materials Science group Shankara, Ram, Guru, Subhradeep, Rajdip, Kodandaram, Gaurav, Nilesh, Naidu, Badri and Suman Sarkar for making my stay in the lab an enjoyable and memorable experience. My special thanks to Shankara, Ram, Guru, Subhradeep, and Rajdip for many a wonderful discussions I had with them. Their suggestions have helped me improve my computational skills and learn various aspects of elasticity.

I would specially like to thank Subhradeep and Rajdip for carefully going through the manuscript of this thesis. Their suggestions have helped me in giving this thesis a better form.

I would like to thank Pratyush (BHU) and Chiru (IIT-M) who visited our lab during the summer months. I spent a nice time discussing with them.

I would like to thank Shankara, Sudhir, Phani, Phaniraj, Ram, Guru, Subhradeep, Sanjit, Subrata, Suman-da, Basani, Kaushik (Roy), Kottada, Sarmistha-di, Eswar, Kodanda, Arindamp, Krishanu, Ranadeep, Arindam (Physics), Twarita, Partha, Debdutt, Victoria, Jeyakumar, Kavitha, Tripti, Kay Thi, Pratap, Santonu, MJNVPrasad, Rejin, Aditi, P. Murali, Rajdip, Somnath, Kaushik, Shibayan, Indrani, Hindol, Somjeet, Ankur, R. Chitra, Geetika, Ujwala, Antara, and innumerable others for making my stay in the campus a memorable and cherishable experience.

I thank my parents for their patience, emotional support and blessings without which this thesis would not have been possible.

Last but not the least, I thank my wife, Joyita, who has been patient and encouraging and has supported me throughout this work.

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Glossary

$A_1, A_2, A_{41}, A_{42}, A_{61}, A_{62}$	Bulk free energy coefficients
A_Z	Zener anisotropy parameter (is unity in elastically isotropic systems)
$B(\mathbf{n})$	Microstructure dependent elastic interaction term calculated in reciprocal space [4]
$\mathbf{C}$	Elastic stiffness tensor (independent of composition and order parameters)
c	Composition (scaled, nondimensional)
c_0	Alloy composition
c_∞	Far-field composition
Δc^α	Deviation from equilibrium composition on the matrix side of the curved interface
Δc_c^α	Deviation from equilibrium composition on the matrix side of the curved interface due to capillarity
Δc_{el}^α	Deviation from equilibrium composition on the matrix side of the curved interface due to elastic stresses
$\frac{\delta}{\delta \theta}$	Variational derivative with respect to θ
∇	Gradient operator
E_n	Characteristic energy used for nondimensionalisation

$\mathbf{E}$	Homogeneous strain tensor
$\boldsymbol{\epsilon}$	Total strain tensor
$\boldsymbol{\epsilon}^0$	Eigenstrain tensor (misfit strain tensor)
$\boldsymbol{\varepsilon}$	Periodic strain tensor
$\boldsymbol{\epsilon}^{(p)}$	Position independent part of eigenstrain tensor $\boldsymbol{\epsilon}^0$ associated with p 'th variant
ϵ_c	Strength of misfit strain associated with inhomogeneity in composition, c (assumed to be zero in our study)
ϵ_η	Characteristic misfit strain produced by ordering
ϵ_{xx}	Principal component of the misfit strain along x axis
ϵ_{yy}	Principal component of the misfit strain along y axis
$\mathcal{F}$	Total free energy of the system
$\mathcal{F}_c$	Chemical contribution to total free energy
$\mathcal{F}_e$	Elastic contribution to total free energy
f_0	Bulk free energy density
f_α	Bulk free energy of disordered α phase
f_β	Bulk free energy of ordered β phase
G	Average shear modulus
γ_i	Energy of the interphase interface between disordered α phase and ordered β phase
γ_a	Energy of the antiphase domain boundary between any two ordered variants of the β phase
$\Delta\mathcal{G}_v$	Change in volume free energy during nucleation
$\Delta\mathcal{G}_e$	Elastic strain energy per unit volume of the nucleus

η_1	Long range order parameter field describing the first variant of the ordered precipitate phase; $\eta_1 = \pm 1$ inside the first variant
η_2	Long range order parameter field describing the second variant of the ordered precipitate phase; $\eta_2 = \pm 1$ inside the second variant
η_3	Long range order parameter field describing the third variant of the ordered precipitate phase; $\eta_3 = \pm 1$ inside the third variant
$\mathfrak{J}$	Planar moment of inertia tensor
J	Imaginary number $\sqrt{-1}$
$\mathbf{J}$	Flux of atoms during diffusion
κ_c	Gradient energy coefficient associated with gradient in composition c (positive scalar constant)
κ_η	Gradient energy coefficient associated with gradient in long range order parameters η_i (positive scalar constant)
χ	Shape parameter defined using the planar moment of inertia tensor
$\mathbf{k}$	Reciprocal space vector
L_c	A characteristic length defined by $\gamma_i/G\epsilon_\eta^2$; represents the transition size between capillarity dominant and elasticity dominant regimes
L_n	A characteristic length used for nondimensionalisation
L	Positive relaxation coefficient in Cahn-Allen equation
M	Mobility
μ	Chemical potential
N_v	Number of atoms per unit volume

ν	Effective Poisson's ratio
Ω^{-1}	Normalized inverse Green tensor
$\mathbf{n}$	Unit vector in reciprocal space
R	Particle size defined as the equivalent radius of a circle having the same area as the precipitate
R_c	Critical particle size at which the particle shape undergoes symmetry-breaking transition
ρ	Normalized particle size; R/L_c
$\boldsymbol{\sigma}^{el}$	Elastic stress tensor
T_n	Characteristic time used for nondimensionalisation
τ	Time
t	Misfit anisotropy parameter defined as $\epsilon_{yy}/\epsilon_{xx}$; varies from +1 to -1
θ_p	Position dependent part of eigenstrain field ($\theta_0 = c$; $\theta_i = \eta_i^2$ ($i = 1, 2, 3$))
$\mathbf{u}$	Displacement field in real space
$\tilde{\mathbf{u}}$	Displacement field in Fourier space
V	Volume of the domain

Chapter 1

Introduction

Many technologically important alloys, such as Ni base superalloys and Ti-Al base alloys, contain coherent precipitates in their microstructures. Phase transformations in these alloys during processing produce ordered precipitate phases that differ from the parent matrix phase, both compositionally and structurally. The structural changes during transformations often lead to precipitates whose crystallographic symmetry is a subgroup of that of the parent phase. As a result, multiple orientational variants of the precipitate phase may form in the microstructure which arrange in complex spatial patterns.

When the lattice parameters of these orientational variants differ from that of the matrix phase, each variant may have its own (anisotropic, crystallographically equivalent) misfit with the matrix. Elastic stresses arising due to this misfit strongly influence the microstructures in these alloys. For example, in Ti-Al-Nb alloys, the variants of the orthorhombic precipitate phase in a hexagonal matrix form self-accommodating patterns to minimize the elastic energy [5–7]. Other examples include tetragonal precipitates in a cubic matrix in Co-Pt alloys and in Y_2O_3 - ZrO_2 [8, 9].

Elastic stresses produce several interesting effects on microstructures, such as changes in particle shape, alignment of particles along certain crystallographic directions, and par-

ticle splitting (see reviews by Doi [10] and Fratzl *et al.* [11]). There have been extensive theoretical attempts to elucidate the effect of elastic stresses during microstructural evolution. Most of these studies have considered systems with dilatational misfit. There are relatively few studies which have taken anisotropy in misfit into account.

The role of an anisotropic misfit was explored systematically by Sankarasubramanian *et al.* [1] who studied equilibrium shapes of isolated precipitates. In particular, they examined symmetry breaking transitions in equilibrium shapes, and their dependence on the anisotropy in misfit. However, due to their emphasis on equilibrium shapes, phenomena such as (a) interactions among multiple particles and variants, and (b) dynamical conditions (e.g., growth and coarsening) were beyond the scope of their study.

The effect of an anisotropic misfit on microstructural evolution was studied by several groups using phase field models [12–14]. However, their studies are restricted to only one particular kind of anisotropy in misfit (specifically, pure shear; see, for example, Ref. [15, 16]). However, when the precipitate-matrix misfit is anisotropic, this anisotropy encompasses a whole range of possibilities; in two dimensions, the anisotropy could lie anywhere between pure dilatation (in which the principal misfit strains are equal) at one end, and pure shear (in which they are equal in magnitude, but have opposite signs) at the other. Thus, in this thesis, we have undertaken a systematic study of the effect of different levels of anisotropy in misfit strain on microstructural evolution.

This work may be seen as sharing a similar spirit with the work of Sankarasubramanian *et al.* [1, 17]. Like them, we too explore systematically the role of anisotropy in misfit. There are, however, several key differences: their work focuses on equilibrium shapes, while ours focuses on evolution of microstructures. Theirs is on a single isolated precipitate, while ours deals with multivariant, multi-particle microstructures. Theirs is based on a sharp-interface model (coupled with the finite element method for solving for elastic stress fields), while ours is based on a phase field model (coupled with a Fourier transform method for solving for both stress and diffusion fields).

Our work may also be seen as building on that of Wen *et al.* [13], who also studied microstructures involving hexagonal and orthorhombic phases using a phase field model. The misfit in their study, however, was restricted to a pure shear; we remove this restriction to explore more fully the other possibilities for anisotropic misfit.

Finally, our work is inspired by the detailed experimental study of Ti-Al-Nb alloys by Muraleedharan *et al.* [2]. In these alloys, too, the microstructure has precipitates of the O phase with an orthorhombic symmetry in a hexagonal α_2 matrix. The misfit between one of the variants of the O phase and the matrix is very close to pure shear. Indeed, one of the cases of misfit anisotropy we have studied – pure shear – is close to that observed in these alloys, thus allowing us to undertake an extensive comparison of the microstructures from our simulations with those reported by Muraleedharan *et al.*

The broad aim of the thesis is to study the effect of different kinds (or, levels) of anisotropy in misfit on the evolution of multivariant microstructures. For this purpose, we have used a phase field model – similar to the one used by Wen *et al.* [13] – in a two dimensional system in which multiple variants of an ordered precipitate phase form in a disordered matrix. We have used this model to study five different systems each with a different anisotropy in misfit. We have studied the microstructural evolution in both single particle and multiparticle settings. The growth shapes from the single particle simulations are compared with the equilibrium shapes reported by Sankarasubramanian *et al.* [1]. And our results from multivariant, multiparticle simulations are compared with those from Muraleedharan *et al.* [2].

The thesis is organized as follows: in Chapter 3 we present the details of the formulation to solve the governing equations. In Chapter 4, we present the results for the dynamical growth shapes. In Chapter 5, we present the results for the evolution of multivariant microstructures and subsequently discuss them. Chapter 6 summarizes the major conclusions from this study.

Chapter 2

Literature Review

The effect of elastic stresses arising from a lattice parameter mismatch during solid state phase transformations has been studied extensively, both experimentally and theoretically. These studies cover both diffusional and displacive phase transformations; see the classic monographs by Khachaturyan [4] and by Mura [18], and several excellent reviews [10, 11, 19, 20]. In this chapter, we restrict ourselves to an overview of the experimental and theoretical studies that show how elastic stresses (arising due to lattice mismatch) influence the morphology of coherent precipitates. We specially emphasize on the microstructural features in systems with anisotropic misfit, where the lattices of the parent and product phases possess different lattice parameters along different directions.

2.1 Experimental observations

There is a significant amount of experimental evidence that shows how stresses can affect the precipitate morphology in several ways. In a classic study, Ardell and Nicholson [21] have shown that coherent Ni_3Al (γ') precipitates in a Ni-6.71 wt% Al alloy change their shape from sphere to cuboids, and later, to plates with increasing size. They have also observed that the precipitates align themselves along elastically soft $\langle 100 \rangle$ directions. In

an appendix to this paper, Eshelby postulated that the preferential alignment of the precipitates resulted from elastic interactions among them. Following their work, a large number of experimental studies have examined the effect of elastic stresses during precipitation of ordered intermetallic γ' -Ni₃X (X=Al, Ge, Si, etc.) (having a $L1_2$ structure) in disordered γ matrix (with an fcc structure) [22–28]. While the misfit strain of the γ' phase is isotropic, the elastic energy of the precipitate is minimized when the precipitate assumes a plate-like shape. In these cases, the particle shape change and alignment is driven by anisotropy in elastic constants.

In coherent solids, externally applied stress can lead to preferential coarsening of precipitates which is also known as “rafting”. Tien and Copley [29] reported such a preferential alignment of γ' precipitates parallel or perpendicular to the direction of applied uniaxial stress in Ni base superalloy single crystals. A combination of several factors, such as misfit strain, elastic modulus mismatch and applied stress, are shown to influence this phenomenon of rafting [30].

Further, elastic stresses can produce morphological instabilities such as particle splitting. The phenomenon of particle splitting, where precipitates split into several smaller ones at a particular critical size, has been reported in several systems [31–35]. The splitting of a single coherent precipitate into doublets of plates or octets of cubes has been attributed to elastic interaction energy.

The effect of elastic stresses on the microstructures is even more interesting in alloys where the precipitate phase has a lower crystallographic symmetry than that of the parent phase. The reduction in crystallographic symmetry results in the formation of several orientational variants of the precipitate phase, each with its own anisotropic misfit. One such example is partially stabilized zirconia (PSZ) where tetragonal precipitates form in a cubic matrix [36]. The cubic phase (c) in ZrO₂-Y₂O₃ system may transform to a tetragonal phase (t) by either a diffusional or a diffusionless transformation. Heuer *et al.* [37, 38] have reported that a displacive transformation in ZrO₂-Y₂O₃ results in the formation of

a tetragonal phase, t' , which has the same composition as the parent cubic phase. On the other hand, if the system is aged in the “ $c + t$ ” two-phase region, precipitation of t phase occurs by diffusional transformation [39, 40]. Transmission electron microscopy studies have shown a characteristic image contrast consisting of fine striations which may arise due to coherency strains around fine tetragonal precipitates [41]. The tetragonal phase has three orientational variants (dictated by the crystallographic symmetry) which arrange to form twin related patterns in the microstructure [42].

Bateman and Notis [43] have studied the effect of coherency strains on precipitate morphology and distribution in two Mg based partially stabilized zirconias as a function of aging time at 1420°C. At short aging time, they observed an alternating band structure, each band containing lens-shaped plates of the same orientation variant. A kinked plate-like morphology of precipitates was observed during coarsening caused by the rotation of habit plane from $\{100\}$ to one of the form $\{hhl\}$. The increase in aspect ratio of the precipitates and the rotation of precipitate habits were explained using Khachaturyan’s microelasticity theory [4]. Alternating band structures, where the variants in adjacent bands are twin related, are also reported in a Ni-Fe-Cr alloy which undergoes cubic $\rightarrow$ tetragonal transformation [44]. Development of kinked plate-like morphology during aging is reported in Ti-Al-Ag system [45] during precipitation of $L1_0$ -TiAl(Ag) phase in $L1_2$ -(Al,Ag) $_3$ Ti matrix.

Moreover, interaction between multiple variants can give rise to complex morphological patterns during diffusional or displacive transformations. In Co-Pt [8] and (CuAu)-Pt alloys [46], a checkerboard-like (or tile-like) microstructure develops during aging. The microstructures in these alloys consist of coherent precipitates of the tetragonal phase in a cubic matrix. The checkerboard-like patterns are formed by a regular arrangement of parallel coherent plates of cubic phase and two variants of the tetragonal phase. The plates are aligned along the $[001]$ cubic axis. The variants of the tetragonal phase are diamond shaped while the cubic phase has a square like shape. Development of “sandwich” like

structures formed by alternating arrangement of regularly spaced tetragonal domains of the precipitate phase is observed in Ti-Al-Ag system [47] during precipitation of DO_{23} $\text{Al}_{11}\text{Ti}_5$ phase in Ag-modified L1_2 - Al_3Ti matrix. In an Al-6Ti-1.5Ni (wt%) alloy [48], the microstructure initially consists of a fine scale distribution of coherent particles of metastable L1_2 phase in disordered fcc matrix. During aging the particles develop a three-dimensional cross-like morphology associated with the structural transformation from metastable L1_2 to equilibrium DO_{22} phase, through an intermediate DO_{23} structure.

Hexagonal to orthorhombic transformation also results in the formation of several unique multivariant patterns; examples include Ti-Al-Nb alloys [2, 6, 7, 49, 50], Mg-Cd alloys [51], Ta_4N [52], and lead orthovanadate [53, 54]. In Ti-Al-Nb system the formation of elastic energy minimizing patterns have been reported for displacive as well as diffusional transformations. Though the modes of transformation are different, the characteristic morphological features were found to be similar in this system. Bendersky *et al.* [6] have used a sequence of crystallographic relationships developed from subgroup symmetry relations to analyze the coherent displacive transformation of b.c.c. high temperature phase to low temperature h.c.p. or orthorhombic phases in Ti-Al-Nb system. They determined the orientation of interfaces between the domains by considering the existence of strain-free interfaces between the domains. They also predicted the formation of multivariant self accommodating patterns which minimize the elastic energy.

In a later paper, Bendersky and Boettinger [7] reported the transmission electron microscopy studies of the microstructures in Ti-Al-Nb system for varying Nb content. They observed self accommodating patterns consisting of three variants of the orthorhombic O phase. Detailed experimental studies were performed by Muraleedharan *et al.* [2] to characterize the orientation relationships, habit planes, antiphase domain boundaries and self accommodation of the orientational variants during precipitation of O phase in hexagonal α_2 matrix in a Ti-28.5at% Al-13at% Nb alloy. They found the interface between the product and the parent phases to be completely coherent. The microstructural features

such as the average habit plane, the orientation relationships and specific combinations of the variants of the product phase were explained in terms of the phenomenological theory of martensitic transformations [55]. They also reported re-transformation of the product phase to parent phase at specific locations in the microstructure during coarsening. Similar morphological features have been reported by Wen *et al.* [13] in their study of precipitation of O phase in Ti-Al-Nb alloys with varying Nb content.

The features of self accommodation that occur during precipitation of an ordered phase with lower crystallographic symmetry than the parent phase are common to martensitic transformations [56, 57]; self accommodating clusters of variants were observed in several alloys undergoing martensitic reaction. In Fe-Pt, Cu-Zn, Ni-Ti shape memory alloys [58–60], the strain is accommodated by the formation of parallel plates or by a zig-zag diamond-like morphology. In Zr-Nb alloys, self accommodation leads to the arrangement of primary variant plates in a triangular morphology; secondary martensite plates arrange in an “indentation mark” morphology [61]. In both martensitic and precipitation transformations, the formation of such patterns are dictated by accommodation of misfit strains between the variants.

2.2 Theoretical studies

Various theoretical and computational studies have investigated the effect of elastic stresses on the morphology of the coherent precipitates and their evolution. Based on the nature of the methods used by them, these studies can be broadly classified into three categories: (a) sharp interface models, (b) atomistic models, and (c) diffuse interface models (also known as phase field models). We provide a brief description of each of these models and techniques which have been used for studying microstructural evolution.

2.2.1 Sharp interface models

In these models, the phases are assumed to be continuous up to the interface, and the interface between them is defined to be geometrically sharp entity with zero thickness [11]. Earlier studies for understanding the effect of elastic stresses on the morphology of the precipitates in coherent solid systems were based on these models [62–64].

The sharp interface models have been extensively used to study the equilibrium shapes of precipitates using an energy based approach. For a given particle size, the equilibrium shape of a coherent particle is obtained by minimization of an appropriate energy of the system. When interfacial energy alone is taken into consideration, the equilibrium shape of a particle is determined by the anisotropy in interfacial energy and is given by Wulff construction [65]. Also, there have been attempts to predict the equilibrium shapes by minimization of elastic energy alone, assuming certain geometrical shapes, such as, ellipsoids, cubes or plates [4, 64, 66, 67]. More recently, several studies have examined the equilibrium shapes taking both interfacial and elastic energies into account [1, 17, 68, 69]; in these studies, the equilibrium shape is obtained by the minimization of the sum of interfacial and elastic energies. These studies show that the equilibrium shapes of precipitates depend crucially on their sizes; interfacial energy dictates the shape when the particle size is smaller and elastic energy dominates when the particle size is larger.

In their study of two dimensional isotropic systems with dilatational misfit, Johnson and Cahn [68] showed that the equilibrium shape undergoes a symmetry breaking transition from circle to ellipse at a critical size. Such transitions have also been observed in two dimensional systems with cubic elastic anisotropy and an anisotropy in misfit; see [70] for a review.

Sharp interface models have also been used to study temporal evolution of microstructures (see [11] and references therein). However, these studies require explicit tracking of the interface at all times. Hence, it is numerically difficult to implement these models

in multiparticle settings where the particles may have arbitrary shapes. Moreover, these models are restricted by their inability to account for splitting, coalescence or dissolution of precipitates.

2.2.2 Atomistic models

These models are based on a microscopic description of the system in terms of atoms; the individual atoms are assumed to be connected to their neighbors through mechanical springs. The interactions between the atoms determine the elastic properties of the system. Further, in these models, the interaction between a pair of atoms depends on the chemical nature of the atoms as well as on their separation distance. Lee has used such a discrete atom model where he assumes the springs to be longitudinal, harmonic ones [71–73]; the stiffness of the springs are also assumed to be different for different orientations. Microstructural evolution is studied in this model using Monte Carlo simulations. Lee and co-workers used this model to study the morphological evolution of coherent precipitates with an arbitrary misfit in elastically anisotropic and inhomogeneous systems. Fratzl and co-workers have extended this model to include additional elastic interactions arising from transverse springs connecting the nearest neighbors and longitudinal springs connecting the next nearest neighbors [74, 75]. Their model is characterized by three elastic constants and is used to study coarsening in systems with cubic anisotropy.

2.2.3 Diffuse interface models

Unlike the sharp interface models, the diffuse interface models do not assume the boundary between the phases to be geometrically sharp interfaces. In these models there is no restriction imposed by an arbitrary assumption of interfacial width, and the properties change smoothly over a definite width from one phase to another.

Cahn and Hilliard [76] introduced a diffuse interface model for studying the free energy

in compositionally nonuniform systems. Cahn used this model to study spinodal decomposition where the interfaces formed during the early stages are indeed compositionally diffuse [77]. A later paper [78] extended this concept to systems with ordered domains, wherein the antiphase domain boundaries are treated as regions of finite width in which the long range order parameters exhibit gradients.

In these models, the temporal evolution of microstructures is governed by the Cahn-Hilliard nonlinear diffusion equation for composition (conserved field) and Allen-Cahn equation for non-conserved order parameter fields [77,78]. The main advantage of these models is that the field variables describing the microstructure are continuous everywhere which makes them more adept at studying complex topological changes (such as splitting) than their sharp interface counterparts. These models are also referred to as phase field models since the field variables involved in these models are also called phase fields. They describe the microstructural features at the mesoscale and can address a broad spectrum of problems: see [79,80] for recent reviews.

The phase field models have been successfully used to study microstructural evolution in elastically stressed solids [81]. Wang *et al.* [82] have used a phase field model to study coherent precipitation of ordered $L1_2$ phase from a disordered f.c.c. solution. Their formulation of the free energy functional of composition and long range order parameters is based on a static concentration wave representation of the ordered state. Their study captures several essential features of the microstructural evolution during precipitation, namely, nucleation, growth and coarsening of the ordered domains. They have also analyzed the effect of antiphase boundary energy and elastic stresses on the growth and coarsening kinetics of ordered domains.

The microstructural evolution becomes more complex during the precipitation of an ordered phase with a reduced crystallographic symmetry than the parent phase. Wang *et al.* addressed such a problem in their study of precipitation of tetragonal phase from Mg-partially-stabilized cubic zirconia [14,83]. Their simulations show correlated nucleation

of different orientational variants of the tetragonal phase with the nuclei forming misfit strain accommodating patterns. They also observe alternating band structures, each band containing lens-shaped particles of the same orientation variant. Their simulation results show a good agreement with the experimental observations. Twin formation during displacive cubic $\rightarrow$ tetragonal transformation in yttria-zirconia system has also been investigated using phase field models [84].

Le Bouar *et al.* [12] have examined the mechanism of formation of chessboard-like structures during coherent precipitation of ordered tetragonal $L1_0$ phase in a cubic matrix. Their simulation results show that these patterns are formed by elastic strain driven self-assembly of the orientation variants of the tetragonal precipitate phase. The simulated microstructures during the various stages of precipitation are also shown to be in excellent agreement with the experimental observations.

Wen *et al.* [13,15] have developed a phase field model to study the dynamic evolution of multivariant microstructures during coherent hexagonal $\rightarrow$ orthorhombic transformation in the presence or absence of applied strain field. Their model can handle transformations which involve pure ordering as well as precipitation. Their simulation results compare well with the experimentally observed morphological patterns. More recently, Cui *et al.* [85] have modeled hexagonal $\rightarrow$ orthorhombic phase transformation in polycrystals using a combination of strain-only theory and phase field modeling.

All these studies listed above demonstrate the usefulness of phase field models to predict the experimental observations. However, most of these dynamic studies have either considered systems with pure dilatational misfit or pure shear misfit. Hence, in this thesis, we undertake a systematic study of the effect of anisotropy in misfit strain on microstructural evolution during precipitation of an ordered orthorhombic phase in a hexagonal matrix. We also examine the complex interplay of interfacial energy, antiphase boundary energy and different levels of anisotropy in misfit on the evolution of multivariant microstructures.

Chapter 3

Formulation

In this chapter, we present a phase field model for studying microstructural evolution in a binary two phase system in which one of the phases can exist in the form of one of several variants. Our formulation closely follows that of Wen *et al.* [13]; it describes the two phase multivariant microstructure using a set of conserved and non-conserved field variables. It accounts for misfit strains arising from compositional inhomogeneity (dilatational misfit strain), and inhomogeneity in structural order parameter (anisotropic misfit strain). However, it is the latter that we have used in our study.

We describe the model system in section 3.1. Section 3.2 describes the energetics of this system; the total free energy of the compositionally and structurally inhomogeneous binary alloy system is expressed as a function of the field variables: composition and the required number of order parameters. The chemical free energy of this system is obtained using a modified Ginzburg-Landau free energy functional; the elastic strain energy of the system is derived using Khachaturyan's microelasticity theory [4]. Then, we present the kinetic equations (Cahn-Hilliard and Allen-Cahn equations) which govern the temporal evolution of the conserved and non-conserved field variables in section 3.3. In section 3.5, we present the details of the numerical technique used for solving the evolution equations.

In what follows, scalars are denoted by lightface letters, while vectors and higher order tensors are denoted by boldface letters. The equations are presented in their non-dimensional forms. All the quantities in the equations are rendered non-dimensional using a characteristic length L_n , a characteristic energy E_n and a characteristic time T_n . The non-dimensionalisation procedure will be discussed at the end of this chapter.

3.1 Model system

We consider a binary alloy system containing two atomic species A and B which undergoes hexagon $\rightarrow$ rectangle order-disorder transformation (equivalent to hexagonal $\rightarrow$ orthorhombic transformation in 3D). Ordered rectangular variants of the precipitate phase (denoted by β) form due to ordering of the 2D disordered hexagonal lattice of the parent phase (denoted by α). A phase diagram for this system is shown in Fig. 3.1.

At a high temperature T_1 , the alloy with composition x_1 (where x_1 is the mole fraction of species B) lies in the single phase α field of the phase diagram. When the alloy is quenched to a lower temperature T_2 , it finds itself inside the two phase $\alpha + \beta$ field. At this temperature, the alloy is a metastable supersaturated solid solution from which the β phase, with an equilibrium composition x_β , precipitates out and a more stable solid solution of α , having the equilibrium composition x_α , is formed.

The 2D disordered hexagonal lattice of the α phase and the ordered rectangular variants of the β phase are shown schematically in Fig. 3.2. The A and B atoms randomly occupy any site of the disordered, hexagonal lattice of the parent α phase and all the lattice sites are equivalent (Fig. 3.2a). When the parent phase undergoes ordering, it transforms into rectangular β phase in which the A and B atoms are preferentially arranged in alternate layers (normal to the y axis in Fig. 3.2b). Since there are three crystallographically equivalent directions in a hexagonal lattice (denoted by y , y' and y'' in the figure), the disordered regions in the system may produce ordered structures on transfor-

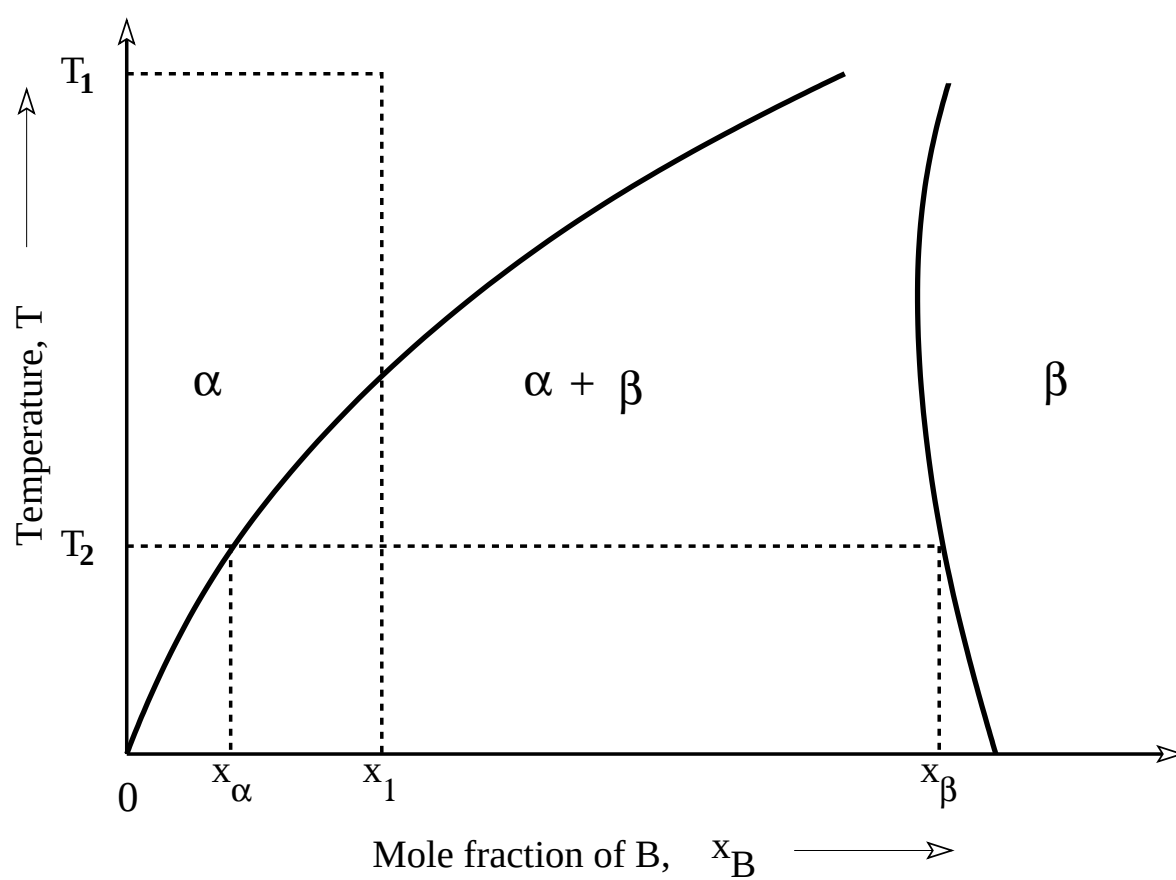


Figure 3.1: Phase diagram of the model system (schematic)

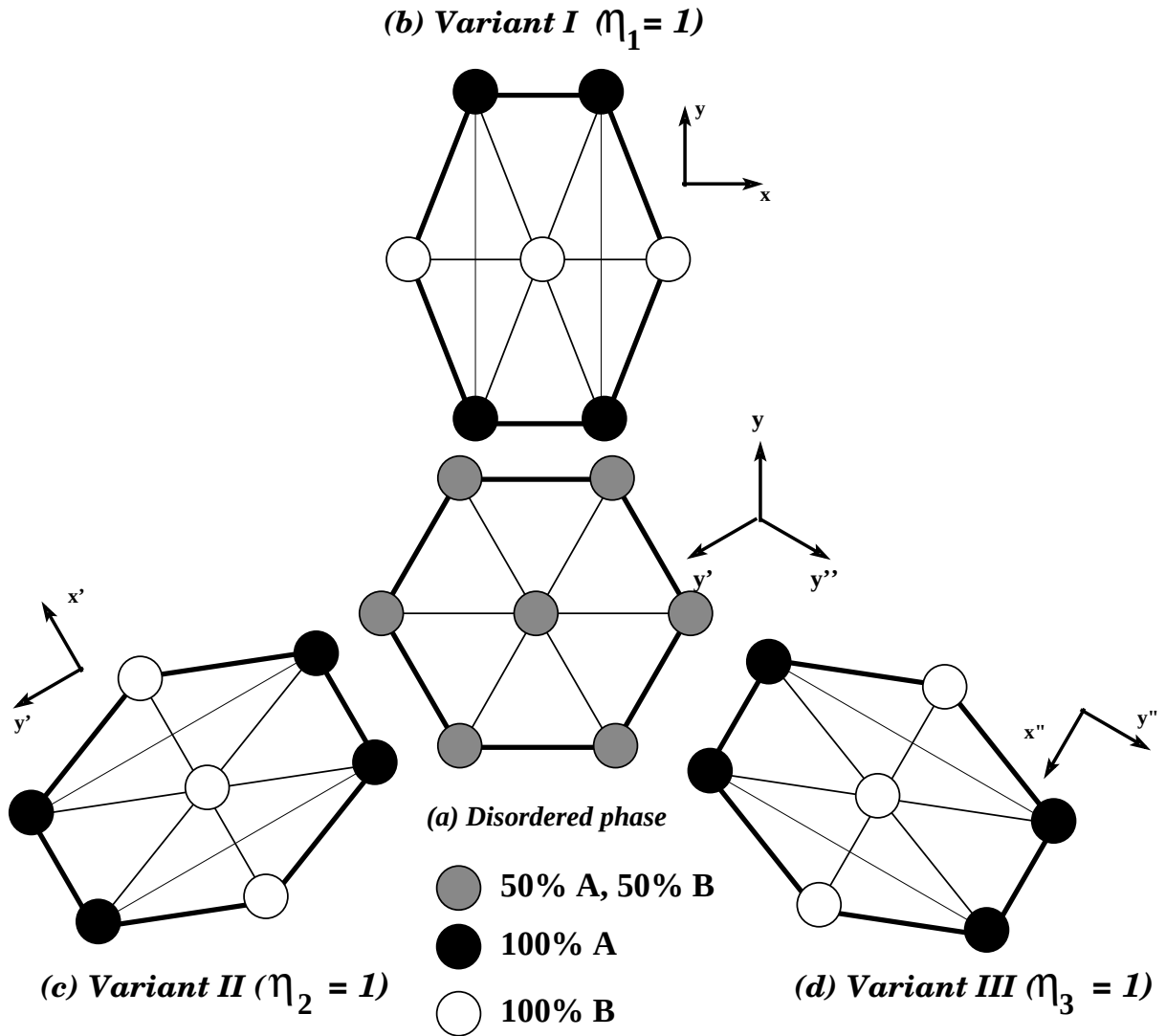


Figure 3.2: Structure of (a) two-dimensional disordered, α phase and (b–d) the ordered rectangular variants of the β phase (schematic)

mation, normal to any of these three directions. The ordered variants, thus formed, are referred to as the orientational variants. We use three scaled non-conserved field variables η_i ($i = 1, 2, 3$) (also referred to as long range order parameter fields) to describe the three orientational variants of the rectangular β phase. Further, we note that the rectangular ordered superlattice can be represented as the superposition of three static concentration waves [4],

$$x_B - \bar{x}_B = (\eta'_1/2)\cos(2\pi y/\lambda) + (\eta'_2/2)\cos(2\pi y'/\lambda) + (\eta'_3/2)\cos(2\pi y''/\lambda),$$

whose wavelength $\lambda = \sqrt{3}a$ and whose wave vectors are parallel to positive y , y' and y'' directions; a is the lattice parameter of the hexagonal lattice, $\bar{x}_B$ is the average composition of the alloy, x_B is the local composition, η'_1 , η'_2 and η'_3 are the long range order parameters which represent the amplitudes of the corresponding concentration waves. Order parameters, defined in this way, are constrained by the inequality $\eta'_1 + \eta'_2 + \eta'_3 \leq 1$.

We define scaled long range order parameters, $\eta_i = \eta'_i/\eta^*$ ($i = 1, 2, 3$), where η^* represents the equilibrium order parameter for a given composition and temperature. Therefore, for any given composition, $(\eta_1, \eta_2, \eta_3) = (000)$ defines the completely disordered, hexagonal structure of the α phase, whereas $(\eta_1, \eta_2, \eta_3) = (100)$, (010) or (001) correspond to each of the three completely ordered orientational variants of the β phase.

The first orientation variant, shown in Fig. 3.2b, is characterized by $\eta_1 = 1$, with the first and the second layers alternately occupied by A and B atoms, respectively. The orientation of this variant with respect to the parent hexagonal lattice remains unchanged if the two layers were occupied in the reverse order by B and A atoms, respectively. Thus, the same orientational variant may be decorated with alternate layers of A and B atoms in two possible ways, shown in Fig. 3.3; the first one is characterized by $\eta_1 = 1$ and the second one is characterized by $\eta_1 = -1$. The ordered structures, thus produced, are referred to as the two translational variants of the same orientational variant. Similarly, each of

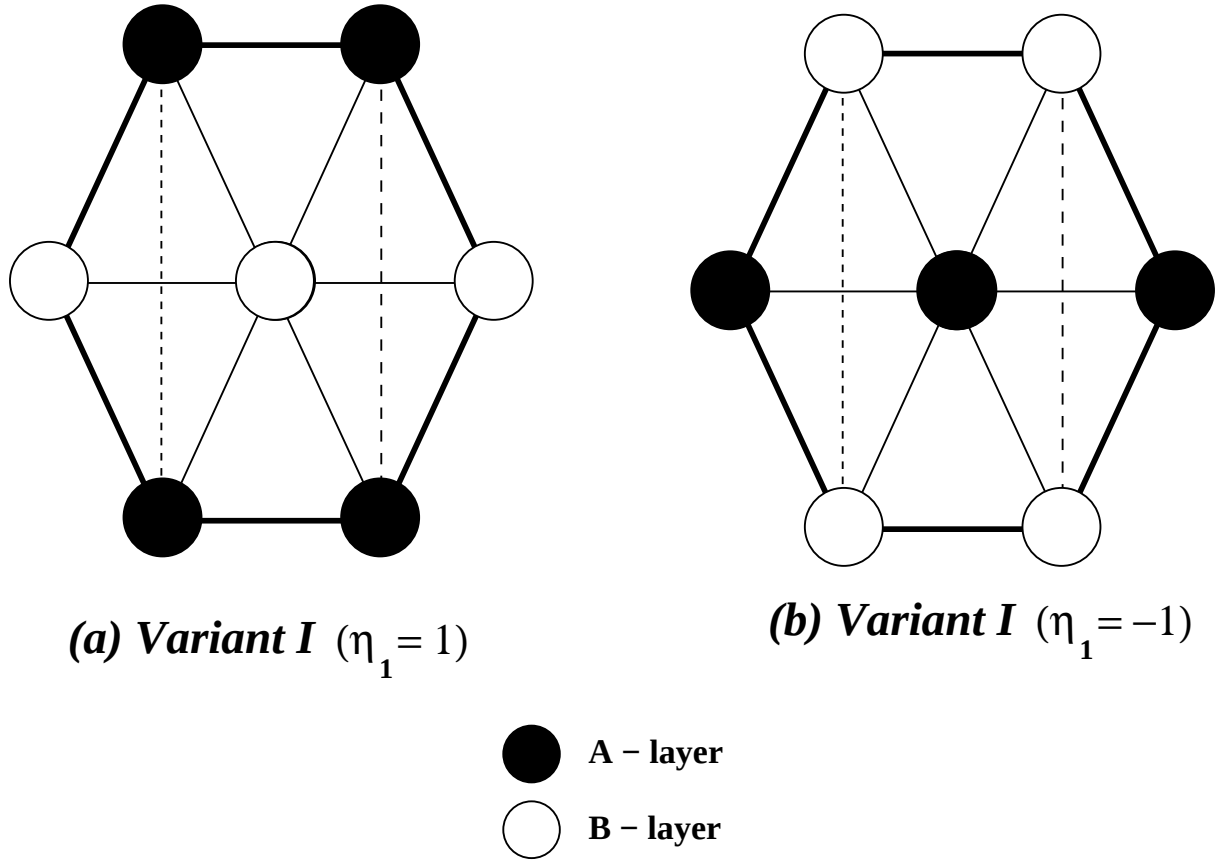


Figure 3.3: Two translational variants of the first orientational variant (schematic)

the other two orientational variants will have two translational variants characterized by $\eta_2 = \pm 1$ and $\eta_3 = \pm 1$. Further, a finite non-zero value of $0 < |\eta_i| < 1$ defines a partially ordered state in which the preferential occupation of A and B atoms has not attained completion, for example, each A-rich layer retains a minor amount of B atoms.

Thus, the two-phase, multivariant microstructure is fully described using the following phase field variables: composition ($c = (x_B - x_\alpha)/(x_\beta - x_\alpha)$, so that $c = 0$ corresponds to the α phase and $c = 1$ corresponds to the β phase), and three long range order parameters (η_1, η_2, η_3) . The order parameter has a finite value inside the precipitate (taken to be unity for convenience) and is zero in the matrix. The interphase interfaces represent those regions of the microstructure where the order parameter and the composition fields exhibit large gradients.

Misfit strain due to compositional inhomogeneity

In our model system, the local stress-free transformation strain, (also known as eigenstrain) $\epsilon_{ij}^0(\mathbf{r})$, is expressed as a function of each of the four phase field variables (viz. c , η_1 , η_2 and η_3).

The eigenstrain field associated with composition is described by

$$\epsilon_{ij}^0(c) = \beta(c)\epsilon_c\delta_{ij}, \quad (3.1)$$

where, ϵ_c is a constant and determines the strength of the eigenstrain associated with composition, δ_{ij} is the Kronecker delta, and $\beta(c)$ is a function of composition. If we choose the strain free matrix phase as the reference state (i.e. $\beta(c) = c(\mathbf{r})$), then $\beta(c_m) = 0$ and $\beta(c_p) = 1$ (c_m and c_p denote the compositions of the matrix and precipitate phases, respectively). On the other hand, if the average alloy composition $\bar{c}$ is used as the reference state (i.e. $\beta(c) = c(\mathbf{r}) - \bar{c}$), then $\beta(c_m) = -\bar{c}$ and $\beta(c_p) = 1 - \bar{c}$. In both forms of $\beta(c)$, we assume that the lattice parameter a of a disordered alloy increases linearly with composition (Vegard's law), and $\epsilon_c = (1/a_0)(da/dc)$ is the Vegard's law coefficient. Compositional inhomogeneity always gives rise to dilatational or isotropic eigenstrain.

Misfit strain due to structural order parameter inhomogeneity

In our model, an eigenstrain field is also associated with each of the order parameter fields. Here the eigenstrain field is anisotropic, and the extent of anisotropy depends on the precipitate variant. For example, let us consider the first variant ($\eta_1 \neq 0$). When ordering of the parent lattice produces this variant, it may result in different changes in lattice parameters along the x and y directions. This implies that the eigenstrain field $\epsilon_{ij}^0(\eta_1)$ associated with the order parameter field $\eta_1(\mathbf{r})$ may be non-dilatational (i.e. $\epsilon_{xx}^0(\eta_1) \neq \epsilon_{yy}^0(\eta_1)$ in 2D). As we have discussed earlier, two translational variants can

occur for the same orientational variant. The formation of two translational variants is similar to rigid body translation where the orientation of the variant remains unchanged; only the sign of the order parameter associated with the variant changes. Therefore, the eigenstrain field associated with inhomogeneity in η_1 must be invariant with respect to the change in sign of η_1 . Hence, in our model the eigenstrain field $\epsilon_{ij}^0(\eta_1)$ is assumed to be proportional to η_1^2 and may be written as

$$\epsilon^0(\eta_1) = \eta_1^2 \begin{pmatrix} \epsilon_{xx} & 0 \\ 0 & \epsilon_{yy} \end{pmatrix} = \epsilon_\eta \eta_1^2 \begin{pmatrix} 1 & 0 \\ 0 & t \end{pmatrix}, \quad (3.2)$$

where $\epsilon_\eta = \epsilon_{xx}$ is the characteristic misfit strain associated with η_1 , and $t = \epsilon_{yy}/\epsilon_{xx}$ represents the difference between the misfits along the two principal directions i.e. the amount of anisotropy in this misfit strain. The factor t may take any value between -1 and $+1$; we refer to this factor as the misfit anisotropy parameter.

In hexagon $\rightarrow$ rectangle transformation, each variant is rotated to the other by 120° . Hence we obtain $\epsilon^0(\eta_2)$ and $\epsilon^0(\eta_3)$ by the transformation of $\epsilon^0(\eta_1)$ by counterclockwise rotations of 120° and 240° , respectively. Therefore

$$\epsilon^0(\eta_2) = \epsilon_\eta \eta_2^2 \begin{pmatrix} \cos 120^\circ & \sin 120^\circ \\ -\sin 120^\circ & \cos 120^\circ \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & t \end{pmatrix} \begin{pmatrix} \cos 120^\circ & -\sin 120^\circ \\ \sin 120^\circ & \cos 120^\circ \end{pmatrix}.$$

Simplifying, we obtain

$$\epsilon^0(\eta_2) = \epsilon_\eta \eta_2^2 \begin{pmatrix} \frac{1+3t}{4} & \frac{\sqrt{3}(1-t)}{4} \\ \frac{\sqrt{3}(1-t)}{4} & \frac{3+t}{4} \end{pmatrix}. \quad (3.3)$$

Similarly,

$$\epsilon^0(\eta_3) = \epsilon_\eta \eta_3^2 \begin{pmatrix} \frac{1+3t}{4} & \frac{-\sqrt{3}(1-t)}{4} \\ \frac{-\sqrt{3}(1-t)}{4} & \frac{3+t}{4} \end{pmatrix}. \quad (3.4)$$

The total eigenstrain field ϵ^0 , which is the sum of the four eigenstrain fields $\epsilon^0(c)$, $\epsilon^0(\eta_1)$, $\epsilon^0(\eta_2)$ and $\epsilon^0(\eta_3)$, can be written in a more compact form as

$$\epsilon_{ij}^0 = \sum_{p=0}^3 \theta_p \epsilon_{ij}^{(p)}, \quad (3.5)$$

where $\theta_0 = c$, $\theta_i = \eta_i^2$ ($i = 1, 2, 3$), and $\epsilon_{ij}^{(p)}$ denotes the position independent part of the eigenstrain field associated with θ_p .

The eigenstrain fields in our model, described in terms of θ_p and $\epsilon_{ij}^{(p)}$, are listed below:

$$\theta_0 = c ; \quad \epsilon_{ij}^{(0)} = \epsilon_c \delta_{ij} \quad (3.6a)$$

$$\theta_1 = \eta_1^2 ; \quad \epsilon_{ij}^{(1)} = \begin{pmatrix} 1 & 0 \\ 0 & t \end{pmatrix} \quad (3.6b)$$

$$\theta_2 = \eta_2^2 ; \quad \epsilon_{ij}^{(2)} = \begin{pmatrix} \frac{1+3t}{4} & \frac{\sqrt{3}(1-t)}{4} \\ \frac{\sqrt{3}(1-t)}{4} & \frac{3+t}{4} \end{pmatrix} \quad (3.6c)$$

$$\theta_3 = \eta_3^2 ; \quad \epsilon_{ij}^{(3)} = \begin{pmatrix} \frac{1+3t}{4} & \frac{-\sqrt{3}(1-t)}{4} \\ \frac{-\sqrt{3}(1-t)}{4} & \frac{3+t}{4} \end{pmatrix} \quad (3.6d)$$

3.2 Energetics

The total free energy, $\mathcal{F}$, of the system is written as a sum of chemical ($\mathcal{F}_c$) and elastic ($\mathcal{F}_e$) contributions.

$$\mathcal{F} = \mathcal{F}_c + \mathcal{F}_e.$$

Each of these contributions is described below.

Chemical free energy

The chemical free energy $\mathcal{F}_c$ of the inhomogeneous system as a function of the field variables is expressed using the coarse grained Ginzburg-Landau free energy functional [13]

$$\mathcal{F}_c = N_v \int_V \left[f_0(c, \eta_1, \eta_2, \eta_3) + \kappa_c (\nabla c)^2 + \kappa_\eta \left(\sum_{i=1}^3 \nabla \eta_i \right)^2 \right] dV, \quad (3.7)$$

where N_v is the number of molecules per unit volume, $f_0(c, \eta_1, \eta_2, \eta_3)$ is the bulk free energy density, κ_c and κ_η are the gradient energy coefficients associated with the gradients in composition and order parameters respectively.

The bulk free energy density f_0 is approximated using a Landau type polynomial expansion and may incorporate any number of terms allowed by the nature of the transformation (first order or higher order transformation) and the symmetry operations with respect to the parent phase. For hexagon $\rightarrow$ rectangle transformation, it suffices to include terms up to the sixth order in η_i [13]. Thus, the local free energy density is approximated by the following expression:

$$\begin{aligned} f_0(c, \eta_i) = & A_1 c^2 + A_2 (1 - c) (\eta_1^2 + \eta_2^2 + \eta_3^2) \\ & + A_{41} (\eta_1^4 + \eta_2^4 + \eta_3^4) + A_{42} (\eta_1^2 \eta_2^2 + \eta_2^2 \eta_3^2 + \eta_1^2 \eta_3^2) \\ & + A_{61} (\eta_1^6 + \eta_2^6 + \eta_3^6) + A_{62} (\eta_1^2 \eta_2^2 \eta_3^2), \end{aligned} \quad (3.8)$$

where the coefficients A_1 , A_2 , A_{41} , A_{42} , A_{61} and A_{62} are the parameters in our model. Since hexagon $\rightarrow$ rectangle transformation is thermodynamically first order, the stability of the ordered phase requires that $A_{41} < 0$ and $A_{61} > 0$.

The above bulk free energy function $f_0(c, \eta_i)$ has seven global minima at $(c, \eta_1, \eta_2, \eta_3) = (0, 0, 0, 0)$, $(1, 1, 0, 0)$, $(1, 0, 1, 0)$, $(1, 0, 0, 1)$, $(1, -1, 0, 0)$, $(1, 0, -1, 0)$ and $(1, 0, 0, -1)$. At these values of the field variables, $f_0(c, \eta_i) = 0$. The free energy surface, plotted as a function of η_1 and η_2 (with $c = 1$ and $\eta_3 = 0$), is shown in Fig. 3.4. The two minima at

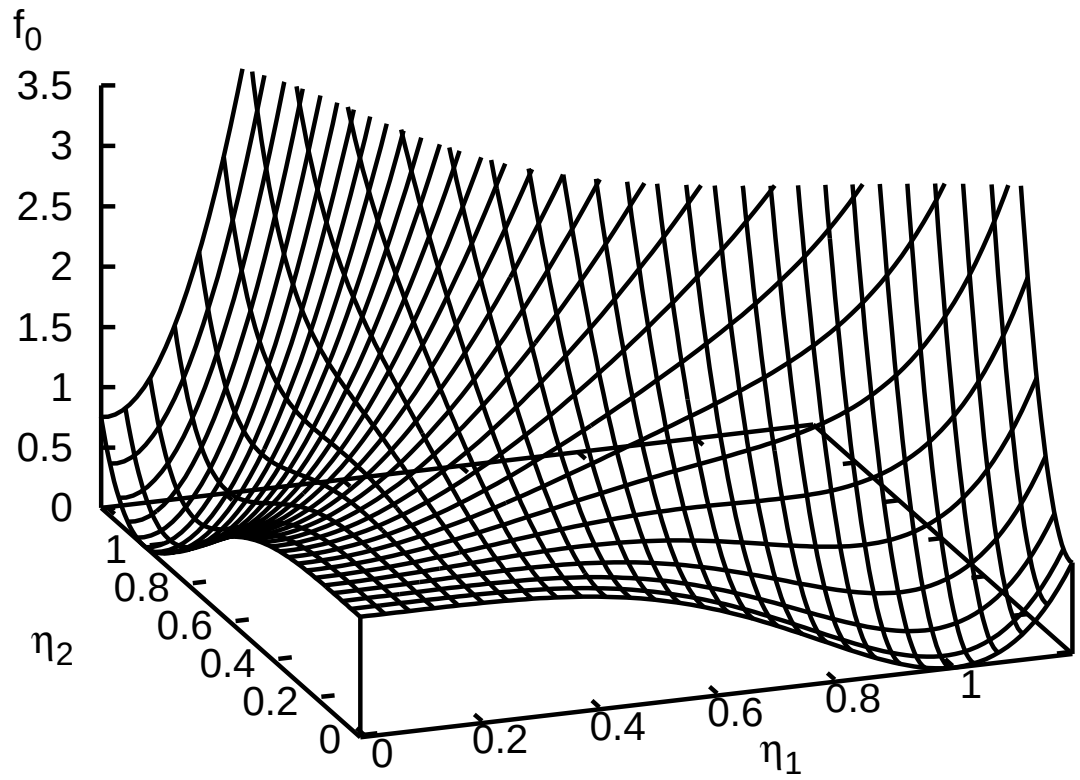


Figure 3.4: Bulk free energy surface f_0 plotted as a function of η_1 and η_2 for an alloy of composition $c = 1.0$ (η_3 is assumed to be zero). The surface shows two minima at $(\eta_1, \eta_2) = (1, 0)$ and $(0, 1)$

$(\eta_1 = 1, \eta_2 = 0)$ and $(\eta_1 = 0, \eta_2 = 1)$ are shown in the figure. f_0 has two more minima symmetrically located at $(\eta_1 = -1, \eta_2 = 0)$ and $(\eta_1 = 0, \eta_2 = -1)$; for clarity, negative values of η_i are not shown.

The bulk free energy of the disordered α phase, f_α , is obtained by setting $\eta_1 = \eta_2 = \eta_3 = 0$ in Eq. 3.8:

$$f_\alpha = A_1 c^2. \quad (3.9)$$

f_α has a simple parabolic form as shown in Eq. 3.9.

To obtain the free energy of the homogeneous, ordered β phase, f_β , we rewrite f_0 in a simplified form as a function of c and only one of the order parameters, say η , by setting $\eta_2 = \eta_3 = 0$ and $\eta_1 = \eta$ in Eq. 3.8:

$$f_0(c, \eta) = A_1 c^2 + A_2(1 - c)\eta^2 + A_{41}\eta^4 + A_{61}\eta^6. \quad (3.10)$$

The equilibrium order parameter ($\eta_e(c)$) is obtained by minimizing $f_0(c, \eta)$ in Eq. 3.10, for each composition c , with respect to η :

$$\frac{\partial f_0(c, \eta)}{\partial \eta} = 2A_2(1 - c)\eta + 4A_{41}\eta^3 + 6A_{61}\eta^5 = 0. \quad (3.11)$$

The above equation has a trivial solution $\eta = 0$ and the other roots are

$$\begin{aligned} \eta_e(c) = & \pm \frac{1}{3} \frac{\sqrt{-3A_{61}(A_{41} - \sqrt{(A_{41}^2 - 3A_{61}A_2 + 3A_{61}A_2c)})}}{A_{61}}, \\ & \pm \frac{1}{3} \frac{\sqrt{-3A_{61}(A_{41} + \sqrt{(A_{41}^2 - 3A_{61}A_2 + 3A_{61}A_2c)})}}{A_{61}}. \end{aligned} \quad (3.12)$$

Of the four roots for η_e given in Eq. 3.12, two roots are physical and give the minimum. These roots are equal in magnitude and opposite in sign. Substituting $\eta_e(c)$ back in

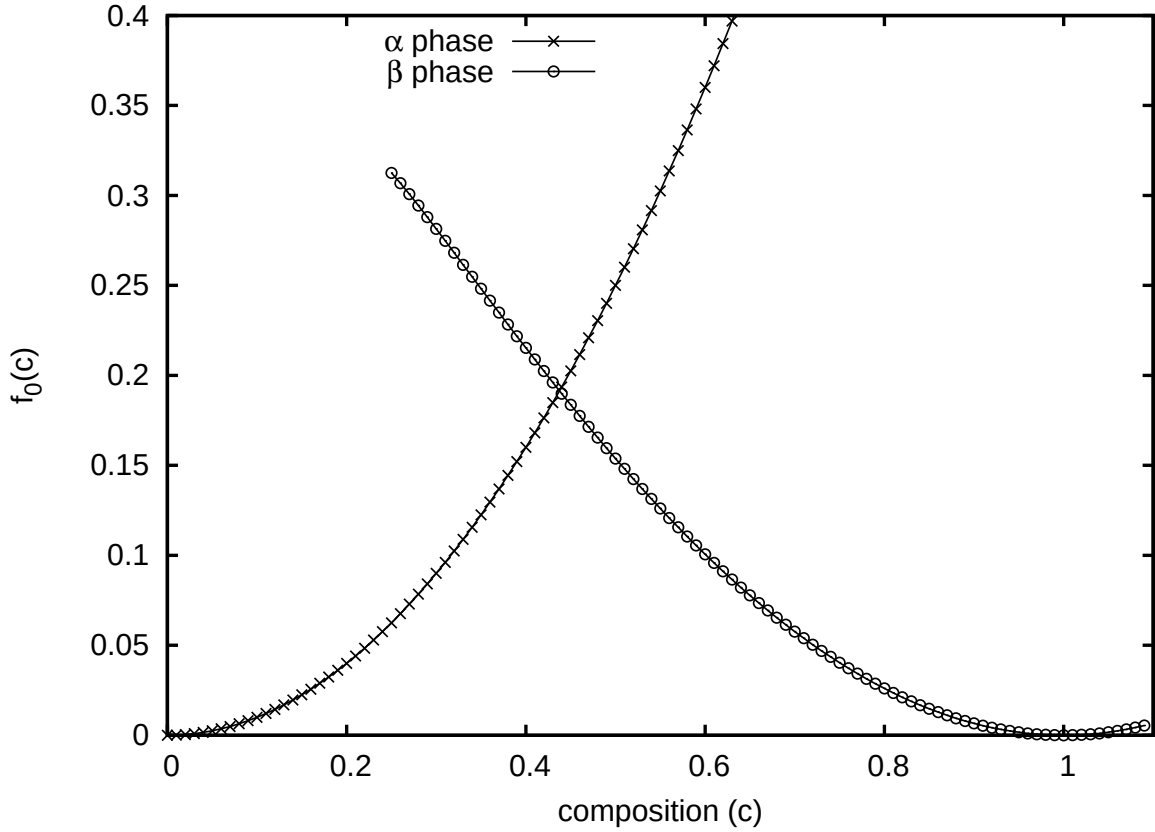


Figure 3.5: Free energy of the disordered α and ordered β phase plotted as a function of alloy composition

Eq. 3.10, we obtain f_β :

$$f_\beta = A_1 c^2 + A_2 (1 - c) \eta_e^2(c) + A_{41} \eta_e^4(c) + A_{61} \eta_e^6(c). \quad (3.13)$$

The free energy vs. composition plots for the α and β phases are shown in Fig. 3.5.

In our model we assume that the gradient energy coefficients κ_c and κ_η are positive (scalar) constants which implies that the interfacial energy between the α and β phases and antiphase domain boundary energy between two orientational variants of the β phase are isotropic. In the chemical free energy functional given in Eq. 3.7, κ_c and κ_η may either be constants or second rank tensors. An anisotropy in interfacial energy may be introduced in the model either by using a tensor form of the gradient energy coefficients,

or by replacing the gradient energy coefficients with an orientation dependent continuous function [86, 87].

Elastic energy

Elastic strain energy for a coherent microstructure (where the lattice directions and planes are continuous across the precipitate and the matrix phases) has contributions due to the misfit strain and inhomogeneous elastic moduli [88, 89]. We assume that our model system is elastically homogeneous i.e. the elastic moduli C_{ijkl} do not depend on c and η_i . The elastic contribution to the free energy is given as

$$\mathcal{F}_e = \frac{1}{2} \int_V \sigma_{ij}^{el}(\mathbf{r}) \epsilon_{ij}^{el}(\mathbf{r}) dV, \quad (3.14)$$

where,

$$\epsilon_{ij}^{el}(\mathbf{r}) = \epsilon_{ij}(\mathbf{r}) - \epsilon_{ij}^0(\mathbf{r}), \quad (3.15)$$

with $\epsilon_{ij}^0(\mathbf{r})$ being the position dependent eigenstrain tensor field, and $\epsilon_{ij}(\mathbf{r})$ the total strain.

The total strain may be represented as a sum of periodic strain, $\varepsilon_{ij}(\mathbf{r})$, and a homogeneous, macroscopic strain E_{ij} :

$$\epsilon_{ij}(\mathbf{r}) = \varepsilon_{ij}(\mathbf{r}) + E_{ij}. \quad (3.16)$$

The periodic strain $\varepsilon_{ij}(\mathbf{r})$ is expressed in terms of the displacement field, $\mathbf{u}(\mathbf{r})$, as:

$$\varepsilon_{ij}(\mathbf{r}) = \frac{1}{2} \left[\frac{\partial u_i(\mathbf{r})}{\partial r_j} + \frac{\partial u_j(\mathbf{r})}{\partial r_i} \right]. \quad (3.17)$$

The homogeneous strain, E_{ij} , is position independent, uniform strain acting on the entire domain and denotes the mean strain tensor of the domain. In the absence of applied

stress, using homogeneous modulus approximation, E_{ij} is obtained as the weighted mean of the total eigenstrain field [90]:

$$E_{ij} = \sum_{p=0}^3 \epsilon_{ij}^{(p)} \bar{\theta}_p \quad (3.18)$$

where $\bar{\theta}_p = \frac{1}{V} \int \theta_p dV$.

Applying Hooke's law (assuming both the phases to be linear elastic), the local elastic stress σ_{ij}^{el} is given by

$$\begin{aligned} \sigma_{ij}^{el}(\mathbf{r}) &= C_{ijkl} \epsilon_{kl}^{el}(\mathbf{r}) \\ &= C_{ijkl} (\epsilon_{kl}(\mathbf{r}) - \epsilon_{kl}^0(\mathbf{r})) \end{aligned} \quad (3.19)$$

where C_{ijkl} is the elastic stiffness tensor of the elastically homogeneous system. Since C_{ijkl} is symmetric ($C_{ijkl} = C_{ijlk}$), using Eq. 3.17 and Eq. 3.18 in Eq. 3.19, σ_{ij}^{el} may be written as:

$$\sigma_{ij}^{el}(\mathbf{r}) = C_{ijkl} \frac{\partial u_k(\mathbf{r})}{\partial r_l} - C_{ijkl} \sum_{p=0}^3 \epsilon_{kl}^{(p)} (\theta_p(\mathbf{r}) - \bar{\theta}_p). \quad (3.20)$$

To obtain the elastic energy of the system, we solve the equation of mechanical equilibrium

$$\nabla \cdot \boldsymbol{\sigma}^{el} = \frac{\partial \sigma_{ij}^{el}}{\partial r_j} = 0. \quad (3.21)$$

Substituting Eq. 3.20 into the equation of mechanical equilibrium (Eq. 3.21) and rearranging the terms, we get

$$\begin{aligned} C_{ijkl} \frac{\partial^2 u_k(\mathbf{r})}{\partial r_j \partial r_l} &= C_{ijkl} \sum_{p=0}^3 \epsilon_{kl}^{(p)} \frac{\partial \Delta \theta_p(\mathbf{r})}{\partial r_j}, \\ &= \sum_{p=0}^3 \hat{\sigma}_{ij}^{(p)} \frac{\partial \Delta \theta_p(\mathbf{r})}{\partial r_j}, \end{aligned} \quad (3.22)$$

where,

$$\hat{\sigma}_{ij}^{(p)} = C_{ijkl}\epsilon_{kl}^{(p)}$$

and $\Delta\theta_p(\mathbf{r}) = \theta_p(\mathbf{r}) - \bar{\theta}_p$. Assuming periodic boundary condition, Eq. 3.22 can be analytically solved in reciprocal space (Fourier space).

In the reciprocal space, the mechanical equilibrium equation takes the form

$$|\mathbf{k}|^2(C_{ijkl}n_jn_k)\tilde{u}_k(\mathbf{k}) = -J \sum_{p=0}^3 n_j \hat{\sigma}_{ij}^{(p)} \Delta\tilde{\theta}_p(\mathbf{k})|\mathbf{k}|, \quad (3.23)$$

where, $\mathbf{k}$ is the Fourier wave vector, $\mathbf{n} = \mathbf{k}/|\mathbf{k}|$ is the unit vector in reciprocal space, $\tilde{u}_k(\mathbf{k})$ and $\Delta\tilde{\theta}_p(\mathbf{k})$ represent the Fourier transforms of $u_k(\mathbf{r})$ and $\Delta\theta_p(\mathbf{r})$, respectively, and J is $\sqrt{(-1)}$.

Defining Ω_{il}^{-1} as $C_{ijkl}n_jn_k$ in Eq. 3.23, the displacements $\tilde{u}_k(\mathbf{k})$ are obtained in Fourier space as follows:

$$\tilde{u}_k(\mathbf{k}) = -J \sum_{p=0}^3 \Omega_{il}(\mathbf{n}) \hat{\sigma}_{ij}^{(p)} n_j \Delta\tilde{\theta}_p(\mathbf{k})|\mathbf{k}|^{-1}, \quad (3.24)$$

In the Fourier space, the periodic strain is given by

$$\tilde{\varepsilon}_{ij}(\mathbf{k}) = -0.5J(k_j\tilde{u}_i(\mathbf{k}) + k_i\tilde{u}_j(\mathbf{k})), \quad (3.25)$$

where $\tilde{\varepsilon}_{ij}(\mathbf{k})$ is the Fourier transform of $\varepsilon_{ij}(\mathbf{r})$ defined in Eq. 3.17.

The elastic strains and stresses in Eqns. 3.16 and 3.19 are obtained using the solutions for the displacement fields given in Eqn. 3.24. Thus, using Khachaturyan's formalism [4], the elastic energy $\mathcal{F}_e$ due to misfit strain (see Eq. 3.5) may be expressed as an integral in the reciprocal space:

$$\mathcal{F}_e = \frac{1}{2} \sum_{p,q=0}^3 \int \frac{d^3\mathbf{k}}{(2\pi)^3} B_{pq}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) \tilde{\theta}_q^*(\mathbf{k}), \quad (3.26)$$

where, $B_{pq}(\mathbf{n}) = C_{ijkl}\epsilon_{ij}^{(p)}\epsilon_{kl}^{(q)} - n_i\hat{\sigma}_{ij}^{(p)}\Omega_{jk}(\mathbf{n})\hat{\sigma}_{kl}^{(q)}n_l$, $\Omega_{il}^{-1}(\hat{\mathbf{n}}) = C_{ijkl}n_jn_k$ is the normalized inverse Green tensor. $\epsilon_{ij}^{(p)}$ is the eigenstrain tensor (defined in Eqns. 3.6a–d) associated with the field θ_p , $\theta_0 = c$, and $\theta_i = \eta_i^2$; ($i = 1, 2, 3$) (defined in Eq. 3.5). In Eq. 3.26, $\tilde{\theta}^*$ refers to the complex conjugate of $\tilde{\theta}$. Also note that a volume of $(2\pi)^3/V$ about $\mathbf{k} = 0$ is excluded from the integration in Eq. 3.26. The details of the derivation of the elastic energy is given in Appendix A.

3.3 Kinetics

The temporal evolution of the field variables is governed by two equations: the Cahn-Hilliard equation for the composition field $c(\mathbf{r}, \tau)$ (conserved field) [77], and the Cahn-Allen equation for the order parameter fields $\eta_p(\mathbf{r}, \tau)$ (non-conserved fields) [78]. (Note that τ represents time.) We solve these evolution equations to study microstructural evolution.

Cahn-Hilliard equation

Following Cahn [77], the net flux $\mathbf{J}$ of B atoms, assumed to be a linear and homogeneous function of the gradient in chemical potential μ , is given by:

$$\mathbf{J} = -M\nabla\mu, \quad (3.27)$$

where M is the mobility. We assume M to be a positive (scalar) constant, independent of the composition and order parameters. The chemical potential μ is given by

$$\mu = \frac{\delta(\mathcal{F}/N_v)}{\delta c} = \frac{\delta(\mathcal{F}_c/N_v)}{\delta c} + \frac{\delta(\mathcal{F}_e/N_v)}{\delta c} \quad (3.28)$$

The temporal evolution of the composition field is described by the continuity equation:

$$\frac{\partial c(\mathbf{r}, \tau)}{\partial \tau} = -\nabla \cdot \mathbf{J} = M \nabla^2 \mu. \quad (3.29)$$

Using Eq. 3.7, we obtain the following expression for $\frac{\delta(\mathcal{F}_c/N_v)}{\delta c}$:

$$\frac{\delta(\mathcal{F}_c/N_v)}{\delta c} = \frac{\partial f_0}{\partial c} - 2\kappa_c \nabla^2 c. \quad (3.30)$$

Substituting μ from Eq. 3.28 into Eq. 3.29 and using Eq. 3.30 we obtain the nonlinear Cahn-Hilliard equation:

$$\frac{\partial c(\mathbf{r}, \tau)}{\partial \tau} = M \nabla^2 \left(\frac{\partial f_0}{\partial c} - 2\kappa_c \nabla^2 c + \frac{\delta(\mathcal{F}_e/N_v)}{\delta c} \right). \quad (3.31)$$

The expression for $\frac{\delta(\mathcal{F}_e/N_v)}{\delta c}$ in the reciprocal space is given in section 3.4.

Cahn-Allen equation

The temporal evolution of each of the order parameters η_p is tracked by solving the Cahn-Allen equation:

$$\frac{\partial \eta_p(\mathbf{r}, \tau)}{\partial \tau} = -L \frac{\delta(\mathcal{F}/N_v)}{\delta \eta_p} = -L \left(\frac{\delta(\mathcal{F}_c/N_v)}{\delta \eta_p} + \frac{\delta(\mathcal{F}_e/N_v)}{\delta \eta_p} \right) \quad (p = 1, 2, 3), \quad (3.32)$$

where L is the relaxation coefficient. L is a positive constant, independent of c and η_p . $\frac{\delta(\mathcal{F}_c/N_v)}{\delta \eta_p}$ is obtained using Eq. 3.7:

$$\frac{\delta(\mathcal{F}_c/N_v)}{\delta \eta_p} = \frac{\partial f_0}{\partial \eta_p} - 2\kappa_\eta \nabla^2 \eta_p. \quad (3.33)$$

The next section 3.4 presents the expressions of $\frac{\delta(\mathcal{F}_e/N_v)}{\delta \eta_p}$ in the reciprocal space.

3.4 Variational derivatives $\frac{\delta(\mathcal{F}_e/N_v)}{\delta c}$ and $\frac{\delta(\mathcal{F}_e/N_v)}{\delta \eta_i}$

To obtain $\frac{\delta(\mathcal{F}_e/N_v)}{\delta \theta_i}$ ($\theta_0 = c, \theta_p = \eta_p^2$ ($p = 1, 2, 3$)), we start with the total elastic energy expression, given in Eq. 3.26:

$$\mathcal{F}_e = \frac{1}{2} \sum_{p,q=0}^3 \int \frac{d^3 \mathbf{k}}{(2\pi)^3} B_{pq}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) \tilde{\theta}_q^*(\mathbf{k}),$$

The variation $\delta \mathcal{F}_e$ corresponding to the variation $\delta \theta_i(\mathbf{r})$ is given by

$$\delta \mathcal{F}_e = \frac{1}{2} \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \sum_{p,i=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) \delta \tilde{\theta}_i^*(\mathbf{k}) + \frac{1}{2} \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \sum_{i,q=0}^3 B_{iq}(\mathbf{n}) \delta \tilde{\theta}_i(\mathbf{k}) \tilde{\theta}_q^*(\mathbf{k}), \quad (3.34)$$

where $\delta \tilde{\theta}_i(\mathbf{k}) (= \int \delta \theta_i(\mathbf{r}) e^{-J\mathbf{k} \cdot \mathbf{r}} d^3 \mathbf{r})$ is the Fourier transform of $\delta \theta_i(\mathbf{r})$.

Evaluating the two terms on the right hand side of Eq. 3.34, we obtain the final expression for $\frac{\delta \mathcal{F}_e}{\delta \theta_i}$:

$$\left[\frac{\delta \mathcal{F}_e}{\delta \theta_i} \right]_{\mathbf{k}} = \sum_{p=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}). \quad (3.35)$$

The derivation of $\frac{\delta \mathcal{F}_e}{\delta \theta_i}$ is shown in detail in Appendix B.

Using Eq. 3.35, $\frac{\delta(\mathcal{F}_e/N_v)}{\delta c}$ and $\frac{\delta(\mathcal{F}_e/N_v)}{\delta \eta_i}$ are obtained in Fourier space as

$$\left[\frac{\delta(\mathcal{F}_e/N_v)}{\delta c} \right]_{\mathbf{k}} = \frac{1}{N_v} \sum_{p=0}^3 B_{p0}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}), \quad (3.36)$$

$$\begin{aligned}
\left[\frac{\delta(\mathcal{F}_e/N_v)}{\delta\eta_i} \right]_{\mathbf{k}} &= \left[\left(\frac{\delta(\mathcal{F}_e/N_v)}{\delta\theta_i} \right)_{\mathbf{r}} \left(\frac{\delta\theta_i}{\delta\eta_i} \right)_{\mathbf{r}} \right]_{\mathbf{k}} \\
&= \left[\left(2\eta_i \frac{\delta(\mathcal{F}_e/N_v)}{\delta\theta_i} \right)_{\mathbf{r}} \right]_{\mathbf{k}} \\
&= \frac{1}{N_v} \left[2\eta_i(\mathbf{r}) \left(\sum_{p=0}^3 B_{pi}(\mathbf{n}') \tilde{\theta}_p(\mathbf{k}') \right)_{\mathbf{r}} \right]_{\mathbf{k}}, \tag{3.37}
\end{aligned}$$

where $\theta_0 = c$ and $\theta_p = \eta_p^2$ ($p = 1, 2, 3$). The notations $[\dots]_{\mathbf{r}}$ and $[\dots]_{\mathbf{k}}$ stand for inverse Fourier transform and Fourier transform, respectively. These expressions of the variational derivatives given in Eqns. 3.36 and 3.37 are used in both Eqns. 3.38a and 3.38b.

3.4.1 Incorporation of fluctuations in the evolution equations

The Cahn-Hilliard and Cahn-Allen equations, Eqns. 3.31 and 3.32, are deterministic in nature. While these equations can describe dissipative processes, such as precipitate growth and coarsening, their deterministic nature prevents them from describing thermally activated processes (such as nucleation) which involves an energy barrier [91]. Modeling the process of nucleation requires incorporation of stochastic terms (Langevin force terms). Hence additional random noise terms are included in the evolution equations to mimic the nucleation process [92, 93].

The modified Cahn-Hilliard and Cahn-Allen equations, after the inclusion of Langevin force terms, are given as:

$$\frac{\partial c(\mathbf{r}, \tau)}{\partial \tau} = M \nabla^2 \left(\frac{\partial f_0}{\partial c} - 2\kappa_c \nabla^2 c + \frac{\delta(\mathcal{F}_e/N_v)}{\delta c} \right) + \xi_c(\mathbf{r}, \tau), \tag{3.38a}$$

$$\frac{\partial \eta_p(\mathbf{r}, \tau)}{\partial \tau} = -L \left(\frac{\partial f_0}{\partial \eta_p} - 2\kappa_\eta \nabla^2 \eta_p + \frac{\delta(\mathcal{F}_e/N_v)}{\delta \eta_p} \right) + \xi_{\eta_p}(\mathbf{r}, \tau), \tag{3.38b}$$

where $\xi_c(\mathbf{r}, \tau)$ and $\xi_{\eta_p}(\mathbf{r}, \tau)$ are the Langevin force terms associated with c and η_p , respectively. These stochastic terms are assumed to be uncorrelated in both space and time (white noise) and are approximated by a Gaussian random number distribution. Below,

we list the properties of $\xi_c(\mathbf{r}, \tau)$ and $\xi_{\eta_p}(\mathbf{r}, \tau)$:

$$\begin{aligned}\langle \xi_c(\mathbf{r}, \tau) \rangle &= 0, \langle \xi_{\eta_p}(\mathbf{r}, \tau) \rangle = 0, \\ \langle \xi_c(\mathbf{r}, \tau) \xi_c(\mathbf{r}', \tau') \rangle &= -2k_B T M \delta(\tau - \tau') \nabla^2 \delta(\mathbf{r} - \mathbf{r}') \\ \langle \xi_{\eta_p}(\mathbf{r}, \tau) \xi_{\eta_p}(\mathbf{r}', \tau') \rangle &= 2k_B T L \delta(\tau - \tau') \delta(\mathbf{r} - \mathbf{r}')\end{aligned}\tag{3.39}$$

The amplitude of the growing fluctuations is determined by a balance between the perturbations (introduced through the Langevin force terms) and the dissipative forces (due to the energy gradients) using the fluctuation-dissipation theorem [94, 95].

We solve the evolution equations (Eqns. 3.38a and 3.38b) numerically, assuming periodic boundary conditions. The details of the numerical procedure is given in section 3.5.

3.5 Numerical implementation

Since we assume periodic boundary conditions, the temporal evolution equations (Eqns. 3.38a and 3.38b) are solved numerically using a semi implicit Fourier spectral method [96, 97]. First, let us consider Eq. 3.38a and take the spatial Fourier transform on both sides of the equation:

$$\frac{\partial \tilde{c}(\mathbf{k}, \tau)}{\partial \tau} = -Mk^2 \tilde{g}(\mathbf{k}, \tau) - Mk^2 \tilde{h}(\mathbf{k}, \tau) - 2M\kappa_c k^4 \tilde{c}(\mathbf{k}, \tau) + \tilde{\xi}_c(\mathbf{k}, \tau).\tag{3.40}$$

Here, $g(\mathbf{r}, \tau) = \frac{\partial f_0}{\partial c(\mathbf{r}, \tau)}$, $h(\mathbf{r}, \tau) = \frac{\delta(\mathcal{F}_e/N_v)}{\delta c(\mathbf{r}, \tau)}$ and $\tilde{c}(\mathbf{k}, \tau)$, $\tilde{g}(\mathbf{k}, \tau)$, $\tilde{h}(\mathbf{k}, \tau)$, $\tilde{\xi}_c(\mathbf{k}, \tau)$ are the Fourier transforms of $c(\mathbf{r}, \tau)$, $g(\mathbf{r}, \tau)$, $h(\mathbf{r}, \tau)$ and $\xi_c(\mathbf{r}, \tau)$ respectively.

In our numerical scheme, the random noise term $\xi_c(\mathbf{r}, \tau)$ is added to the composition field $c(\mathbf{r}, \tau)$ in real space at each time step; the resultant composition field is then Fourier transformed (i.e. $\tilde{c}(\mathbf{k}, \tau) = \left[\tilde{c}(\mathbf{r}, \tau) + \tilde{\xi}_c(\mathbf{r}, \tau) \right]_{\mathbf{k}}$).

Thus, the semi implicit discretization of the above equation is obtained as

$$\frac{\tilde{c}(\mathbf{k}, \tau + \Delta\tau) - \tilde{c}(\mathbf{k}, \tau)}{\Delta\tau} = -Mk^2\tilde{g}(\mathbf{k}, \tau) - Mk^2\tilde{h}(\mathbf{k}, \tau) - 2M\kappa_c k^4\tilde{c}(\mathbf{k}, \tau + \Delta\tau) \quad (3.41)$$

Rearranging the above equation, we get

$$\tilde{c}(\mathbf{k}, \tau + \Delta\tau) = \frac{\tilde{c}(\mathbf{k}, \tau) - Mk^2\Delta\tau\tilde{g}(\mathbf{k}, \tau) - Mk^2\Delta\tau\tilde{h}(\mathbf{k}, \tau)}{(1 + 2M\kappa_c k^4\Delta\tau)} \quad (3.42)$$

We proceed in a similar way to solve the Cahn-Allen equations. Taking Fourier transform on both sides of Eq. 3.38b we get

$$\frac{\partial\tilde{\eta}_p(\mathbf{k}, \tau)}{\partial\tau} = -L\tilde{v}(\mathbf{k}, \tau) - L\tilde{w}(\mathbf{k}, \tau) - 2L\kappa_\eta k^2\tilde{\eta}_p(\mathbf{k}, \tau) + \tilde{\xi}_{\eta_p}(\mathbf{k}, \tau), \quad (3.43)$$

where $v(\mathbf{r}, \tau) = \frac{\partial f_0}{\partial\eta_p(\mathbf{r}, \tau)}$, $w(\mathbf{r}, \tau) = \frac{\delta(\mathcal{F}_e/N_v)}{\delta\eta_p(\mathbf{r}, \tau)}$, and $\tilde{\eta}_p(\mathbf{k}, \tau)$, $\tilde{v}(\mathbf{k}, \tau)$, $\tilde{w}(\mathbf{k}, \tau)$, $\tilde{\xi}_{\eta_p}(\mathbf{k}, \tau)$ are the Fourier transforms of $\eta_p(\mathbf{r}, \tau)$, $v(\mathbf{r}, \tau)$, $w(\mathbf{r}, \tau)$ and $\xi_{\eta_p}(\mathbf{r}, \tau)$ respectively.

The noise terms ξ_{η_p} associated with η_p are treated in a similar manner to ξ_c : they are incorporated into $\eta_p(\mathbf{r}, \tau)$ following which the Fourier transforms of the resultant order parameter fields are taken. Thus, the semi implicit discretization of Eq. 3.43 yields the following:

$$\tilde{\eta}_p(\mathbf{k}, \tau + \Delta\tau) = \frac{\tilde{\eta}_p(\mathbf{k}, \tau) - L\Delta\tau\tilde{v}(\mathbf{k}, \tau) - L\Delta\tau\tilde{w}(\mathbf{k}, \tau)}{(1 + 2L\kappa_\eta k^2\Delta\tau)} \quad (3.44)$$

At a given time τ , $c(\mathbf{r})$ and $\eta_i(\mathbf{r})$ are known; we use them in the reciprocal space in Eqns. 3.42 and 3.44 for obtaining the fields at a future time $(\tau + \Delta\tau)$. This time integration procedure is repeated to study how the microstructure, starting from a given initial configuration, evolves with time.

All our simulations are carried out on a discretized 2D domain, $L_x \times L_y$. The domain is discretized with a grid spacing Δx along the x -axis and Δy along the y -axis, and the reciprocal lattice vectors are given by $\mathbf{k}_x = \frac{2\pi}{L_x}\hat{\mathbf{x}}$ and $\mathbf{k}_y = \frac{2\pi}{L_y}\hat{\mathbf{y}}$. The discrete Fourier

transforms in our simulations are implemented using the FFTW package developed by Frigo and Johnson [98].

3.6 Non-dimensionalisation

All the equations presented in the preceding sections are non-dimensional (though they may contain parameters which would take a value of unity during non-dimensionalisation). All the variables in the model are rendered non-dimensional using a characteristic energy, E_n , length L_n and time T_n . E_n and T_n are chosen such that the non-dimensional values of parameter A_1 (in Eq. 3.8) and the mobility M (in Eq. 3.31) assume a value of unity. L_n is estimated using the interfacial width. Using primed quantities to indicate dimensional quantities, we choose L_n , T_n and E_n as the following:

$$E_n = A'_1, \quad (3.45)$$

$$L_n = \left(\frac{\kappa'_c}{A'_1} \right)^{\frac{1}{2}}, \quad (3.46)$$

$$T_n = \frac{L_n^2 (c'_p - c'_m)^2}{M' E_n}. \quad (3.47)$$

The non-dimensional $\delta(F_e/N_v)/\delta c$ and $\delta(F_e/N_v)/\delta \eta_i$ are obtained by using non-dimensional $C_{ijkl}^S = C_{ijkl}/N_v$ (note that N_v is absorbed in C_{ijkl}^S). The non-dimensional energy, time and length units correspond to $3.125 \times 10^{-21} J$, $1.28 s$ and $0.8 nm$, respectively. The details of the non-dimensionalisation procedure are given in Appendix C.

3.7 Energy of interphase interface and antiphase domain boundaries

In this section, we present the procedure to calculate interfacial energy, γ_i , between α and β phases, and antiphase boundary energy, γ_a , between any two variants of the ordered β phase.

For the α - β interface, a one-dimensional simulation is set up with one half of the system with disordered α ($(c, \eta_1) = (0, 0)$) and the other half with ordered β ($(c, \eta_1) = (1, 1)$), as the initial configuration; the other two order parameters are assumed to be zero. On the other hand, the antiphase boundary energy is calculated for an alloy of composition $c = 1$; in this case, the initial configuration is prepared with the first variant of β phase with $(\eta_1, \eta_2, \eta_3) = (1, 0, 0)$ in one half of the system, and the second variant with $(\eta_1, \eta_2, \eta_3) = (0, 1, 0)$ in the other half. In both cases, the Cahn-Hilliard and Cahn-Allen equations, Eqns. 3.38a and 3.38b, are solved to the steady state to obtain the equilibrium composition and order parameter profiles.

The appropriate dimensionless boundary energy (for a one-dimensional planar interface) is obtained from the equilibrium composition and order parameter profiles using the relation [76]

$$\gamma = N_v \int \left[f_0(c, \eta_i) + \kappa_c \left(\frac{\partial c}{\partial x} \right)^2 + \kappa_\eta \left(\frac{\partial \eta_i}{\partial x} \right)^2 \right] dx, \quad (3.48)$$

where γ represents γ_i or γ_a .

The equilibrium profiles of composition and order parameter fields for the two systems are shown in Figs. 3.6 and 3.7. Further, we use a dimensionless grid spacing of $\Delta x = 0.5$ to ensure that there are at least four grid points in the interfacial region. This number of grid points is sufficient to capture the essential features of the interface in these profiles. The dimensionless values of γ_i and γ_a , calculated for different values of κ_c and κ_η , are listed in Table 3.1.

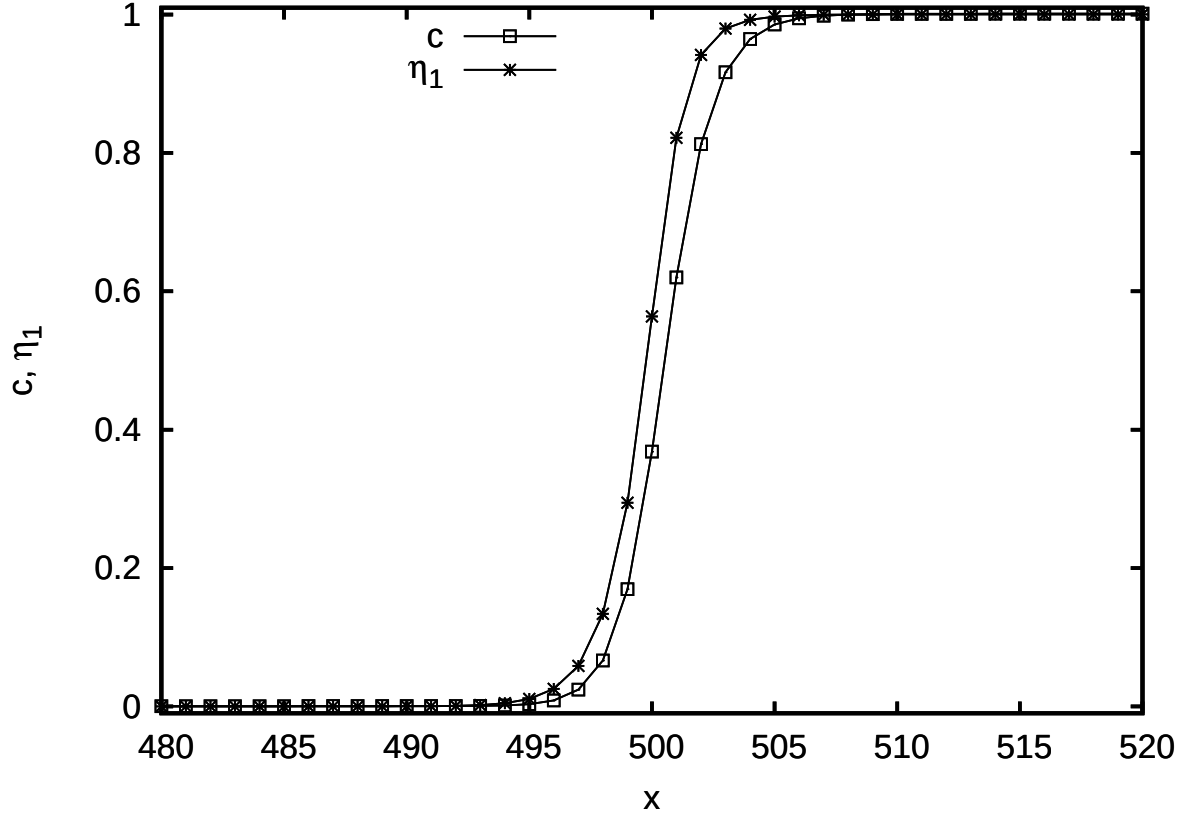


Figure 3.6: Profiles of c and η_1 at equilibrium for a system having an interphase boundary between the disordered α phase and the first variant of the ordered β phase. Results are shown for a grid spacing of $\Delta x = 0.5$.

Table 3.1: Interfacial and APB energies for different values of κ_c and κ_η

κ_c	κ_η	γ_i	γ_a
0.2	0.7	20.48	40.64
0.01	0.7	18.56	38.40
1.0	0.1	18.56	20.48

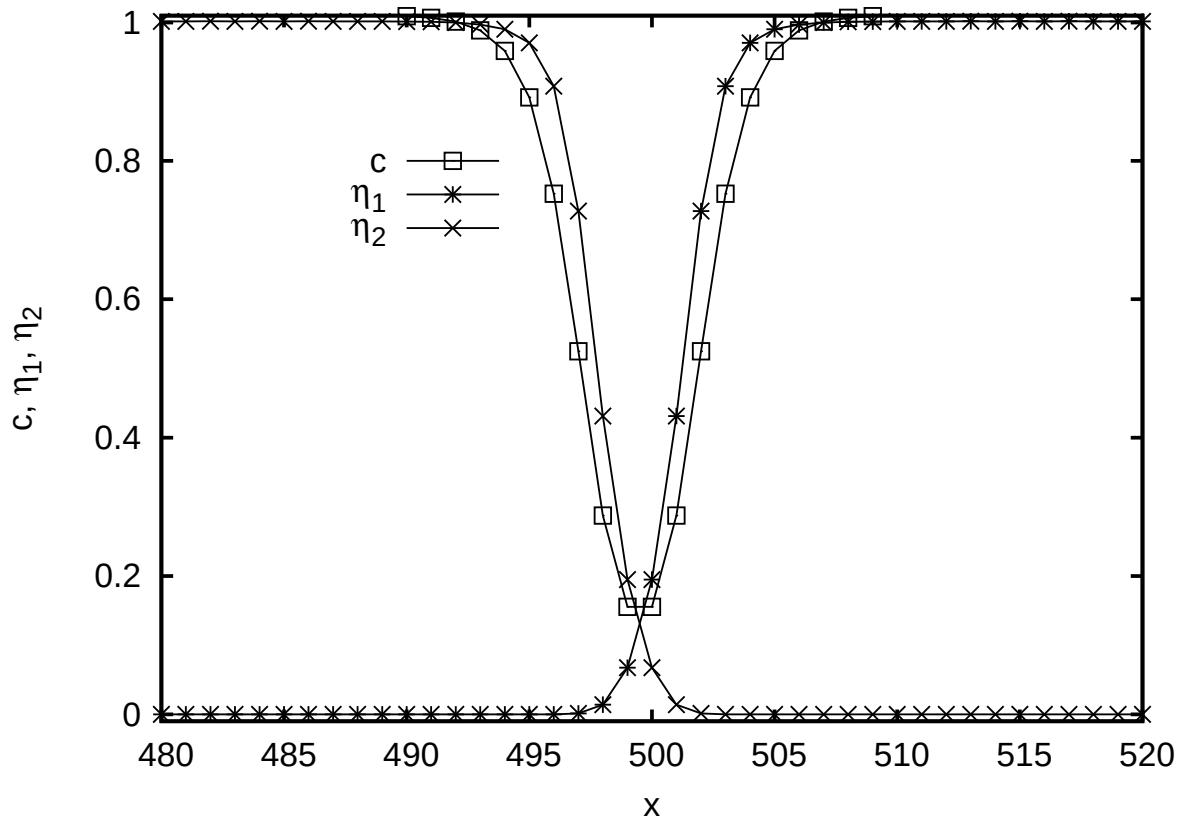


Figure 3.7: Profiles of c , η_1 and η_2 at equilibrium for a system having an antiphase boundary between variants I and II of the ordered β phase in an alloy of composition $c = 1$. Results are shown for a grid spacing of $\Delta x = 0.5$.

Chapter 4

Dynamic Growth Shapes

In this chapter we present the results on the shape of a single coherent misfitting precipitate embedded in a matrix in a kinetic setting. The purpose behind this study is to test the model developed in the previous chapter on a simple system for which some results are known. In particular, we focus our attention on the symmetry of the growing particle and on any symmetry-breaking transitions that might occur.

We use this study to describe the effect of anisotropic misfit strain on the growth shapes of the misfitting particle. We classify the systems in this study based on the nature of the anisotropy in misfit strain, examine whether they exhibit symmetry-breaking transition, and if they do, then compare the results on similar transitions in equilibrium shapes [1].

4.1 Simulation Details

The single particle simulations are done on a two-dimensional system of size 1024×1024 . A circular coherent precipitate of a single variant ($c = 1$ and $\eta_1 = 1$) with a prescribed initial radius is placed in the center of the simulation cell, and allowed to grow. The matrix phase is assumed to have a constant composition, c_∞ . A sustained Gaussian noise

of 1% was introduced during the first hundred steps in order to allow the growing particle to change its symmetry.

For a supersaturated alloy with a composition c_∞ , the critical radius of the nucleus, r^* , (assuming the shape of the critical nucleus to be circular) is given by:

$$r^* = \frac{-\gamma_i}{\Delta\mathcal{G}_v + \Delta\mathcal{G}_e} \quad (4.1)$$

where $\Delta\mathcal{G}_v$ is the change in volume free energy during nucleation and $\Delta\mathcal{G}_e$ is the coherency strain energy per unit volume of the precipitate. We outline the procedure for calculating $\Delta\mathcal{G}_v$, $\Delta\mathcal{G}_e$ and r^* in Appendix D. The interfacial energy, γ_i , between the matrix and the precipitate phases is assumed to be isotropic. To ensure that our results do not depend on the initial size of the particle, we start with a radius which is larger than that predicted from the above calculations.

The matrix and the precipitate phases are assumed to have the same elastic moduli (homogeneous modulus approximation). Further, the elastic stiffness tensor is assumed to be isotropic. The stress-free transformation strain, ϵ^0 , can be isotropic (dilatational) or anisotropic. In our model, we do not consider the effect due to applied stress or interfacial stresses. Thus, the misfit is the only source of elastic stresses in our system.

Following Schmidt and Gross [69], for a system with cubic symmetry, the elastic moduli are defined in terms of an average shear modulus, G , an effective Poisson's ratio, ν , and an anisotropy parameter, A_Z .

$$G = \frac{1}{2\pi} \int_0^{2\pi} C_{1212}(\phi) d\phi, \quad (4.2)$$

$$\nu = \frac{\bar{\lambda}}{2(\bar{\lambda} + G)}, \quad (4.3)$$

$$A_Z = \frac{2C_{44}}{C_{11} - C_{12}}, \quad (4.4)$$

where

$$\bar{\lambda} = \frac{1}{2\pi} \int_0^{2\pi} C_{1122}(\phi) d\phi$$

is the average Lamé constant in the (001) plane. The cubic elastic constants C_{11} , C_{12} and C_{44} are related to G , ν and A_Z through the following equations:

$$C_{11} = G \left(2 \frac{2 + A_Z}{1 + A_Z} - \frac{1 - 4\nu}{1 - 2\nu} \right), \quad (4.5)$$

$$C_{12} = G \left(\frac{2A_Z}{1 + A_Z} - \frac{1 - 4\nu}{1 - 2\nu} \right), \quad (4.6)$$

$$C_{44} = G \frac{2A_Z}{1 + A_Z}. \quad (4.7)$$

The system is elastically isotropic when the anisotropy parameter A_Z is equal to unity.

The anisotropic misfit strain tensor is defined in Eq. 3.2 in the previous chapter. We have studied the following systems with five different values of misfit anisotropy parameter, t .

1. **Case A:** $t = 1$; ϵ_{xx} and ϵ_{yy} are equal; i.e. the misfit is dilatational.
2. **Case B:** $t = 0.5$; ϵ_{xx} and ϵ_{yy} are unequal, but have the same sign. In particular, the misfit along y direction (ϵ_{yy}) is less than that along the x direction (ϵ_{xx}).
3. **Case C:** $t = 0$; the misfit along y direction (ϵ_{yy}) is zero.
4. **Case D:** $t = -0.5$; ϵ_{xx} and ϵ_{yy} have opposite signs and their magnitudes are unequal; the misfit along y direction (ϵ_{yy}) is less than that along the x direction (ϵ_{xx}).
5. **Case E:** $t = -1$; ϵ_{xx} and ϵ_{yy} have opposite signs but their magnitudes are equal; $\epsilon_{yy} = -\epsilon_{xx}$.

The other parameters which we have used in our calculation are listed in Table 4.1.

Further, it is convenient to define a characteristic length L_c :

$$L_c = \frac{\gamma_i}{G\epsilon_\eta^2}, \quad (4.8)$$

where γ_i is the interfacial energy and ϵ_η is the strength of the misfit strain (Eq. 3.2). L_c represents a transition size between capillarity dominant regime and elasticity dominant regime.

Table 4.1: Parameters used in the single particle simulations

Parameters	Non-dimensional values
Bulk free energy coefficients ($A_1, A_2, A_{41}, A_{42}, A_{61}, A_{62}$)	1.0, 2.0, -3.0, 36.0, 2.0, 4.0
Gradient energy coefficient for composition c (κ_c)	0.2
Gradient energy coefficient for order parameter η_1 (κ_η)	0.7
Mobility (M)	1.0
Kinetic Coefficient (L)	1.0
Strength of the misfit strain associated with η_1 (ϵ_{η_1})	0.03
Strength of the misfit strain associated with composition (ϵ_c)	0.0
Misfit anisotropy parameter (t)	1.0, 0.5, 0.0, -0.5, -1.0
Average shear modulus (G)	
Case A	300.0
Cases B–E	500.0, 750.0, 1000.0
Poisson's ratio (ν)	0.3
Zener anisotropy parameter (A_Z)	1
Far field composition (c_∞)	0.2, 0.3, 0.4
Grid spacing (Δx)	0.5
Time step ($\Delta \tau$)	0.01
Noise strength	1.0%

For a precipitate with an arbitrary shape in two-dimensions, its effective radius, R , is defined as the radius of an equivalent circle occupying the same area as that of the precipitate. Since the composition c of the precipitate phase differs from that of the matrix phase, R is defined by:

$$\pi R^2 = \int c_p dA; \quad c_p = \begin{cases} c & \text{if } c > c_i \\ 0 & \text{if } c < c_i \end{cases}, \quad (4.9)$$

where $c_i = 0.5$ is the composition at the interface between the precipitate and the matrix. This radius R may be normalized using L_c to yield a normalized radius $\rho = R/L_c$.

Symmetries exhibited by the system

We aim to study the dynamic growth shapes of the precipitates and their size dependence in terms of symmetry breaking transitions (i.e. whether the shape of the growing precipitate undergoes a transition, beyond a critical size, from a higher symmetry one to a shape with lower symmetry). This aim is motivated by the work of Sankarasubramanian *et al.* [1] who described such transitions in equilibrium shapes.

For describing the symmetry of particle shapes we have used the terminology adopted by Sankarasubramanian *et al.* [1,17]. The symbols I , T , O and M are used to denote isotropic, tetragonal, orthorhombic and monoclinic symmetries in two dimensions, respectively. For convenience, we reproduce the table given by them, to describe the symmetry of the particle shape and that of interfacial energy, elastic modulus and misfit strain. The symmetry elements are listed in Table 4.2. We use plane strain conditions to carry out all our simulations. Therefore, the particles are assumed to be cylindrical, extended infinitely along the z -axis. Hence, all the systems in our study possess a mirror plane perpendicular to the z -axis.

The interfacial energy and elastic moduli have isotropic (I) symmetry in all the cases (in appendix E, some cases with a cubic anisotropy (T) in elastic moduli are considered). When the misfit strain, ϵ_{ij}^0 , is dilatational (Case A), it has isotropic (I) symmetry and when it is anisotropic (Cases B, C, D and E) it has an orthorhombic (O) symmetry. The symmetry of the intersection group is obtained by the intersection of the symmetry elements of the interfacial energy, elastic modulus and the misfit strain.

We restrict our study to those sizes of the growing precipitate where the elastic field and the diffusion field from the neighboring domains (due to periodic boundary conditions) do

Table 4.2: Symmetries of the particle shape

Symbol	Symmetry Notation	Details of symmetry elements in 2D	Example
I	$\frac{\infty}{m}mm$	$\left(\frac{\infty}{m}\right)_z \left(\frac{2}{m}\right)_{\langle 10 \rangle} \left(\frac{2}{m}\right)_{\langle 11 \rangle}$	Circle
T	$\frac{4}{m}mm$	$\left(\frac{4}{m}\right)_z \left(\frac{2}{m}\right)_{\langle 10 \rangle} \left(\frac{2}{m}\right)_{\langle 11 \rangle}$	Square
O	$\frac{2}{m}mm$	$\left(\frac{2}{m}\right)_z \left(\frac{2}{m}\right)_{[10]} \left(\frac{2}{m}\right)_{[01]}$	Ellipse, Rectangle, Diamond
M	$\frac{2}{m}$	$\left(\frac{2}{m}\right)_z$	S-shape

not affect the morphology of the precipitate. We stop the simulations when the diffusion field extends to the boundary of the simulation box.

In what follows, we present instantaneous gray scale snapshots of the system obtained from our simulations at different stages of growth. In these figures, the regions with $c = 1$ are represented by black and the regions with $c = 0$ are represented by white, with the intermediate compositions being represented by intermediate gray scales.

4.2 Case A: Dilatational misfit ($t = 1$)

In this case we have used $G = 300$ and a far-field composition $c_\infty = 0.3$. We restrict ourselves to a lower value of G (compared to the other cases) because the particle dissolves with a further increase in G (see Appendix D). The morphology of the precipitate at an earlier and a later time is shown in Fig. 4.1. As shown in the figure, the particle shape

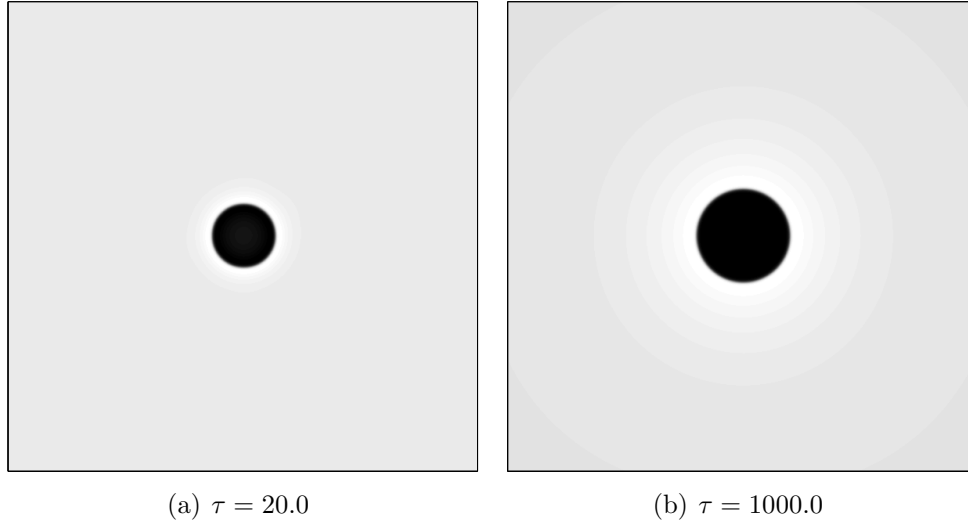


Figure 4.1: Shape of a growing precipitate in a system with dilatational misfit (Case A, $t = 1$). The growth time is indicated in each figure. The size of the box is 600×600 length units. ($G = 300.0$, $c_\infty = 0.3$)

remains circular at all times, retaining its isotropic (I) symmetry at all sizes; in other words, there is no symmetry breaking transition during growth. This symmetry is also the same as the intersection of the symmetries of interfacial energy, elastic moduli, and misfit strain.

4.3 Case B ($t = 0.5$)

For this case, the morphology of the precipitate at different times is shown in Fig. 4.2. In this case we have used a shear modulus $G = 750$, and a far-field composition $c_\infty = 0.3$. In contrast to the previous case (where the misfit is dilatational), anisotropy in misfit in this case leads to non-circular shape. In particular, the particle elongates along the y direction. This is as expected since the misfit along that direction is lower in magnitude.

The particle shape has an O (orthorhombic) symmetry, consistent with the symmetry of the misfit strain. This O symmetry remains unbroken at larger sizes.

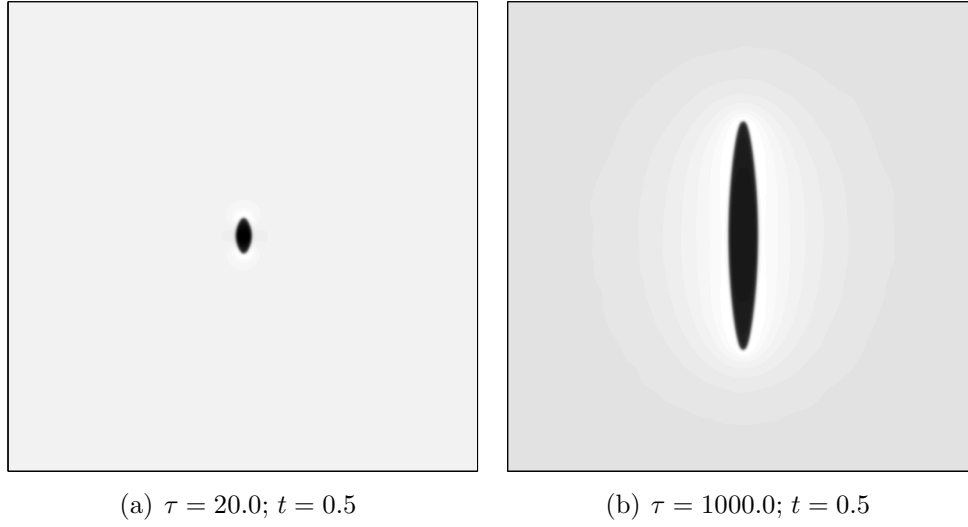


Figure 4.2: Shape of a growing precipitate in a system with non-dilatational misfit (Case B, $t = 0.5$). The growth time is indicated in each figure. The size of the box is 600×600 length units. ($G = 750$, $c_\infty = 0.3$)

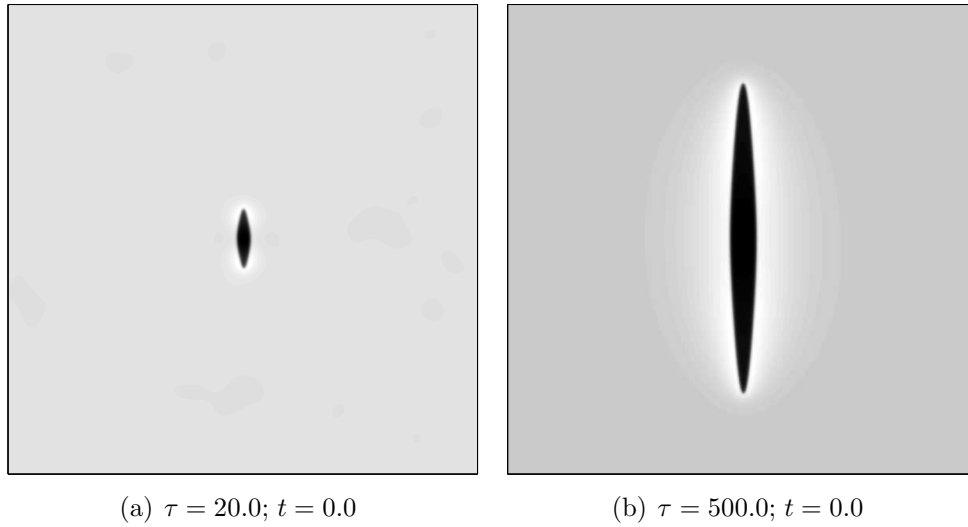


Figure 4.3: Shape of a growing precipitate in a system with non-dilatational misfit (Case C, $t = 0$). The growth time is indicated in each figure. $G = 750$ and $c_\infty = 0.3$ The size of the box is 600×600 length units. ($G = 750$, $c_\infty = 0.3$)

4.4 Case C ($t = 0$)

The morphology of the precipitate at the early and late stages of growth is shown in Fig. 4.3. As in case B, the particle elongates along the y axis, the direction of lower misfit (in fact, $\epsilon_{yy} = 0$ in this case), and it retains its O symmetry during growth.

4.5 Case D: $t = -0.5$

The particle growth shapes for this case (with $G = 750$ and $c_\infty = 0.2$) are shown in Fig. 4.4. In this figure, the particle shapes are outlined using contours at $c = 0.5$. As in Cases B and C, the particle elongates along the y axis (the direction of lower absolute misfit) during early stages, and the shape possesses an O (orthorhombic) symmetry. As the particle becomes larger, however, the long axis of the particle deviates from the y -

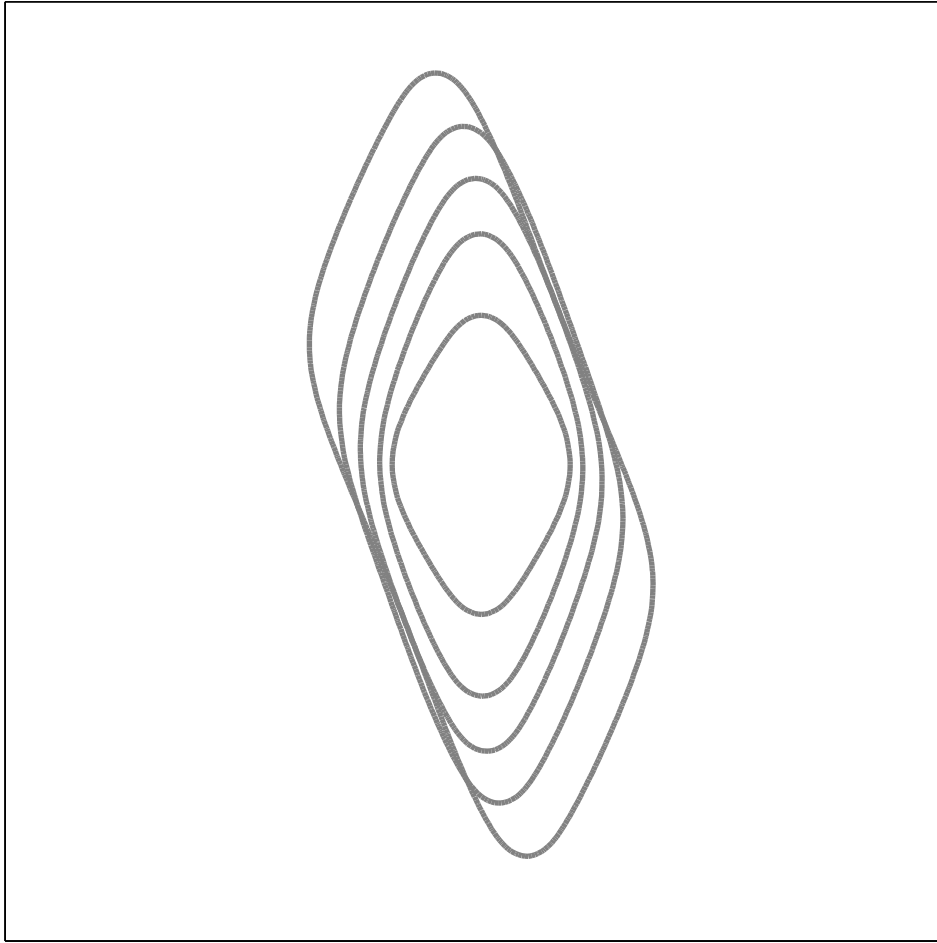


Figure 4.4: Shapes of a growing precipitate in a system with $t = -0.5$ (Case D) at successive times $\tau = 20.0, 260.0, 500.0, 760.0, 980.0$ (represented using contours drawn using $c = 0.5$). Here, $G = 750.0$ and $c_\infty = 0.2$. The dimensions of the box correspond to 140×140 length units.

axis. With this deviation, the particle loses its O symmetry (mirror planes normal to x and y axes), and retains only an M (monoclinic) symmetry with just a two-fold rotation

symmetry about the z axis.

Since the misfit strain components in this case have opposite signs, there is an invariant line along which the misfit strain is zero. The invariant line forms an angle ϕ with the x -axis, given by [4, 73]:

$$\phi = \tan^{-1} \sqrt{-t}, \quad (4.10)$$

where t is the misfit anisotropy parameter (in this case $t = -0.5$).

In the absence of interfacial energy, a precipitate in the form of an infinitely long plate extended along the invariant line would have no elastic energy associated with it. Thus, as the precipitate becomes larger, its direction of elongation is expected to become closer to the invariant direction. Sankarasubramanian *et al.* showed that the equilibrium shape does indeed show this behavior [1]; it initially elongates along the y direction, and at a critical size it undergoes a symmetry-breaking $O \rightarrow M$ transition after which the direction of elongation keeps shifting towards the invariant line. Our results in Fig. 4.4 show that the growth shapes also exhibit this $O \rightarrow M$ transition.

In order to locate the transition size, we need a parameter to describe the symmetry of the shape. We define such a parameter χ , using the planar moment of inertia tensor, $\mathfrak{I}$:

$$\mathfrak{I} = \begin{pmatrix} \mathfrak{I}_{xx} & \mathfrak{I}_{xy} \\ \mathfrak{I}_{xy} & \mathfrak{I}_{yy} \end{pmatrix}, \quad (4.11)$$

where $\mathfrak{I}_{xx}$, $\mathfrak{I}_{yy}$, and $\mathfrak{I}_{xy}$ are evaluated using the following integrals:

$$\begin{aligned} \mathfrak{I}_{xx} &= \int y'^2 dA, \\ \mathfrak{I}_{yy} &= \int x'^2 dA, \\ \mathfrak{I}_{xy} &= \int x'y' dA, \end{aligned} \quad (4.12)$$

where y' and x' are the distances along the y and x directions, respectively, from the center of mass of the particle to the point in the particle/matrix interface.

Since $\mathfrak{I}$ is a symmetric second rank tensor, its principal directions are obtained using the following relation:

$$\tan 2\theta = -\frac{2\mathfrak{I}_{xy}}{\mathfrak{I}_{xx} - \mathfrak{I}_{yy}}, \quad (4.13)$$

where the two values of θ are the angles made by the principal vectors with the x and y directions. Its principal values, $\mathfrak{I}_X$ and $\mathfrak{I}_Y$, are obtained as follows:

$$\begin{aligned} \mathfrak{I}_X &= \frac{\mathfrak{I}_{xx} + \mathfrak{I}_{yy}}{2} + \sqrt{\left[\frac{\mathfrak{I}_{xx} - \mathfrak{I}_{yy}}{2}\right]^2 + \mathfrak{I}_{xy}^2}, \\ \mathfrak{I}_Y &= \frac{\mathfrak{I}_{xx} + \mathfrak{I}_{yy}}{2} - \sqrt{\left[\frac{\mathfrak{I}_{xx} - \mathfrak{I}_{yy}}{2}\right]^2 + \mathfrak{I}_{xy}^2}. \end{aligned} \quad (4.14)$$

Since we wish to track an $O \rightarrow M$ transition, we may identify the symmetry parameter $\chi = \pm(\theta - \pi/2)$ where θ denotes the angle of the first principal direction of $\mathfrak{I}$ with the x axis. A plot of χ with normalized particle size ρ yields the supercritical bifurcation diagram in Fig. 4.5. From this plot, we estimate the $O \rightarrow M$ transition size ρ_c to be ≈ 12 . The bifurcation diagram in Fig. 4.5 does not show the unstable solution branch that must emanate from the instability point. The unstable solution branch is associated with unstable higher symmetry shapes which correspond to energy maxima. Using our method, we cannot determine the shapes corresponding to maxima in energy.

Effect of shear modulus and supersaturation

We have obtained the critical size for $O \rightarrow M$ transition in systems with different shear modulus values ($G = 750, 1000$), and different levels of supersaturation ($c_\infty = 0.2, 0.3, 0.4$). These results are presented in Fig. 4.6 in which we plot the critical size

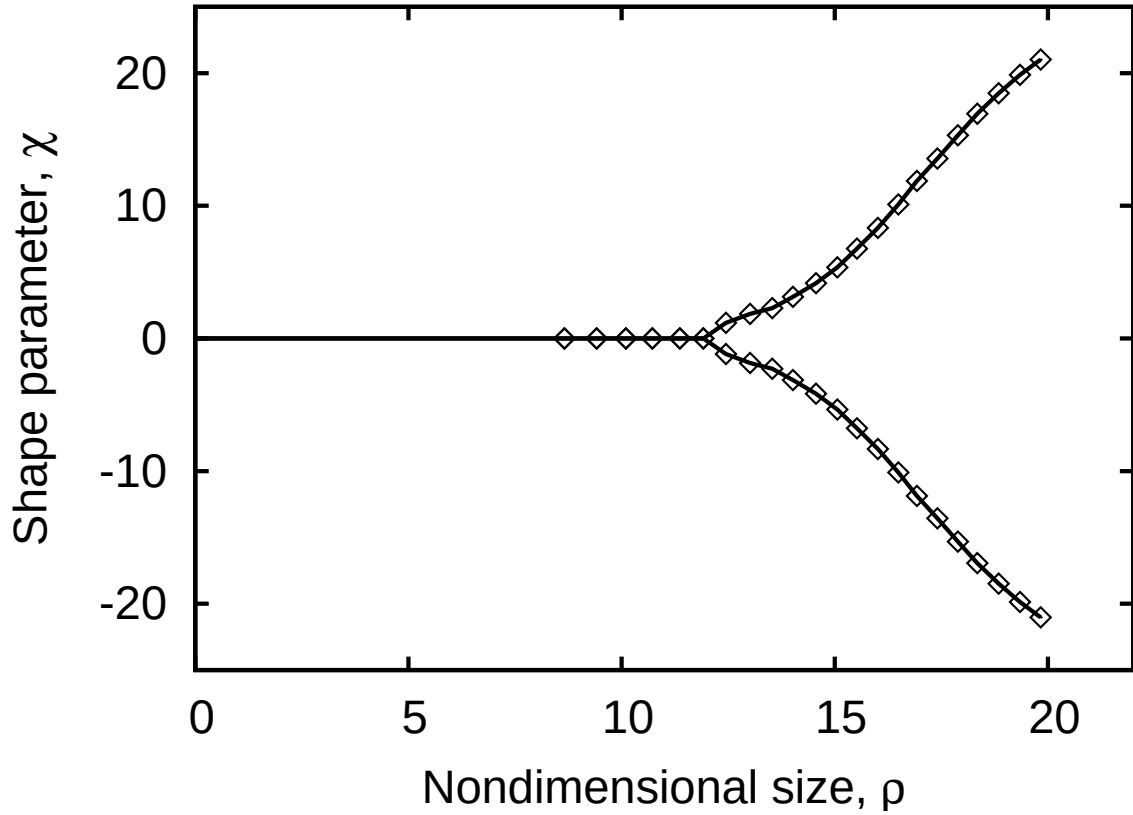


Figure 4.5: Bifurcation diagram showing $O \rightarrow M$ transition in a system with $G = 750$, $c_\infty = 0.2$ in Case D ($t = -0.5$). The line through the data points is only a guide to the eye.

R_c as a function of $L_c = \gamma/G\epsilon_\eta^2$ for different supersaturations. The results for equilibrium shapes, from the work of Sankarasubramanian *et al.* [1, 17], is also shown for easy comparison.

First, R_c for transition in growth shapes is far higher than that for equilibrium shapes. Further, R_c is larger for systems with larger supersaturation and lower shear modulus (higher L_c). The effect of shear modulus is easy to explain: in a system with a larger G (smaller L_c), the effect of elastic stresses manifests itself at a smaller size. The role of supersaturation may be explained using a competition between particle growth rate (which is higher in systems with higher c_∞) and rate of equilibration of particle shape (which requires redistribution of solute by transport of solute from one region of the interface to

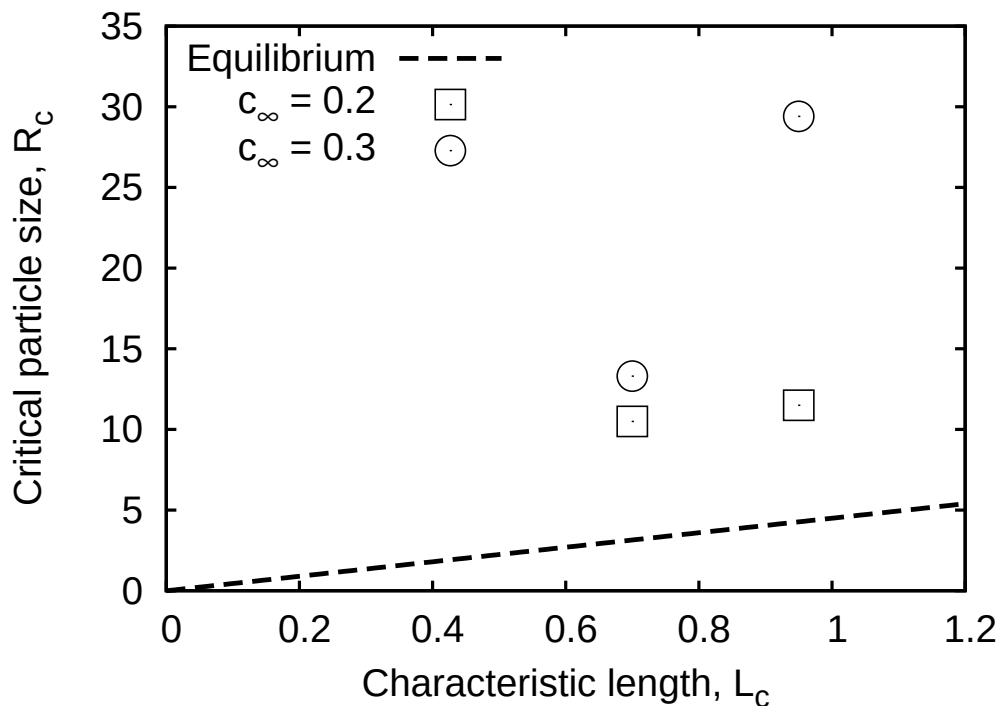


Figure 4.6: Variation of critical size, R_c , for the $O \rightarrow M$ symmetry-breaking shape transition for a growing particle, as a function of L_c for systems with different supersaturations. The dashed line ($R_c = 4.5L_c$) is for the same symmetry-breaking transition in equilibrium shape.

another). In systems with larger c_∞ , the former dominates leading to a delayed transition. In fact, when $c_\infty = 0.4$, the particle kept growing without breaking its symmetry until the end of the simulation.

4.6 Case E: ($t = -1$)

The shapes of the particle at different times during growth (in a system with $G = 500$ and a supersaturation $c_\infty = 0.2$) are shown in Fig. 4.7. The particle has a square-like shape, with T (tetragonal) symmetry, at the initial stages of growth. During the later stages, the particle elongates along $[\bar{1}1]$ direction and acquires a rectangular shape (with O symmetry). Thus, the T symmetry (four-fold rotation symmetry about z axis) of the particle shape is broken during growth, and an O symmetry (mirrors perpendicular to $\langle 11 \rangle$ directions) is retained. This $T \rightarrow O$ transition was reported in equilibrium shapes by Sankarasubramanian *et al.* [1].

In this case, the misfit strains along x and y directions are equal in magnitude but opposite in sign. Here, the misfit strain tensor has invariant lines along directions which make an angle of $\pm 45^\circ$ with the x -axis. In other words, the invariant lines lie along either of the $\langle 11 \rangle$ directions, which are also the directions of elongation of the particle.

Similar to Cases B, C and D, the symmetry of the misfit strain tensor is orthorhombic. Since the interfacial energy and elastic moduli have I (isotropic) symmetry, the symmetry of the intersection group is orthorhombic (O). However, during the early stages the particle shape has a T (tetragonal) symmetry, which is higher than the point group intersection symmetry.

To estimate the critical size of the particle for this $T \rightarrow O$ symmetry breaking transition, we keep track of the change in aspect ratio of the particle shape. Hence, we define χ as

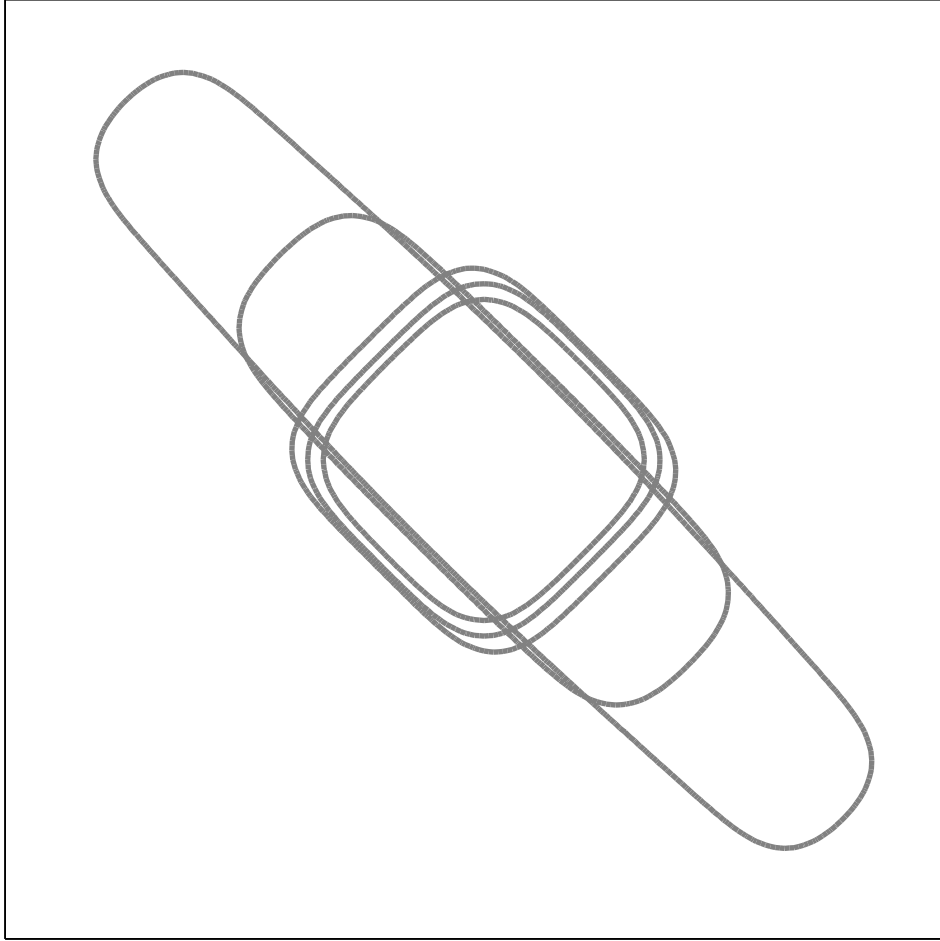


Figure 4.7: Shapes of a growing precipitate in an elastically isotropic system with non-dilatational misfit ($t = -1$, Case E), at successive times $\tau = 20.0, 260.0, 500.0, 760.0, 980.0$ (represented using contours drawn at $c = 0.5$). Here, $G = 500.0$ and $c_\infty = 0.20$. The dimensions of the box correspond to 140×140 length units.

follows:

$$\chi = \pm \frac{\mathfrak{I}_X - \mathfrak{I}_Y}{\mathfrak{I}_X + \mathfrak{I}_Y}, \quad (4.15)$$

where $\mathfrak{I}_X$ and $\mathfrak{I}_Y$ are the principal values of $\mathfrak{I}$ defined in Eq. 4.14. This definition of χ ensures that it is zero when the particle shape has T symmetry, and becomes non-zero when the shape changes to one with O symmetry. The bifurcation diagram, where the shape parameter χ is plotted as a function of normalized particle size ρ , is shown in Fig. 4.8. Our results for this case do agree with those obtained for the equilibrium shapes [1],

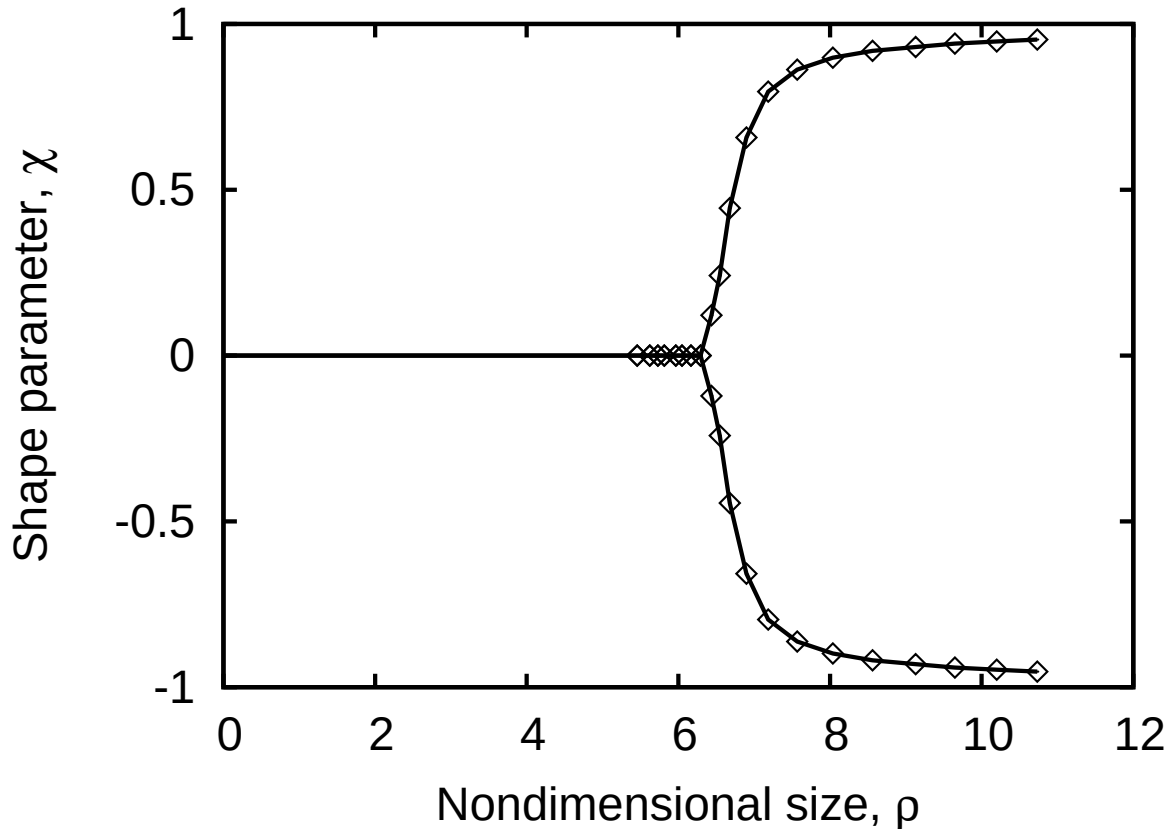


Figure 4.8: Bifurcation diagram showing $T \rightarrow O$ transition in a system with $G = 500$ and $c_\infty = 0.2$ in Case E ($t = -1$). The line through the data points is only a guide to the eye.

although the critical size of the growing particle for this $T \rightarrow O$ symmetry-breaking transition ($\rho_c \approx 6.4$) is larger than that obtained for the corresponding equilibrium shape ($\rho_c^{equi} \approx 2.2$).

Effect of shear modulus and supersaturation

Similar to the previous case, we have studied the effect of G and c_∞ on the critical particle size R_c for the $T \rightarrow O$ symmetry breaking transition. In this case, R_c is obtained for $G = 500$ and 750 , and supersaturations of $c_\infty = 0.2, 0.3$ and 0.4 . We plot the variation of R_c as a function of L_c in Fig 4.9. As shown in the figure, the critical size R_c decreases

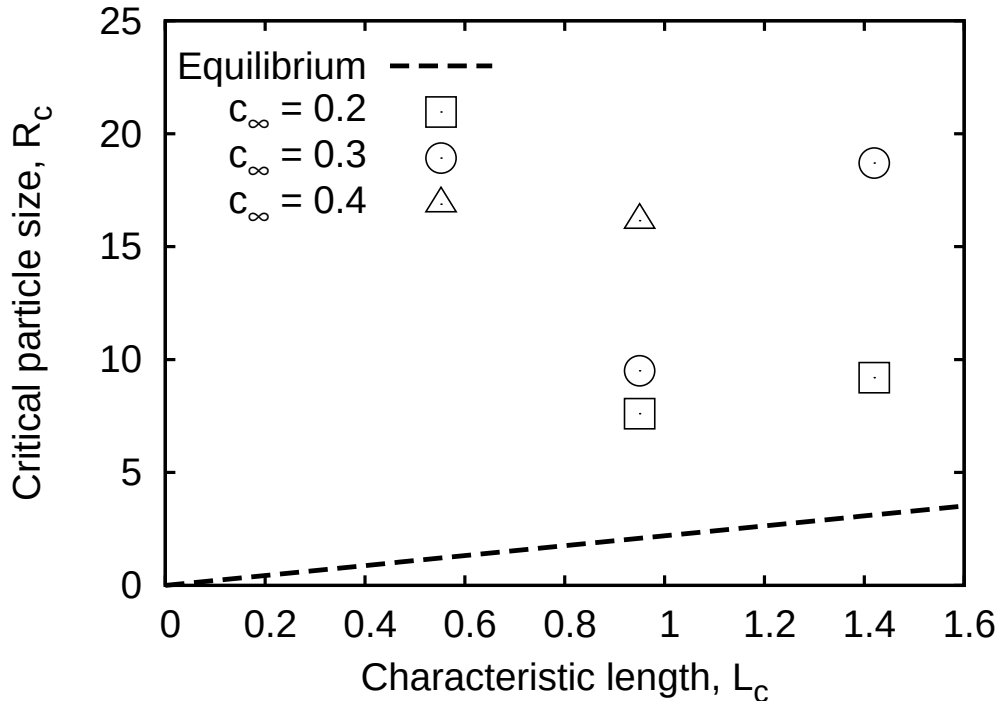


Figure 4.9: Variation of critical size, R_c , for the $T \rightarrow O$ symmetry-breaking shape transition for a growing particle, as a function of L_c for systems with different supersaturations. The dashed line ($R_c = 2.2L_c$) is for the same symmetry-breaking transition in equilibrium shape.

with increasing shear modulus and decreasing supersaturation.

In both Cases D and E, the trends are similar: the growing precipitate breaks the symmetry more easily in systems with low supersaturation and high elastic energy; the larger the particle size, more difficult it becomes to break the symmetry.

4.7 Discussion

The major conclusion from our study of dynamic growth shapes is that symmetry-breaking transition occurs only in those cases where the principal components of the misfit strain tensor have opposite sign (Cases D and E); in the other cases (A, B and C), there is no transition. These results are in agreement with those of Sankarasubramanian *et al.* on equilibrium shapes in systems with homogeneous elastic moduli. The transition size for growth shapes, however, is higher than that for equilibrium shapes. Our results also show that shape transitions occur at smaller particle sizes in systems with lower L_c (i.e., high G , ϵ_η , and low γ_i) and lower supersaturations.

In Cases A and B, Sankarasubramanian *et al.* observed various symmetry breaking transitions for the equilibrium shapes when the particle and matrix phases have different elastic moduli and also when the moduli have a cubic symmetry. Several other studies also report symmetry-breaking transition of the equilibrium shapes in these cases [69, 99, 100]. Since our model does not take into account the effect of elastic inhomogeneity, it is not possible for us to observe these transitions in our study. In fact, Johnson and Cahn [68] have analytically shown that a circle $\rightarrow$ ellipse ($I \rightarrow O$) symmetry-breaking transition occurs only when the precipitate is softer than the matrix.

Since this thesis examines hexagonal to orthorhombic transformation in two dimensions and the elastic constants of the hexagonal parent phase have an isotropic symmetry, in this chapter we have presented our results only for systems that are elastically homogeneous and isotropic. However, it is easy to incorporate anisotropy in elastic constants in our model. Our study of Cases D and E with cubic anisotropy in elastic constants is presented in Appendix E. As shown in the appendix, our results in these cases are in broad agreement with those obtained by Sankarasubramanian *et al.* for equilibrium shapes. Also, in agreement with the results presented in this chapter, the transition sizes are significantly higher for dynamical growth shapes than for equilibrium shapes.

Lee and coworkers [101] have studied the morphological evolution of coherent second-phase particles with different levels of anisotropy in misfit strain using a discrete atom model. Since the critical sizes for symmetry breaking transitions are not reported in their study, a comparison of the critical sizes obtained in our study with theirs is not possible.

Thompson *et al.* [99] observed nonconvex equilibrium shapes in two dimensions on addition of large tetragonal misfit. The concave equilibrium shapes, reported by them, correspond to local minima in energy, and hence are metastable. However, the equilibrium shapes associated with the global minima in energy are always convex [1, 17, 99]. In the kinetic setting, with increasing supersaturation and increasing shear modulus, the growth shapes exhibit concave interfaces. For example, in a system with $G = 750$ and $c_\infty = 0.3$ the growth shapes of the particle in Case E show concavities in the interfaces (Fig. 4.10) at the earlier times, before it breaks its symmetry. During the later stages, the interfaces tend to become convex when the shape attains a lower O symmetry. Wang and Khachaturyan [102] have also reported concave interfaces in their study of precipitate growth in a system with cubic elastic anisotropy and dilatational misfit. They interpret these concavities as kinetic instabilities which may ultimately lead to splitting of the particle. However, our results suggest that the concave interfaces may be indicative of an impending symmetry breaking transition in their simulations.

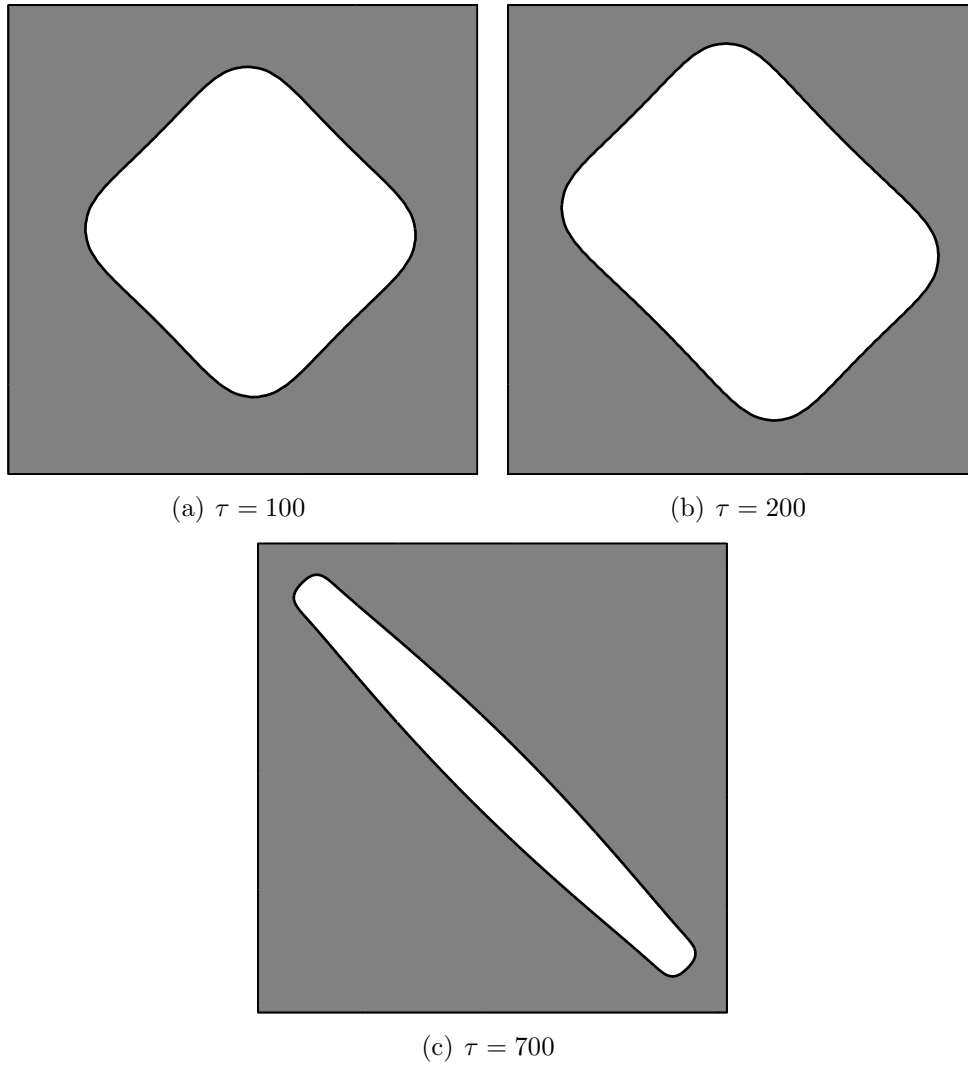


Figure 4.10: Shapes of a growing precipitate in Case E (represented using filled contours drawn at $c = 0.5$). at times $\tau = 100, 200, 600$ Here, $G = 750.0$ and $c_\infty = 0.30$. The shapes are drawn to different scales to emphasize the interfacial features at different times.

4.8 Summary

We examined the shapes of isolated misfitting particles as they grow diffusionally into a supersaturated matrix. Particle shapes are shown to depend on the anisotropy in misfit (represented by the parameter t). In particular when $t < 0$, growing particles undergo symmetry breaking transitions just like equilibrium shapes do. The size at which such a transition occurs, however is higher for dynamical shapes than for equilibrium ones.

Chapter 5

Microstructural Evolution

In this chapter we present the results from our simulations of two-phase, multi-variant microstructural evolution where the three variants of the precipitate phase nucleate and grow in the hexagonal parent phase. Specifically, we have studied systematically the role played by the anisotropy in misfit strain, characterized by parameter t (see Eq. 3.2). Further we have also studied the effect of alloy composition and elastic energy on the morphology of the phases.

In the next section, we present the details of the microstructural simulations. In the subsequent sections, we present the morphological patterns obtained for each of the different cases of anisotropic misfit strain, the influence of alloy composition on precipitate morphology in each of these cases, the effect of characteristic length L_c (i.e. the effect of shear modulus), and for $t = -1$, the effect of antiphase domain boundary energy. Finally, we conclude this chapter with a discussion of the results.

5.1 Simulation details

All the parameter values used in our simulations are listed in Table 5.1. We carry out the

Table 5.1: Parameters used in the multiparticle simulations

Parameters	Non-dimensional values
Bulk free energy coefficients ($A_1, A_2, A_{41}, A_{42}, A_{61}, A_{62}$)	1.0, 2.0, -3.0, 36.0, 2.0, 4.0
Gradient energy coefficient for composition c (κ_c)	0.2
Gradient energy coefficient for order parameter η_1 (κ_η)	0.7
Mobility (M)	1.0
Kinetic Coefficient (L)	1.0
Strength of the misfit strain associated with η_1 (ϵ_{η_1})	0.03
Strength of the misfit strain associated with composition (ϵ_c)	0.0
Misfit anisotropy parameter (t)	1.0, 0.5, 0.0, -0.5, -1.0
Average shear modulus (G)	
Case A	300.0
Cases B–E	500.0, 1500.0
Poisson's ratio (ν)	0.3
Zener anisotropy parameter (A_Z)	1.0
Alloy compositions (c_∞)	0.3, 0.5
Grid spacing (Δx)	0.5
Time step ($\Delta \tau$)	0.01
Noise strength	20.0% (during nucleation)

simulations on a 1024×1024 square grid with a dimensionless grid spacing of $\Delta x = 0.5$ using periodic boundary conditions.

As in Chapter 3, the misfit anisotropy parameter t is systematically varied from $t = +1$ (pure dilatation) to $t = -1$ (pure shear). While we have used $G = 1500$ in all the other cases, we used $G = 300$ in the system with $t = 1$, since higher values of G ($G > 500$) inhibits the formation of stable nuclei (see appendix D). We have studied two alloy compositions, $c_0 = 0.3$ and 0.5 . Further, the characteristic length L_c ($= \gamma_i/G\epsilon^2$) can be increased either by increasing the interfacial energy γ_i or by lowering the value of shear modulus G . Hence, we have used two different non-dimensional values of the shear modulus: $G = 500$ and $G = 1500$ for Cases B–E.

The precipitation of β phase from the initial metastable hexagonal α phase takes place through a nucleation and growth mechanism. In the simulations, nucleation is induced by introducing sustained random fluctuations in c and η_i with a large amplitude (20%) at each grid point. The noise is switched off after a certain time, when a sufficient number of nuclei are produced. Further evolution of the microstructure occurs via growth and coarsening of the precipitate phase. Reduction in noise strength results in a delayed formation of nuclei, though the morphology of the resultant nuclei remain unaffected.

The results of microstructural evolution are presented as time snapshots using a gray scale representation of the local values of the long range order parameters. Such a representation allows us to distinguish between the three pairs of orientation variants of the β phase. In the gray scale map, the shades of gray represent the values of $(\eta_1^2 - \eta_3^2 - 2\eta_2^2 + 2)/3$. Thus, the variants with $\eta_1 = \pm 1$ are represented by a pure white shade, $\eta_2 = \pm 1$ a pure black shade, $\eta_3 = \pm 1$ a dark gray shade, and the matrix phase ($\eta_1 = \eta_2 = \eta_3 = 0$) by a light gray shade. The partially ordered (interfacial) regions are represented by intermediate gray shades that continually vary from pure white to pure black. A similar gray scale representation has also been used by Wen *et al.* [13].

5.2 Effect of t , the misfit anisotropy

In all the five cases we have studied, the microstructural evolution is shown in Figs. 5.1 & 5.2 for $c_0 = 0.3$, and in Figs. 5.3 & 5.4 for $c_0 = 0.5$, respectively. The microstructures of these alloys during the early ($\tau = 120$) and the late stages ($\tau = 1000$) are presented for each case. The general features, common to all the cases, are described below.

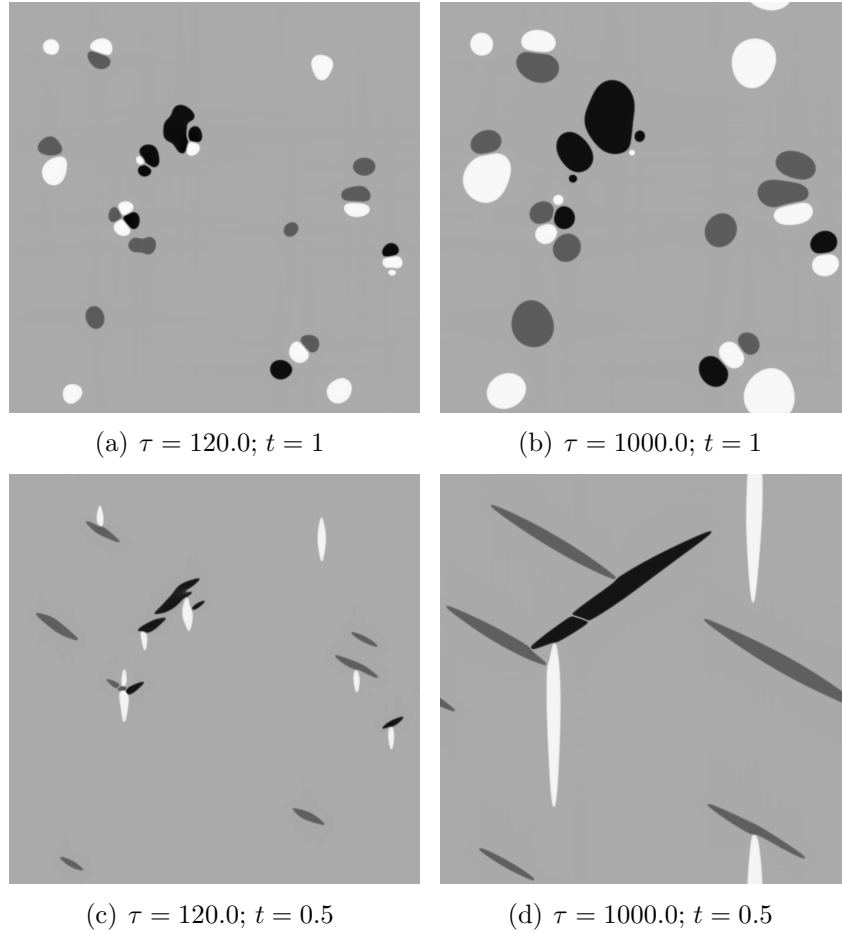


Figure 5.1: Microstructural evolution of an alloy with composition $c_0 = 0.3$ in Cases A ($t = 1$) and B ($t = 0.5$). The evolution time and values of parameter, t , are indicated in each figure.

In both alloys nucleation produces particles of all three orientational variants. Some particles are isolated and some appear in clusters, with the proportion of clusters being higher in the alloy with a higher concentration. By $\tau = 1000$ the microstructural evolution takes the alloys well into the coarsening regime, as evidenced by a smaller number of

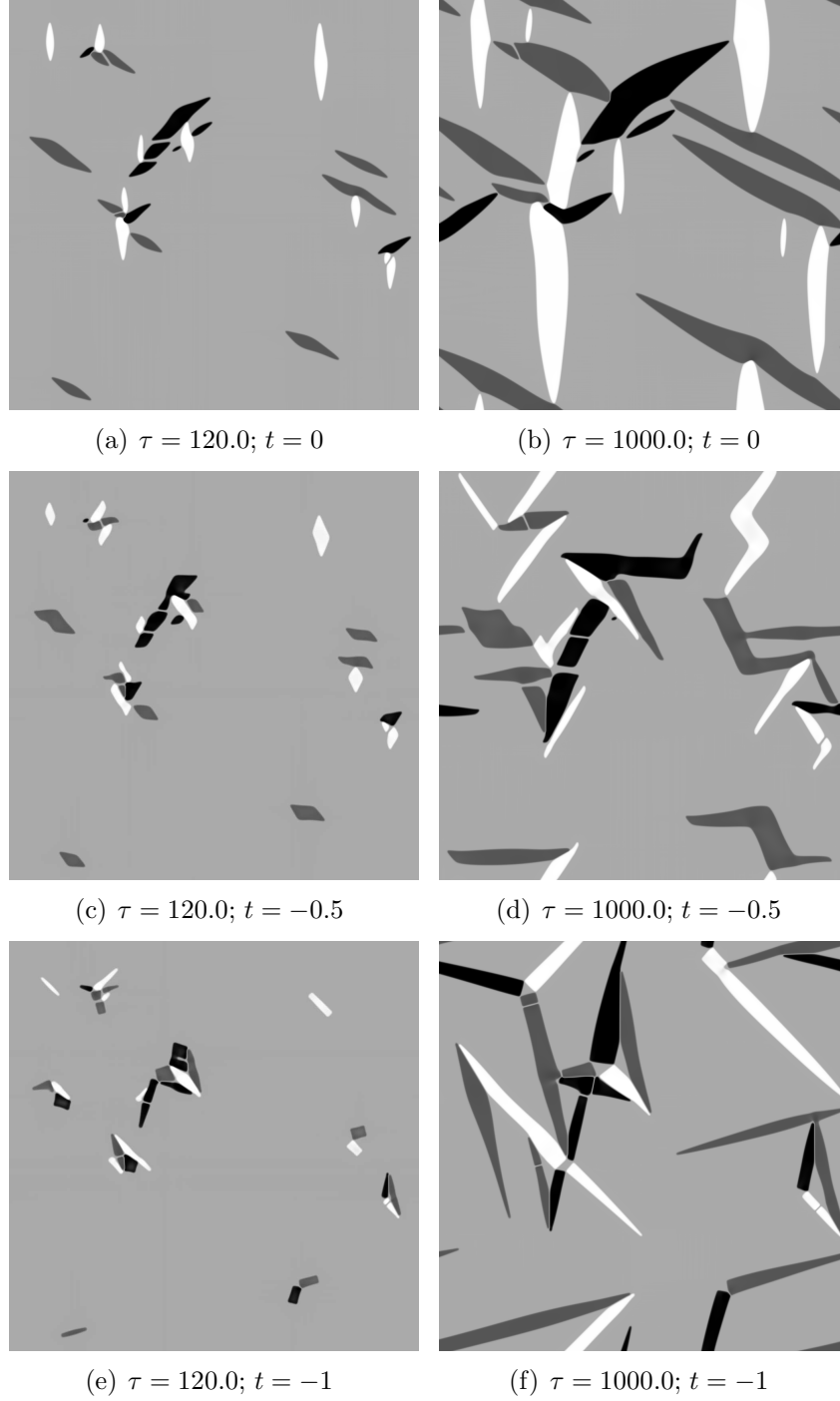


Figure 5.2: Microstructural evolution of an alloy with composition $c_0 = 0.3$ for Cases C ($t = 0$), D ($t = -0.5$) and E ($t = -1$). The evolution time and values of parameter, t , are indicated in each figure.

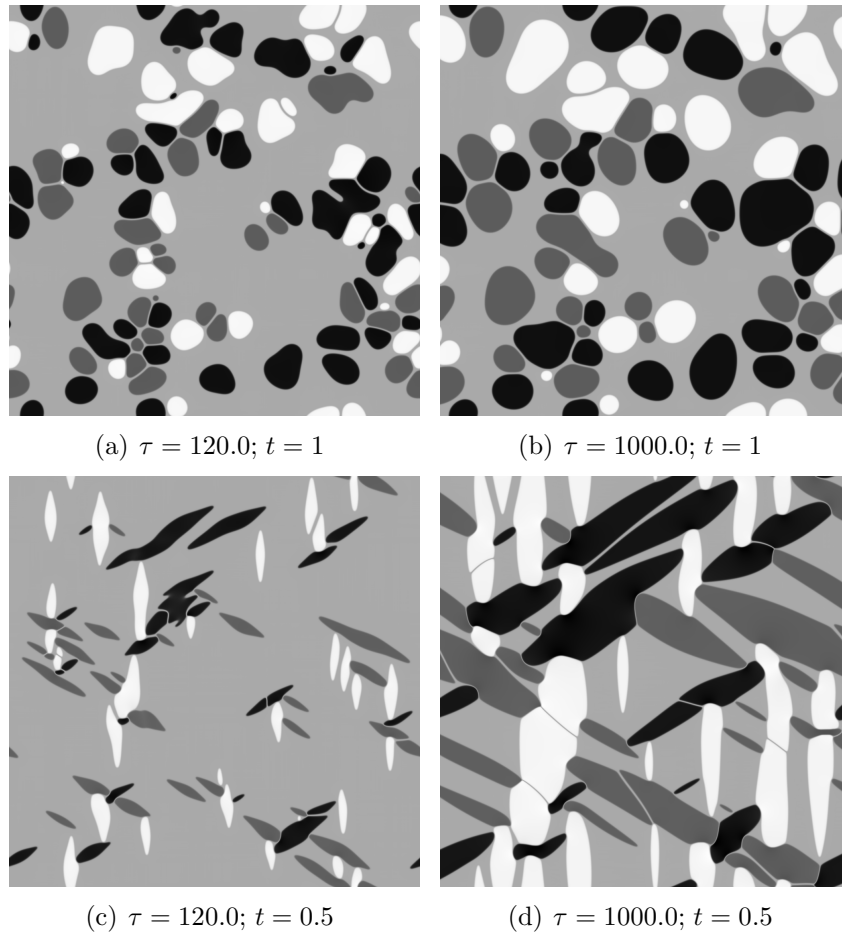


Figure 5.3: Microstructural evolution of an alloy with composition $c_0 = 0.5$ in Cases A ($t = 1$) and B ($t = 0.5$). The evolution time and values of parameter, t , are indicated in each figure.

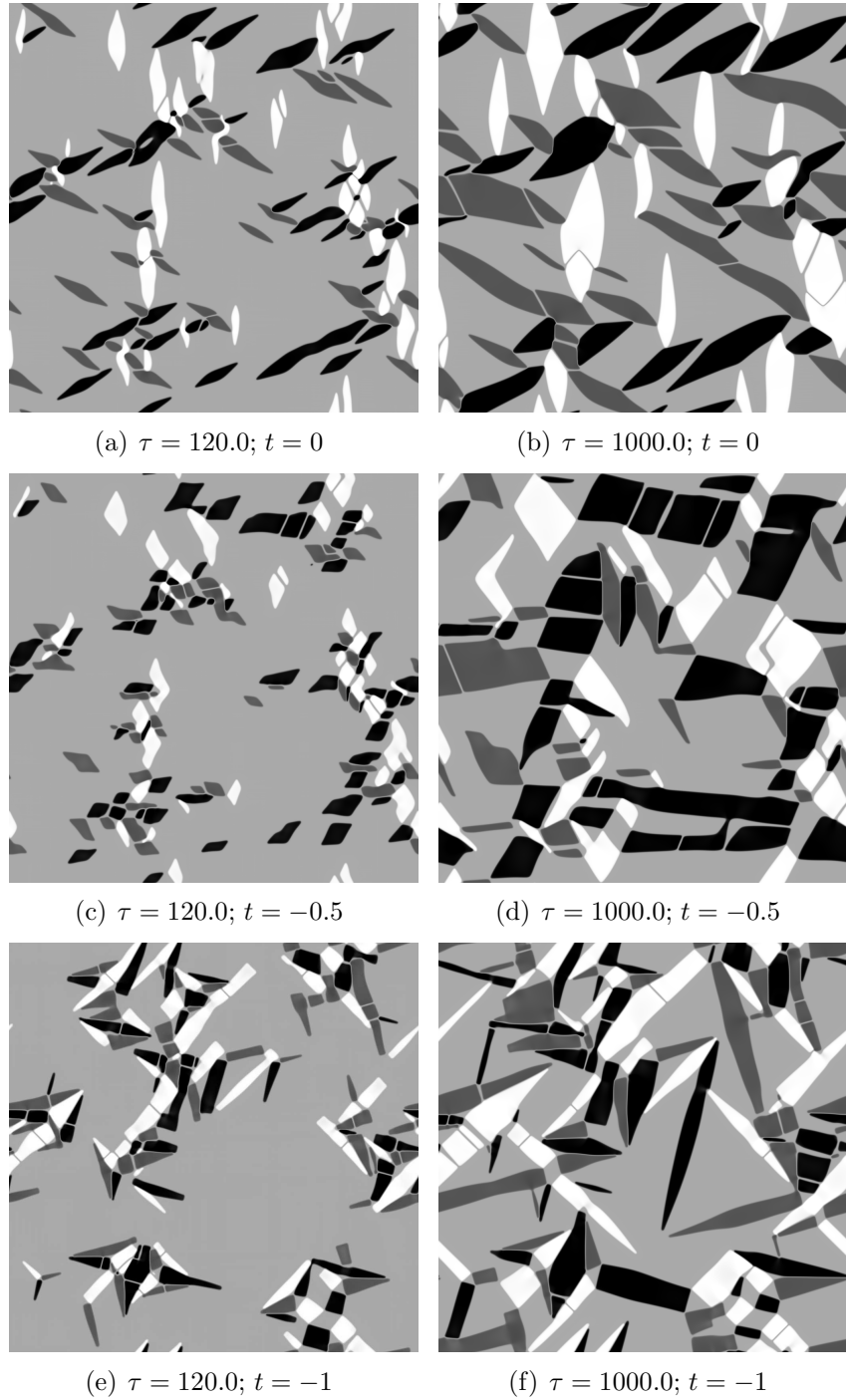


Figure 5.4: Microstructural evolution of an alloy with composition $c_0 = 0.5$ in Cases C ($t = 0$), D ($t = -0.5$) and E ($t = -1$). The evolution time and values of parameter, t , are indicated in each figure.

particles at this time than at $\tau = 120$. The volume fraction of the particles is higher in the alloy with higher concentration.

The shapes and orientations of particles differ for each case and are described separately in the following sections.

5.2.1 Case A: Dilatational Misfit ($t = 1$)

The evolution of microstructures in this system is as depicted in Fig 5.5. The top row is for an alloy with $c_0 = 0.3$ at $\tau = 120$ and $\tau = 1000$; the bottom row is for an alloy with a higher supersaturation $c_0 = 0.5$.

In the alloy with $c_0 = 0.3$, in Fig. 5.5b, the isolated particles appear circular; however, those that have other particles in the vicinity appear non-circular, due to overlap of diffusion fields. In the more concentrated alloy, this overlap is more predominant leading to the microstructure in Fig. 5.5d where almost all the particles appear non-circular.

The dependence of volume fraction on time in these two alloys is as shown in Fig. 5.6. The volume fraction of the precipitates increases to approximately 0.05 at $\tau = 1000$ in the alloy with $c_0 = 0.3$. On the other hand, in the more concentrated alloy ($c_0 = 0.5$) the volume fraction increases to approximately 0.58 which, surprisingly, is greater than that expected (0.5) in this alloy.

The volume fractions of precipitates in both alloys can be rationalized using the effect of elastic stress on equilibrium compositions of the two phases (generalized Gibbs-Thomson effect) [3, 103]; see Section 5.5.

5.2.2 Case B ($t = 0.5$)

The microstructures at different stages of evolution for the two alloys ($c_0 = 0.3$ and 0.5) are shown in Fig. 5.7.

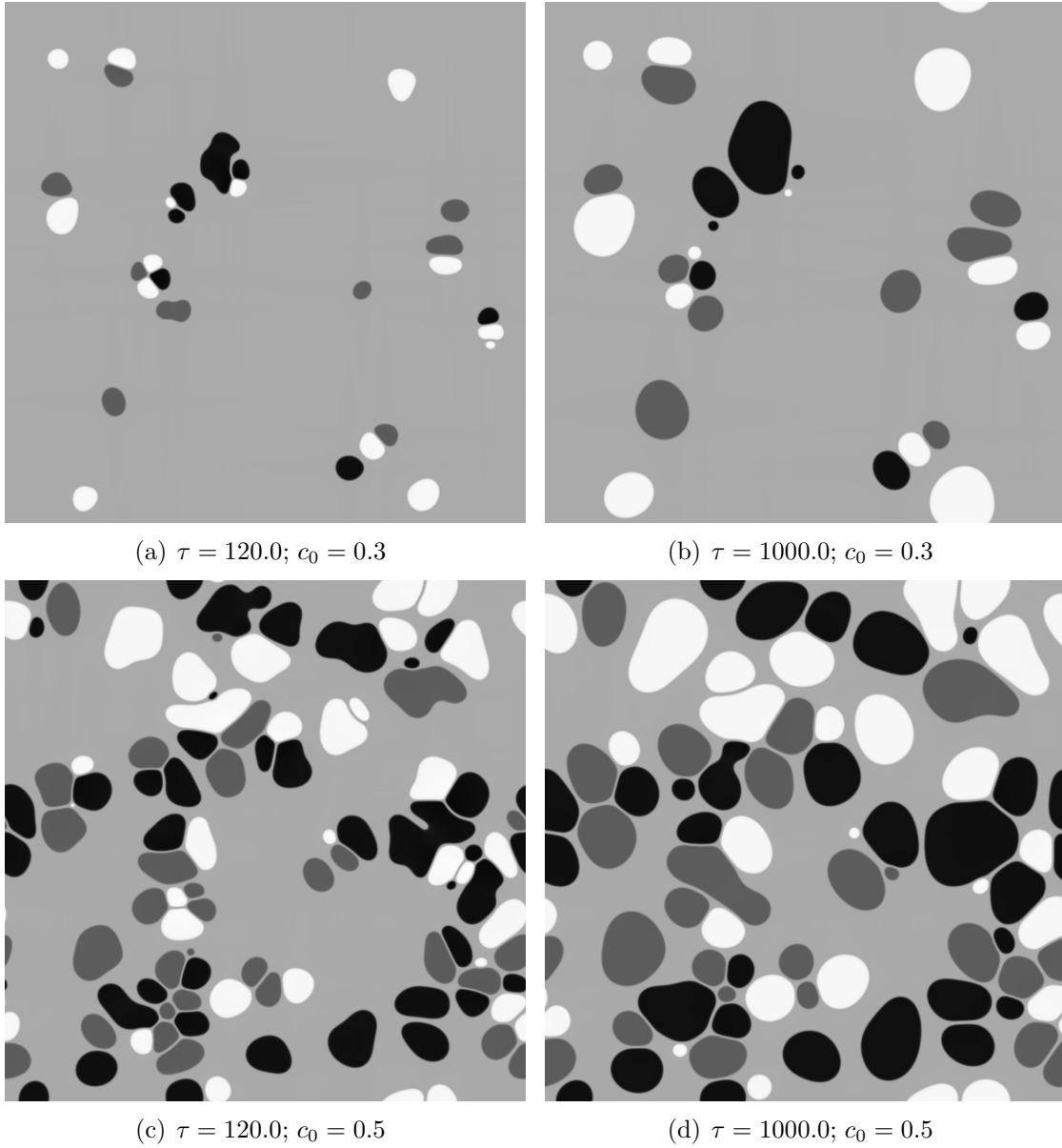


Figure 5.5: Microstructural evolution in a system with dilatational misfit for two alloys with $c_0 = 0.3$ and 0.5 , respectively. The evolution time and the alloy composition is indicated below each figure.

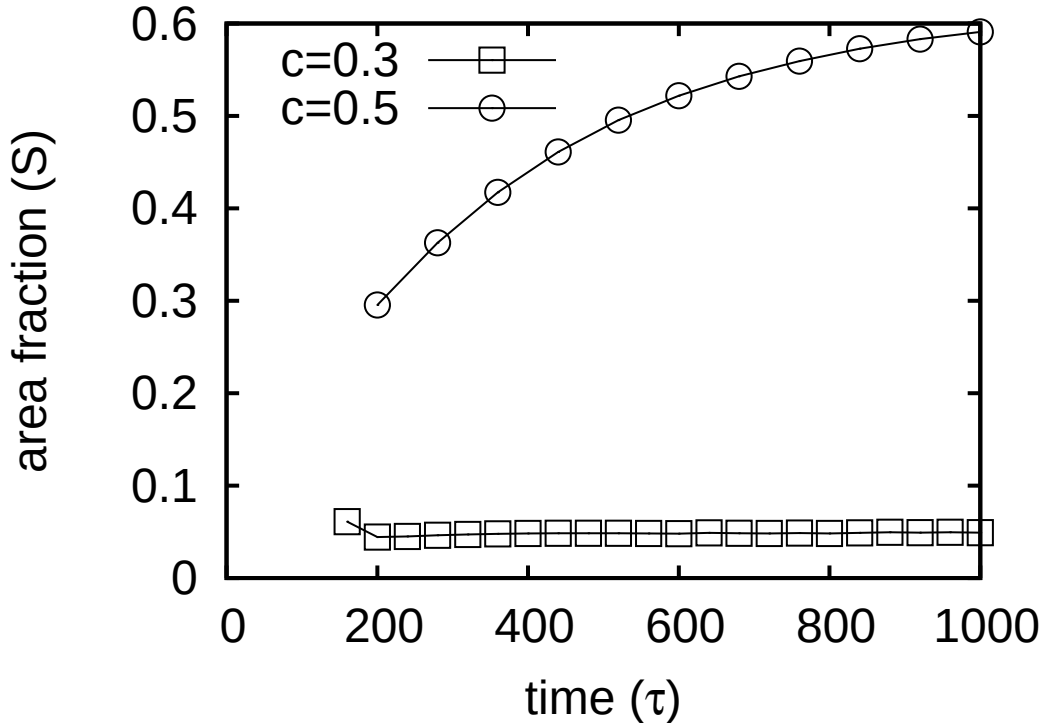


Figure 5.6: Precipitate volume fraction as a function of time for the two alloy compositions ($c_0 = 0.3, 0.5$) in a system with dilatational misfit ($t = 1$).

In this case, $\epsilon_{yy} = 0.5\epsilon_{xx}$ for the first variant (shown in white in the microstructures) i.e. the principal misfit along y -axis is lower than that along the x -axis. In both alloys, particles have a lens-shape with its long axis along the directions of lower misfit (y axis for the white variant and directions at $\pm 120^\circ$ with the y axis for the black and dark gray variants).

Although the orientations of the plates remain roughly the same in both alloys at $\tau = 1000$, the shapes of the individual plates differ significantly; in the alloy with $c_0 = 0.3$, the microstructure at $\tau = 1000$ contains thin elongated plates (Fig. 5.7b), while in that with $c_0 = 0.5$ particles appear with jagged or irregular interfaces. This is clearly a volume fraction effect: since the more concentrated alloy has a greater density of particles, their elongation leads to impingement with other particles at shorter distances. In fact, in Fig. 5.7d, interface irregularities appear wherever the leading edge of a particle impinges on the flat side of another.

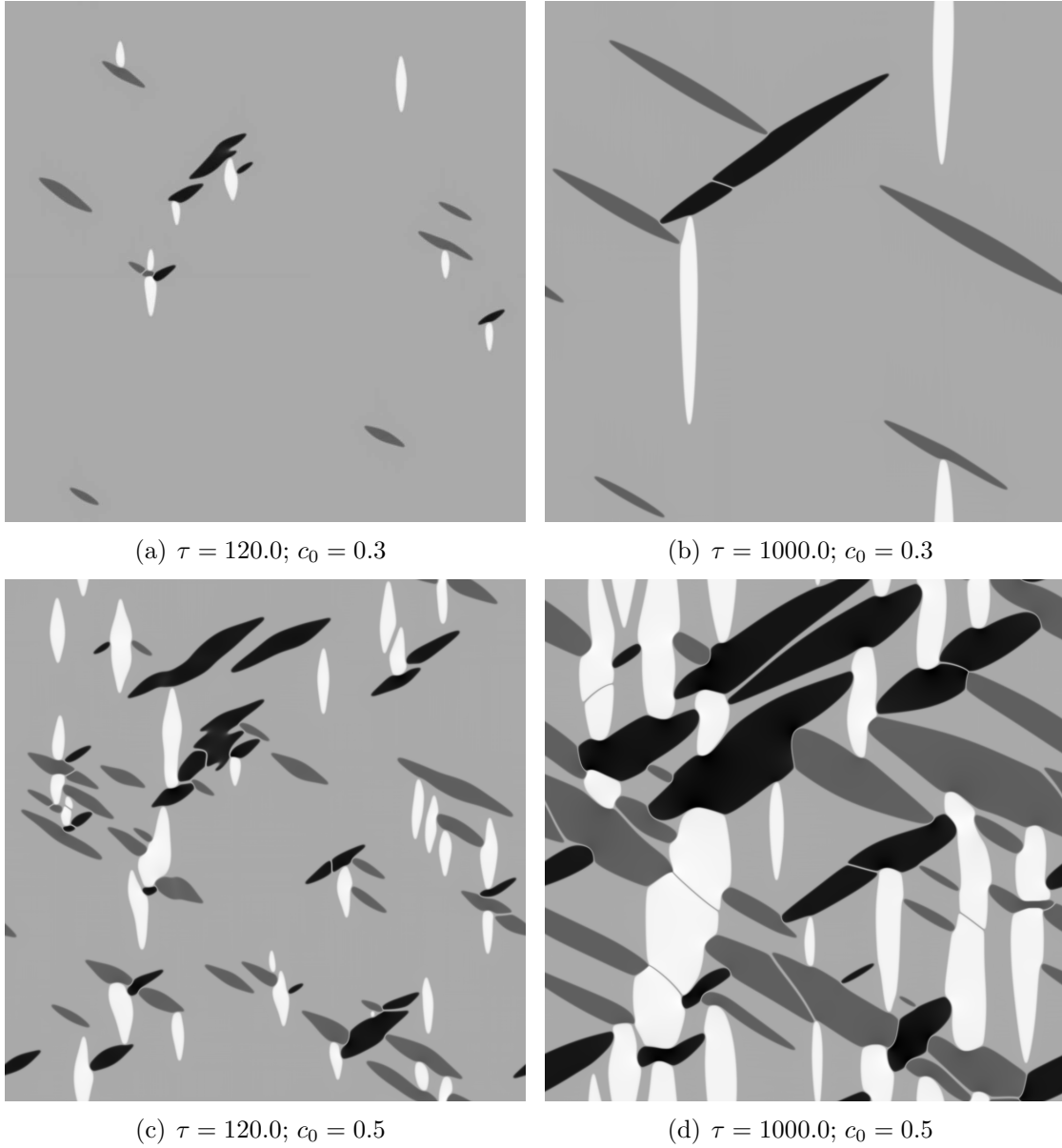


Figure 5.7: Microstructural evolution in Case B for two alloys with $c_0 = 0.3$ and 0.5 , respectively. The evolution time and the alloy composition is indicated below each figure.

The volume fraction of the precipitate phase as a function of time is shown in Fig. 5.8 for the two alloys. The small volume fraction of the particles in the alloy with lower concen-

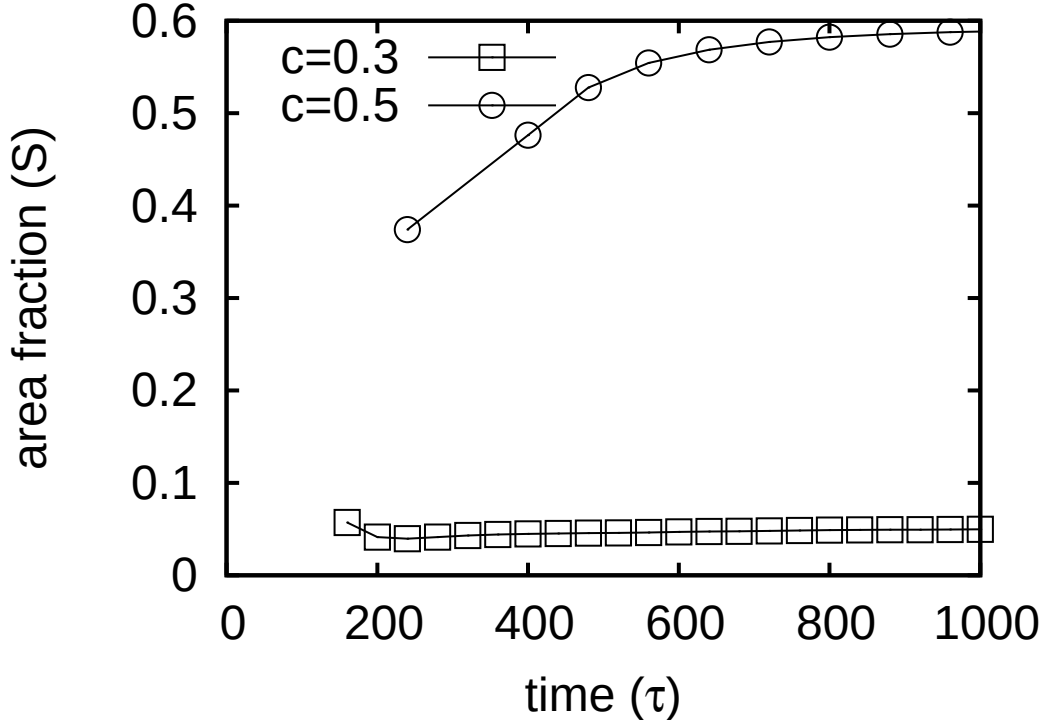


Figure 5.8: Precipitate volume fraction as a function of time for the two alloy compositions ($c_0 = 0.3, 0.5$) in a system with anisotropic misfit ($t = 0.5$)

tration and a significant increase in volume fraction in the alloy with higher concentration can be explained using the generalized Gibbs-Thomson effect; it is discussed further in section 5.5.

5.2.3 Case C ($t = 0$)

The microstructures at different stages of evolution in this system for two alloys with concentrations $c_0 = 0.3$ and 0.5 are shown in the top and bottom rows of Fig. 5.9. In this case, $\epsilon_{yy} = 0$ i.e. the misfit along y axis is zero for the first variant (depicted in white). Thus we find that the white variant is elongated with its long axis along the y axis, and the black and dark gray variants are elongated along directions at 120° and -120° with

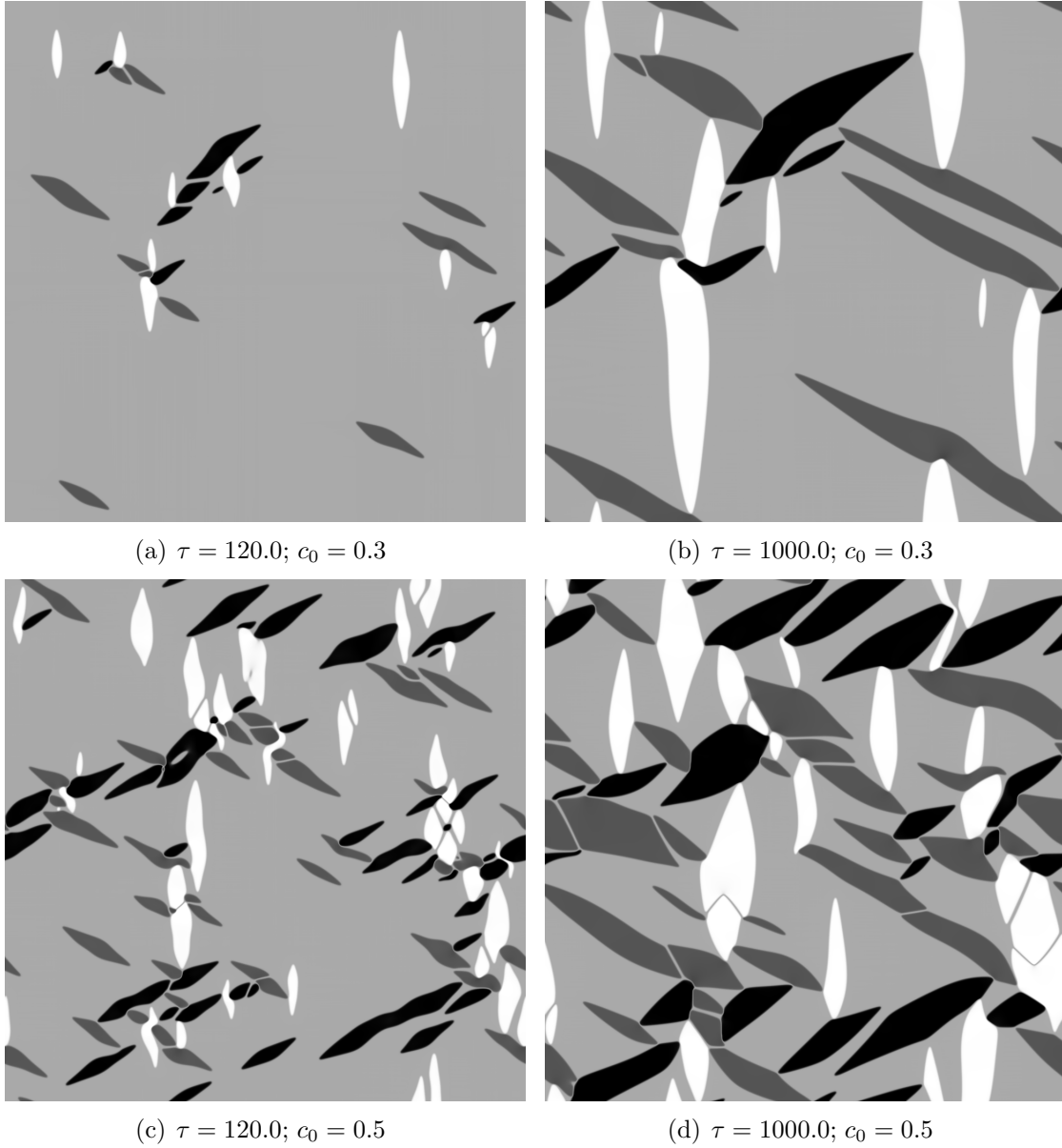


Figure 5.9: Microstructural evolution in Case C for two alloys with $c_0 = 0.3$ and 0.5 , respectively. The evolution time and the alloy composition is indicated below each figure.

respect to the white variant.

The particles in both alloys are broadly lens-shaped with the interfaces exhibiting irregularities at sites of impingement. The microstructure in the alloy with higher concentration at $\tau = 1000$ (Fig. 5.9d) show thicker and shorter plates due to a higher volume fraction.

The evolution of volume fraction of the precipitate phase with increasing time for the two alloys is shown in Fig. 5.10. In both alloys, the volume fraction of the particles appear

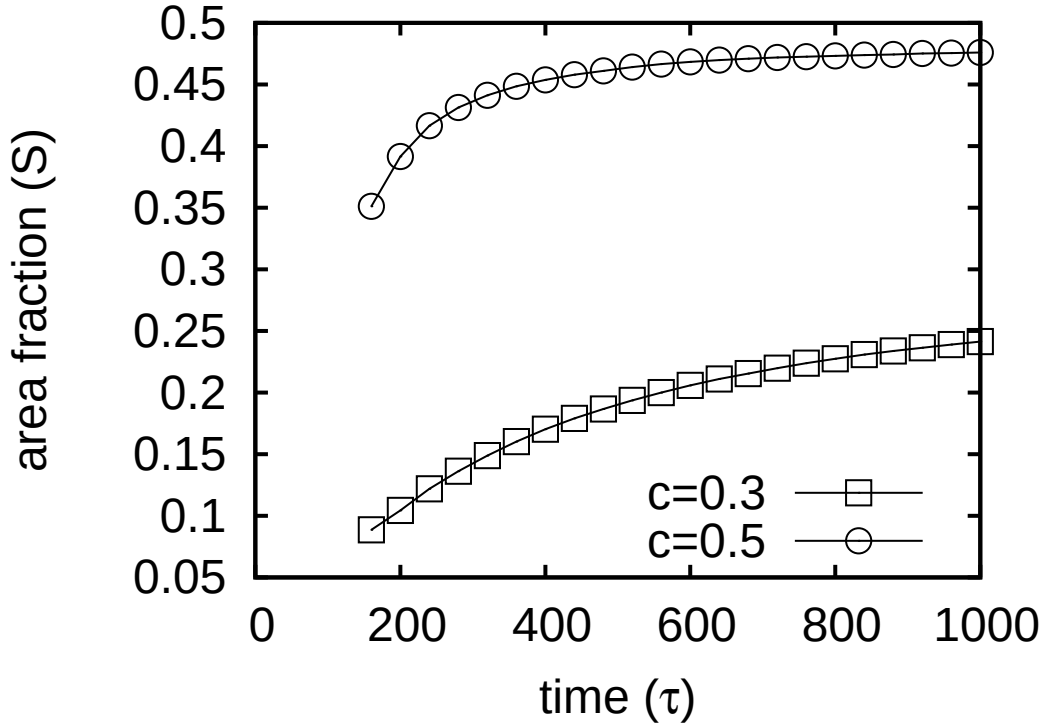


Figure 5.10: Precipitate volume fraction as a function of time for the two alloy compositions ($c_0 = 0.3, 0.5$) in a system with anisotropic misfit ($t = 0$).

to approach asymptotically their respective equilibrium values.

5.2.4 Case D ($t = -0.5$)

In this case, the principal values of the misfit strain are unequal in magnitude and opposite in sign. Hence, each orientational variant in this system is associated with an invariant line along which the normal misfit strain is zero. The invariant line orientations of each

variant with respect to the x axis are obtained as

$$\begin{aligned}\theta(\text{variant 1}) &= \pm 54.72^\circ, \\ \theta(\text{variant 2}) &= \{65.28^\circ, -5.28^\circ\}, \\ \theta(\text{variant 3}) &= \{-65.28^\circ, 5.28^\circ\}.\end{aligned}\tag{5.1}$$

The microstructural evolution in the two alloys (with $c_0 = 0.3$ and 0.5) is as shown in Fig. 5.11. In both alloys the particles are elongated initially along their directions of lower absolute misfit. Thus, the white variant is elongated along the y -axis and black and dark gray variants are elongated along 120° and -120° to the y -axis. At later stages ($\tau = 1000$) the individual variants can be seen to change their orientation and direction of elongation towards their respective invariant lines (Fig. 5.11 b and d).

As a result of this reorientation, we often encounter transient ‘Z’-shaped domains, where each arm of the Z is oriented along the invariant directions (Fig. 5.11 b). For example, consider the white ‘Z’-shaped domain at the top right corner in Fig. 5.11b. The angles subtended by the two arms and the central stem of this domain with the x -axis are approximately $\pm 54^\circ$ which are indeed close to an invariant directions for this variant (see Fig 5.12).

In the alloy with higher concentration, the particles are smaller and thicker than those in the alloy with lower concentration, especially at the later stages of evolution; as in the previous cases, this is clearly the effect of a higher volume fraction in the alloy with $c_0 = 0.5$.

For both alloys the evolution of volume fraction of the precipitate phase with increasing time is shown in Fig. 5.13. Similar to Case C, the volume fraction of the particles increase uniformly with time and appear to approach asymptotically their respective equilibrium volume fractions.

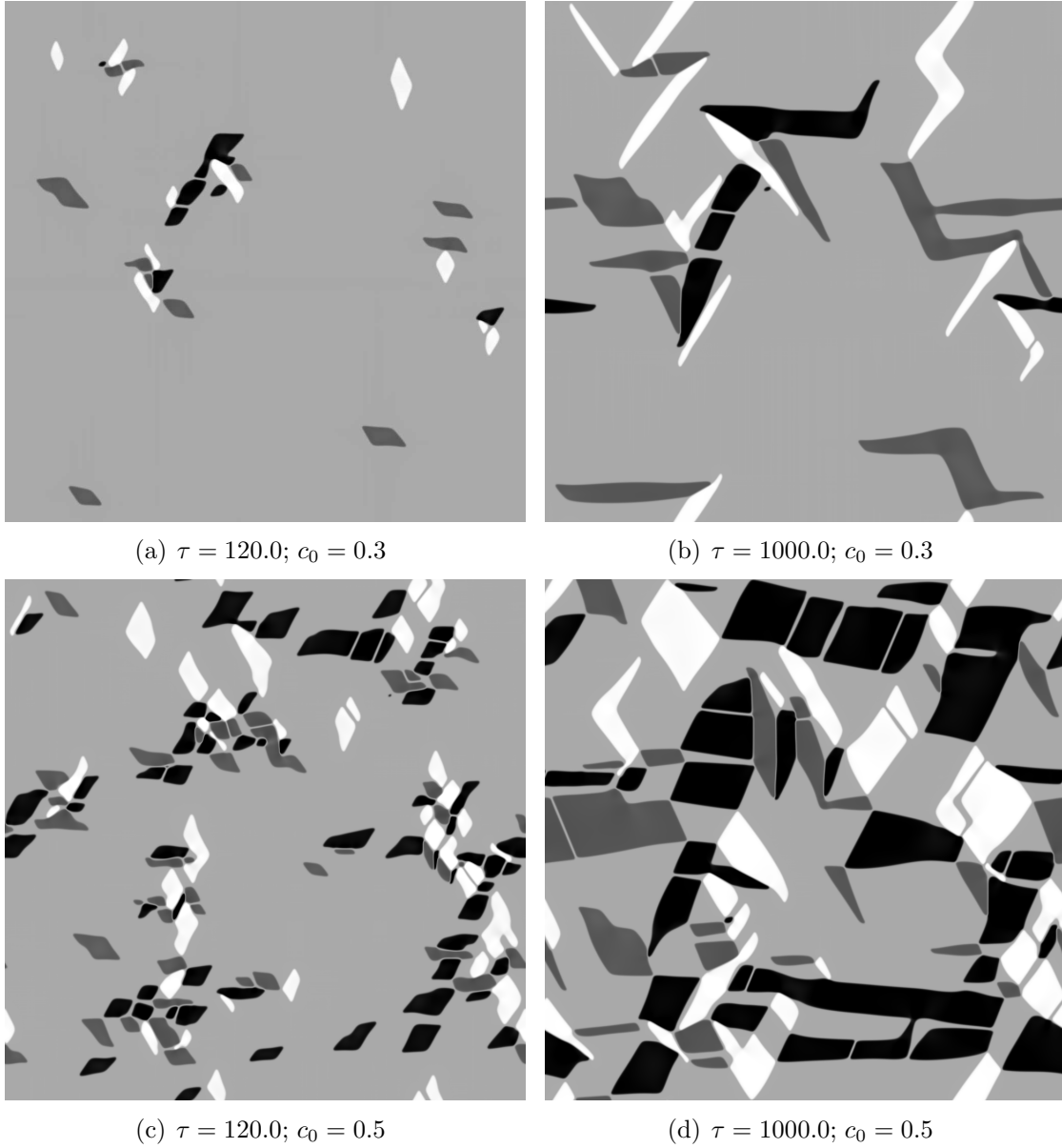


Figure 5.11: Microstructural evolution in a system with $t = -0.5$ (Case D) for two alloys with $c_0 = 0.3, 0.5$. The alloy composition and evolution time are indicated below each figure.

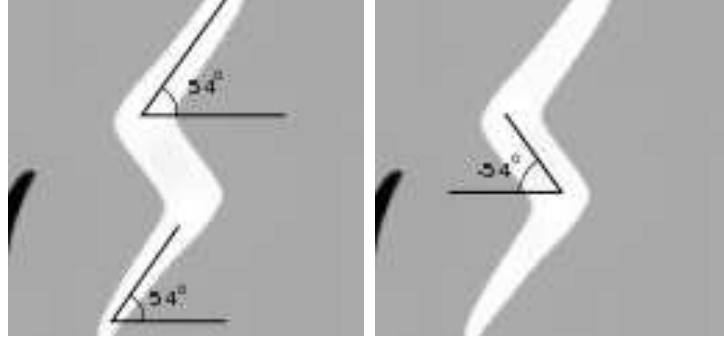


Figure 5.12: Angle subtended by the two arms and the central region of the ‘Z’-shaped domain at $\tau = 1000$.

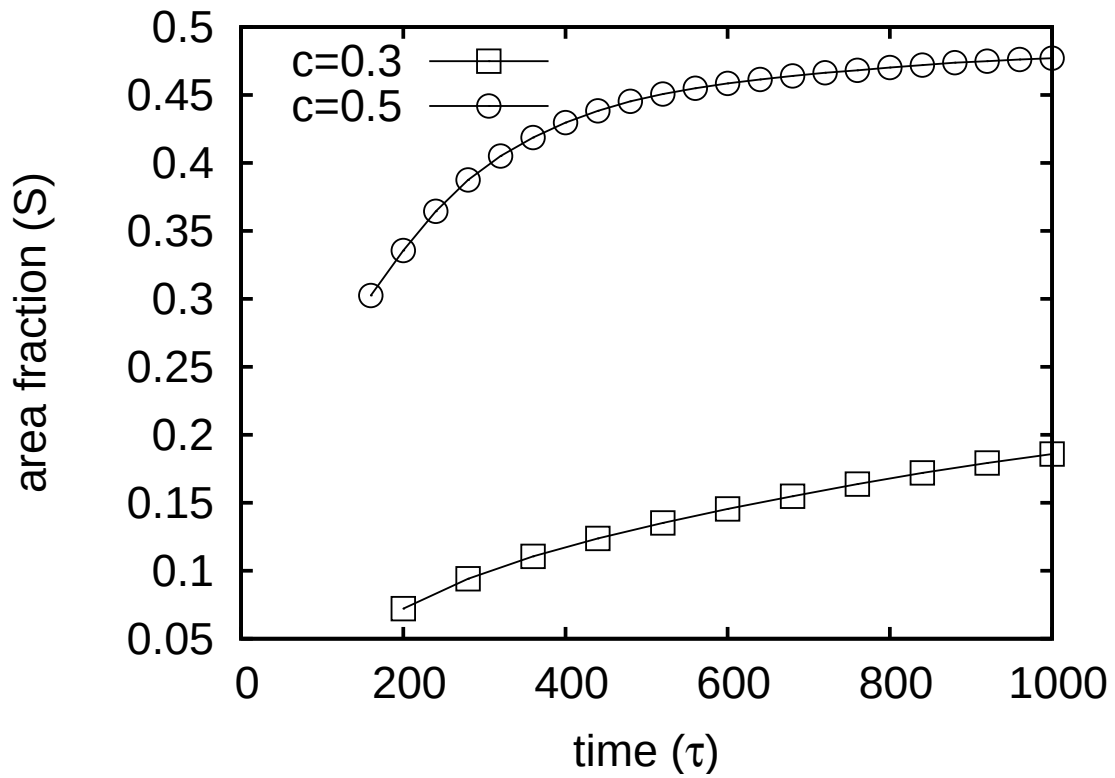


Figure 5.13: Precipitate volume fraction as a function of time for the two alloy compositions ($c_0 = 0.3, 0.5$) in a system with anisotropic misfit ($t = -0.5$).

5.2.5 Case E ($t = -1$)

In this case, the principal misfits are equal in magnitude and opposite in sign. Similar to the previous case, the system contains invariant lines along which the normal misfit is zero. The invariant directions for the three orientation variants are given by

$$\begin{aligned}\theta(\text{domain 1}) &= \pm 45^\circ, \\ \theta(\text{domain 2}) &= \{-15^\circ, 75^\circ\}, \\ \theta(\text{domain 3}) &= \{-75^\circ, 15^\circ\}.\end{aligned}\tag{5.2}$$

The microstructures for two different alloy compositions at various stages of evolution are shown in Fig. 5.14.

In both alloys the domains are plate-shaped. These plates orient along the invariant directions from the very early stages. Some of these plates arrange in the form of stars or triangles.

In the alloy with $c_0 = 0.3$ the domains grow to a very large extent along the invariant directions producing thin elongated plates in the microstructure at $\tau = 1000$ (Fig 5.14 b). For example, the white domain at the top right corner in Fig 5.14b subtends an angle of 45° with the x -axis (see Fig 5.15). In the alloy with higher concentration, due to increased proximity of the plates, the elongation is restricted and some of these plates thicken, leading to shorter plates with lower aspect ratios.

The interaction between these plates often results in the formation of three-sided star patterns, the central region of the star being occupied by the matrix phase. Several other patterns, such as compound stars, triangles, wedges, or checkerboards also form as a result of interaction between the variants (see Fig. 5.16)

Interaction between variants may also result in retransformation of the precipitate phase to the matrix phase. The checkerboard pattern in the left central region of the microstruc-

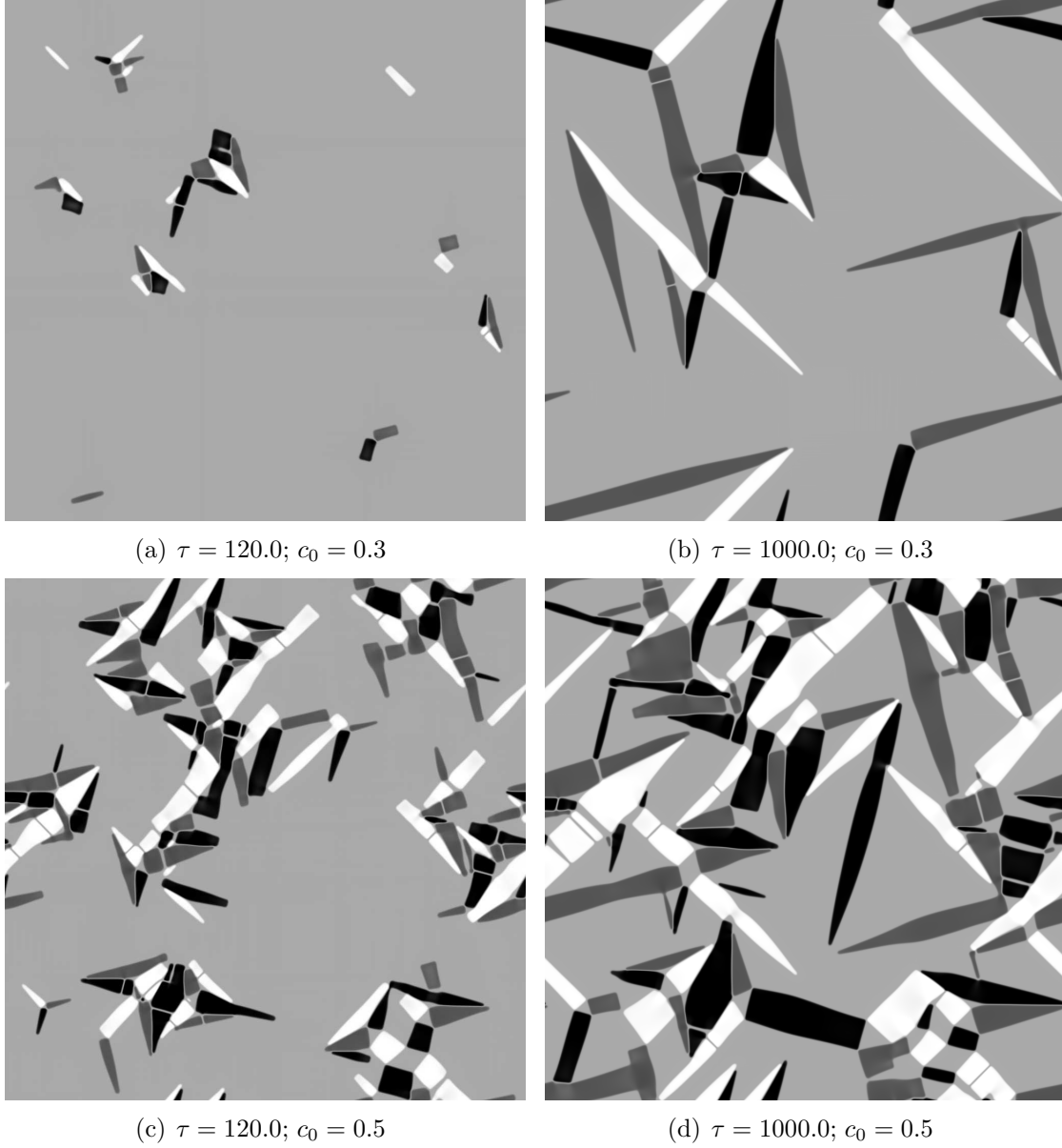


Figure 5.14: Microstructural evolution in a system with $t = -1$ (Case E) for two alloys with compositions $c_0 = 0.3$ and 0.5 . The evolution time is indicated below each figure.

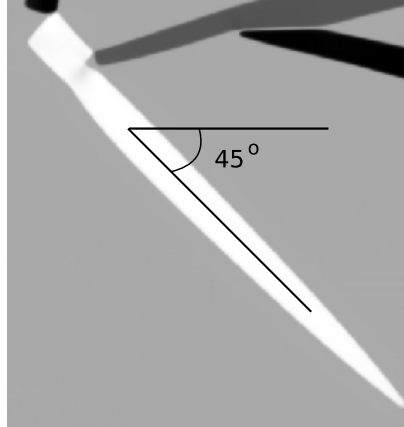


Figure 5.15: Angle subtended by the white domain at $\tau = 1000$.

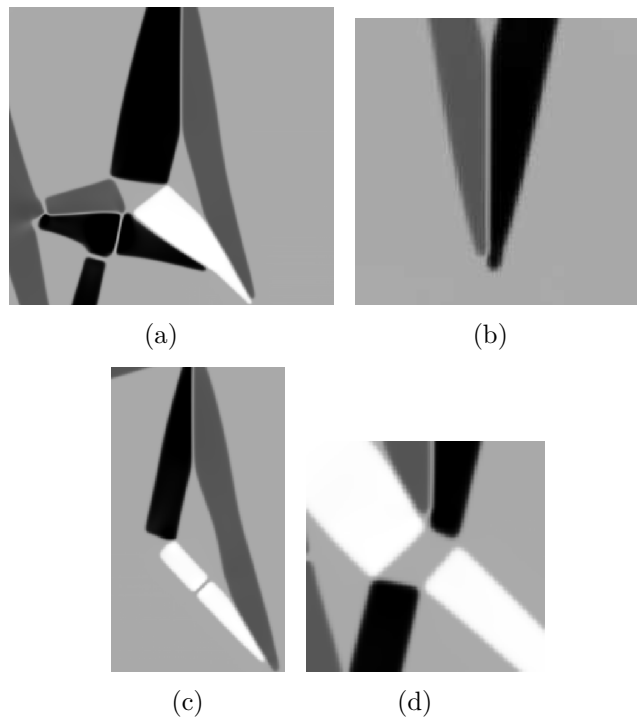


Figure 5.16: Different microstructural patterns observed in Case E: (a) compound star, (b) wedge, (c) triangle, and (d) checkerboard.

ture in Fig 5.14b provides an example. At the early stages, the white variant is continuous (Fig. 5.17a) and finds itself between two black variants that impinged on it from either side. In Fig. 5.17b it is clear that the white variant loses its continuity with the precipitate phase retransforming into the matrix phase at the impingement site.

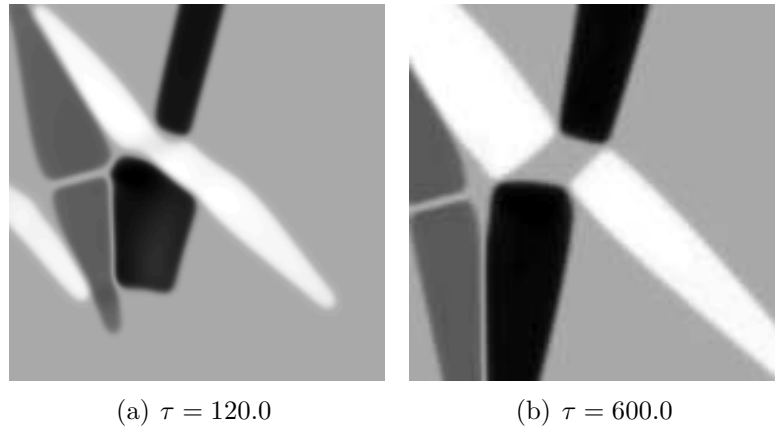


Figure 5.17: Retransformation of precipitate phase to matrix phase giving rise to a checkerboard pattern.

Similarly, when three variants intersect at the center of three-sided stars, the matrix phase is observed at overlapping region. Consider the star in the top left portion of the microstructure in Fig. 5.14a. Fig. 5.18 shows the star at different times of evolution; the area of the central region occupied by the matrix phase increases with increasing time. The increase in area occupied by the matrix gives yet another evidence of retransformation.

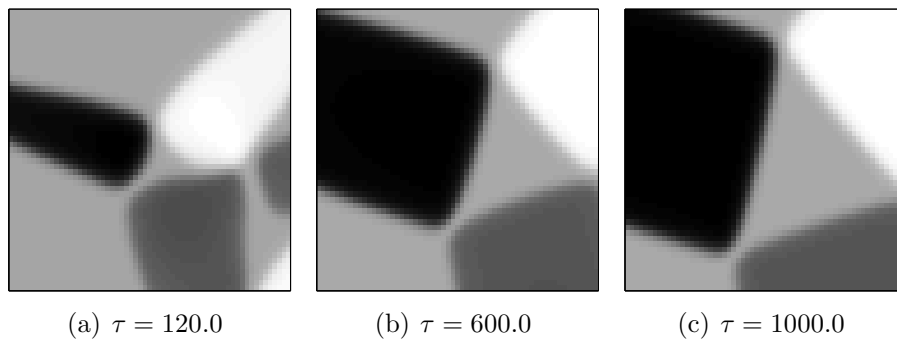


Figure 5.18: Increase in area of the central region occupied by matrix in a star pattern.

Moreover, elastic interactions may also lead to a lateral movement of the variants. An

example is shown in Fig. 5.19 (bottom right of Fig. 5.14b for later times) where the black plate has migrated normal to itself by a distance comparable to its thickness. We discuss a possible mechanism in Section 5.5.

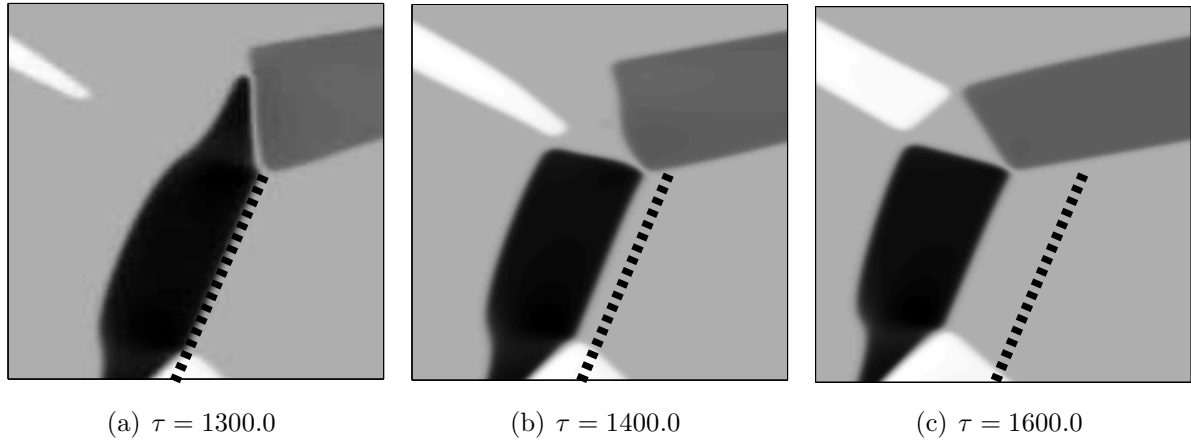


Figure 5.19: Lateral movement of the black variant at different times. The black dotted line has been drawn to show the location of the original matrix/black variant interface at $\tau = 1300$.

The evolution of volume fraction of the precipitate phase in these alloys with increasing time are compared in Fig. 5.20. In both alloys, the volume fractions occupied by the precipitate phase increase with time and approach the respective alloy compositions. Similar to Cases C and D, the precipitate composition in both alloys are closer to the equilibrium values.

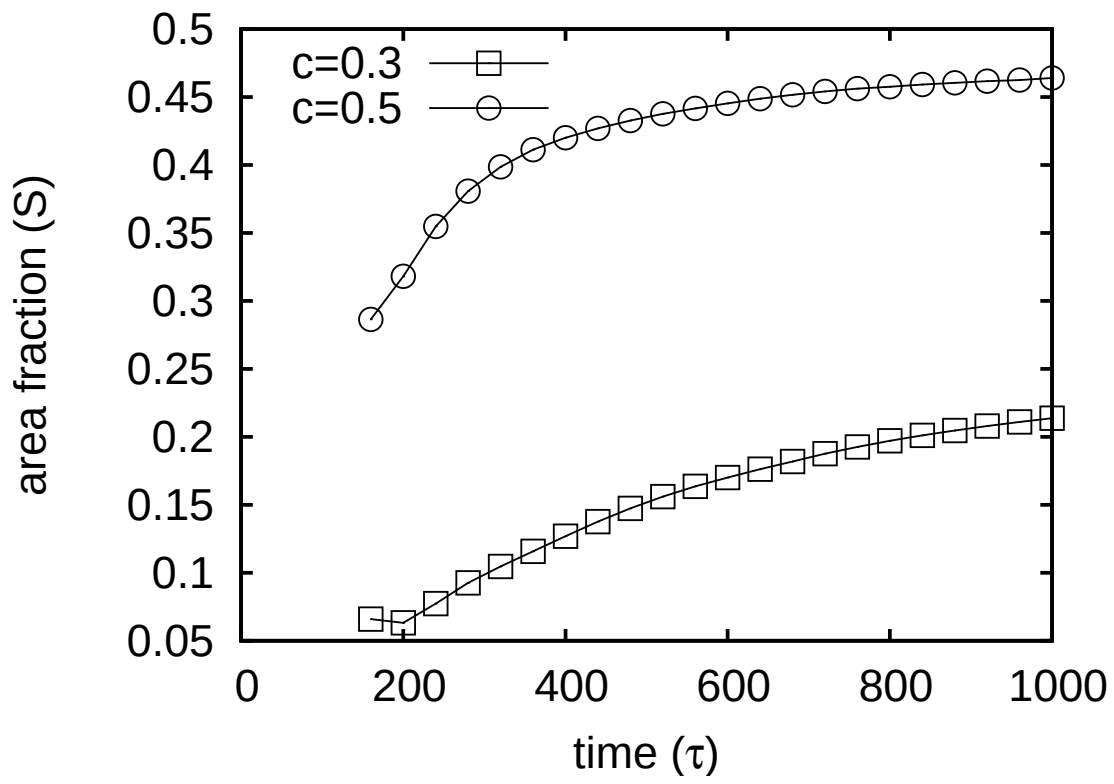


Figure 5.20: Precipitate volume fraction as a function of time for the two alloy compositions ($c_0 = 0.3, 0.5$) in a system with anisotropic misfit ($t = -1$).

5.3 Effect of shear modulus

To examine the effect of elastic energy on morphological evolution, we have carried out a set of simulations with lower shear modulus ($G=500$). We have chosen Cases B, C, D and E for this study. For these cases, the microstructures of an alloy with composition $c_0 = 0.3$ are shown in Figs. 5.21-5.24, wherein the left figure is for the lower shear modulus. It is clear from these figures that reduction in shear modulus results in thicker and smaller domains in all cases.

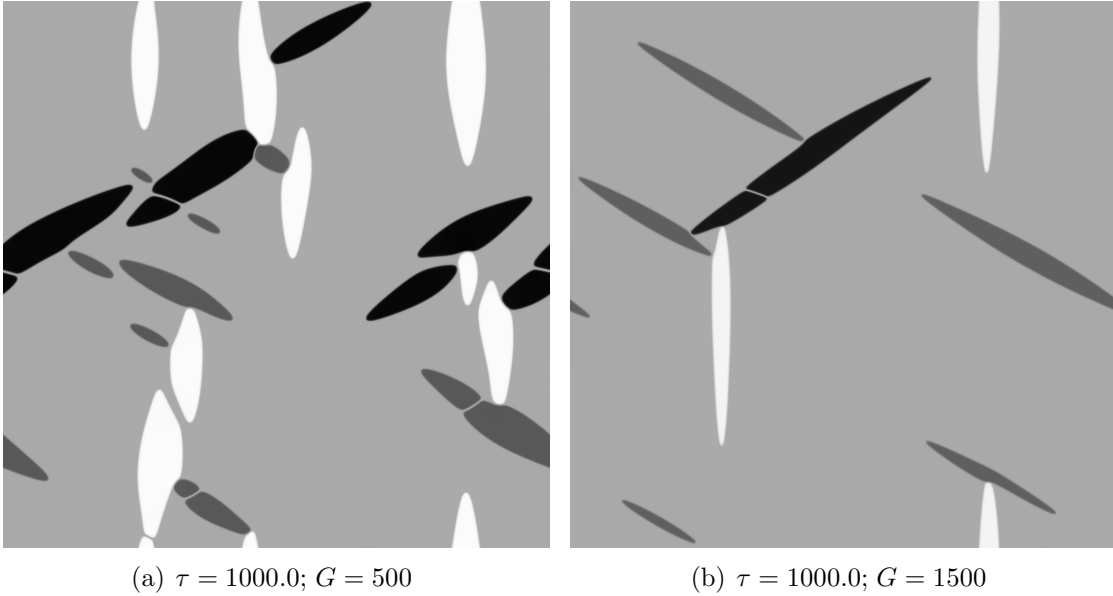


Figure 5.21: Effect of shear modulus G on microstructural evolution in a system with $t = 0.5$. The alloy composition is $c_0 = 0.3$. Microstructures after 1000 time units are shown. The shear modulus, G , is 500 (left) and 1500 (right).

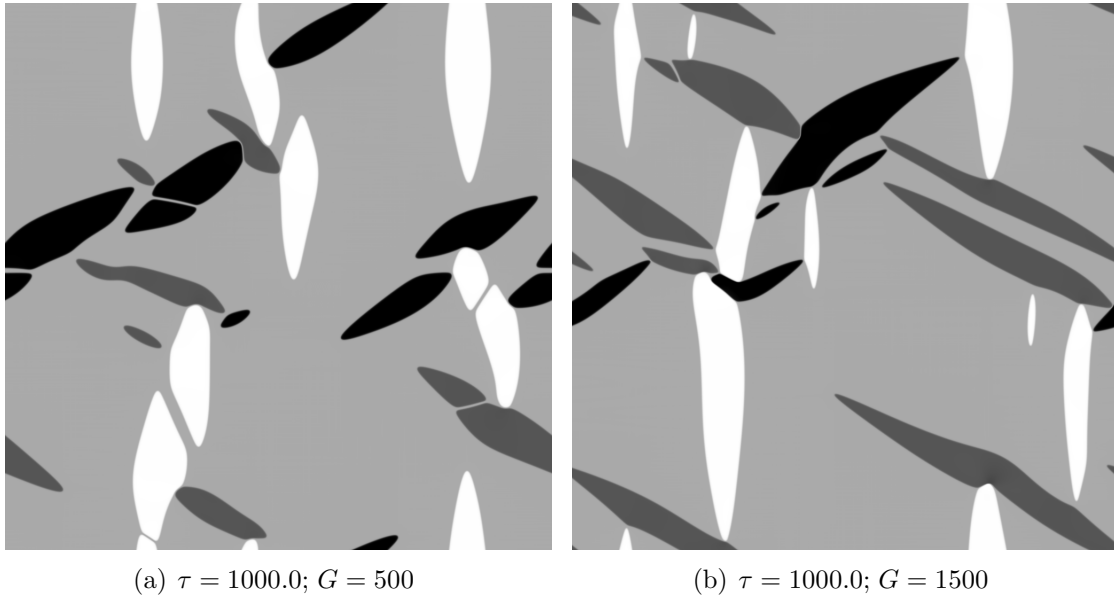


Figure 5.22: Effect of shear modulus G on microstructural evolution in a system with $t = 0$. The alloy composition is $c_0 = 0.3$. Microstructures after 1000 time units are shown. The shear modulus, G , is 500 (left) and 1500 (right).

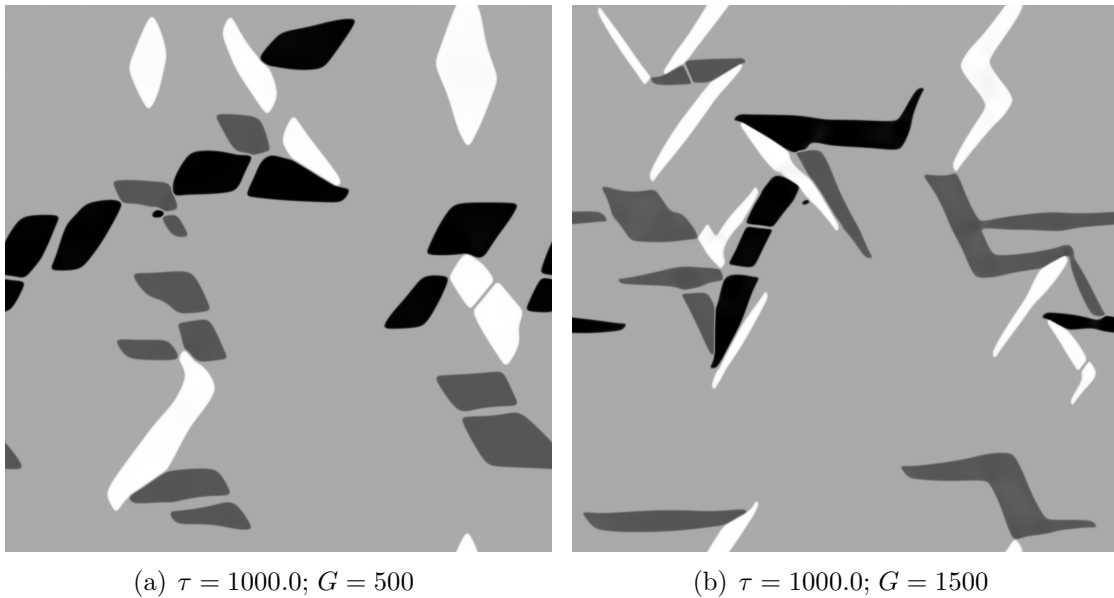


Figure 5.23: Effect of shear modulus G on microstructural evolution in a system with $t = -0.5$. The alloy composition is $c_0 = 0.3$. Microstructures after 1000 time units are shown. The shear modulus, G , is 500 (left) and 1500 (right).

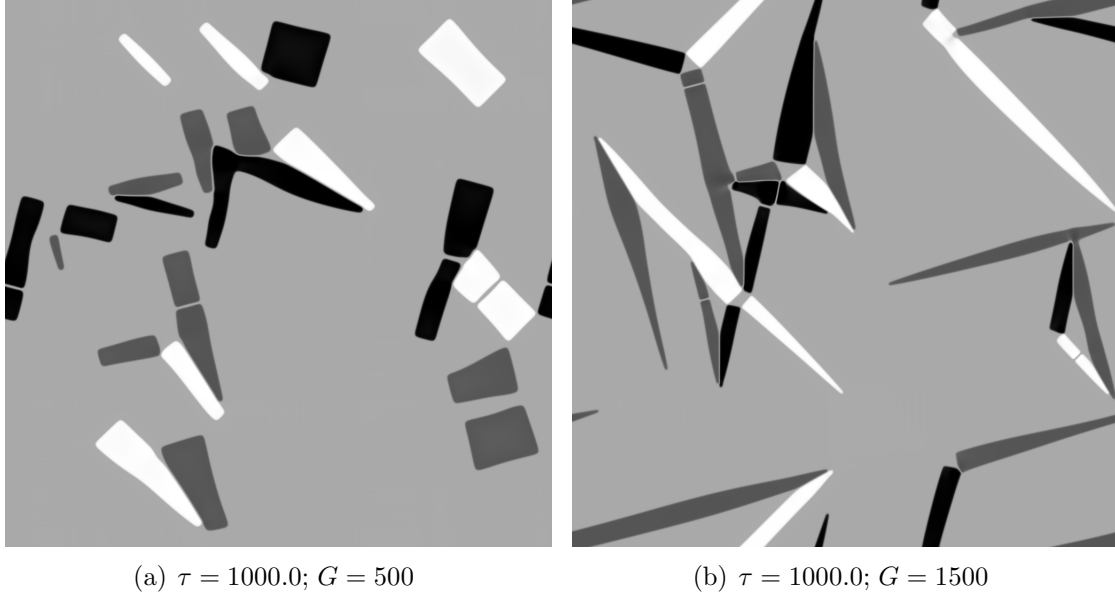


Figure 5.24: Effect of shear modulus G on microstructural evolution in a system with $t = -1$. The alloy composition is $c_0 = 0.3$. Microstructures after 1000 time units are shown. The shear modulus, G , is 500 (left) and 1500 (right).

5.4 Effect of APB energy

Among all the cases that we have studied, Case E is special; since the misfit strain for each variant has only a shear component, and since the shear in the three variants are along directions separated by 120° , the three misfit strains sum to zero. In other words,

$$\sum_{p=1}^3 \epsilon_{ij}^{(p)} = 0. \quad (5.3)$$

So this property may play an important role during nucleation by creating all the three variants in close proximity, thereby reducing the effective misfit. Such patterns that reduce the elastic strain energy are said to be self accommodating.

However, the creation of self accommodating patterns requires the domains in such close proximity that they share an antiphase domain boundary. In the model that we considered so far, the APB energy is nearly twice as much as the interfacial energy, thus making self accommodating patterns energetically nonviable during nucleation.

Therefore, we have studied two more systems that differ only in their APB energies (with the interfacial energy remaining the same). In Table 5.2, we list the values of the gradient energy coefficients and the corresponding interfacial and APB energies for these two systems. The APB energy in system E_2 is more than two times the interfacial

Table 5.2: Gradient energy parameters and corresponding interfacial and APB energies in systems E_2 and E_3

System	κ_c	κ_η	γ_i	γ_a
E_2	0.01	0.7	18.56	38.40
E_3	1.0	0.1	18.56	20.48

energy (thus, this system is similar to that used for Case E in Section 5.2.5); while the APB energy in system E_3 , at about 1.1 times the interfacial energy, is far less than that in system E_2 . In order to accentuate the elastic stress effects, simulations of these two systems used a shear modulus value of 3000 (twice as much as that used in the simulations reported in Sections 5.2 and 5.3).

The microstructures for the early and the late stages, for systems E_2 and E_3 , are presented in Figs. 5.25 and 5.26, respectively.

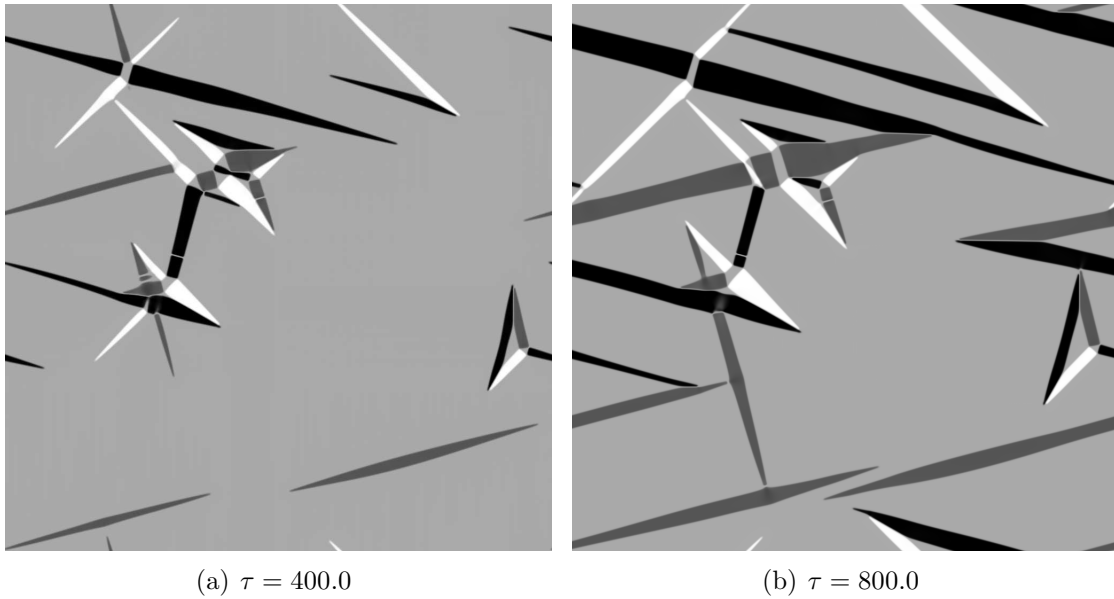


Figure 5.25: Microstructural evolution in system E_2 ($t = -1$). The alloy composition $c = 0.3$. In this system $\gamma_a \approx 2\gamma_i$. The evolution time is indicated below each figure.

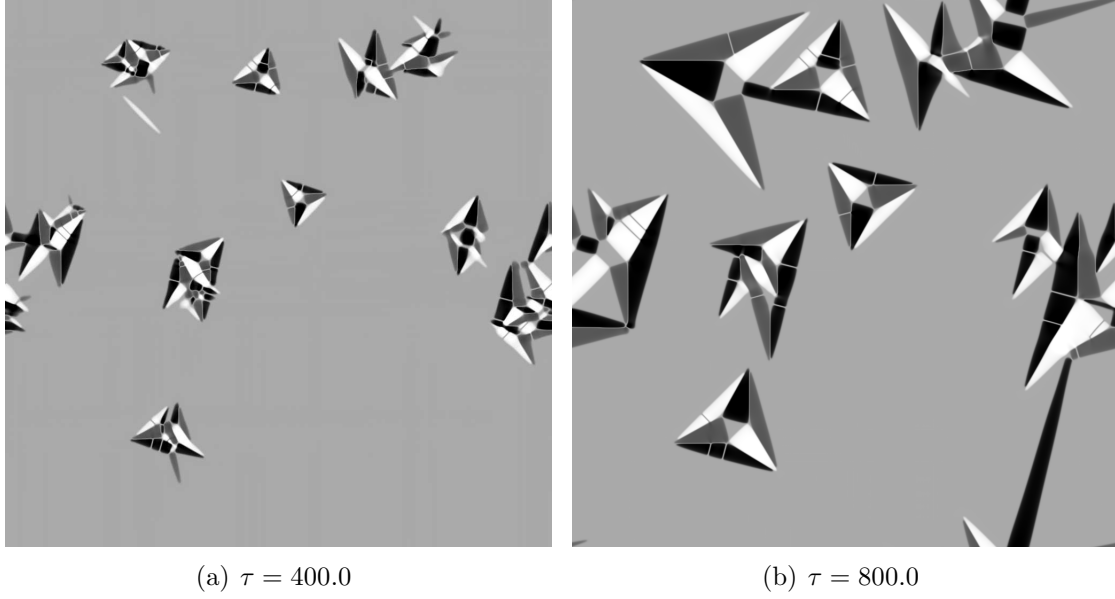


Figure 5.26: Microstructural evolution in system E_3 ($t = -1$). The alloy composition $c = 0.3$. In this system $\gamma_a \approx 1.1\gamma_i$. The evolution time is indicated below each figure.

The microstructures for system E_2 exhibits thin long plates of all three variants oriented along their respective invariant directions. During early stages, many of these plates appear to form in clusters. These features are largely similar to those in Section 5.2 for Case E.

In system E_3 , on the other hand, the microstructures are drastically different. Right from the early stages, almost all the particles appear in compact clusters; within each cluster one finds all the variants packed to produce an external triangular shape (or, combinations of triangles). Clearly, the organization of variants within each triangle is self accommodating, with plate shaped variants sharing their edges (APBs). These compact clusters are long lived; with increasing time the cluster size increases but the pattern inside remains roughly the same. We see such self accommodating clusters in system E_3 primarily because the APB energy is small.

5.5 Discussion

The major conclusion from our study of evolution of multivariant microstructures is that the precipitate morphology depends crucially on the sign of the parameter t .

Shapes and orientations

In Case A ($t = 1$), i.e. for the case of dilatational misfit, the shapes of the domains are circular. For Cases B ($t = 0.5$) and C ($t = 0$), the particles are in the form of lens-shaped plates with their long axis elongated along the directions of lowest absolute misfit. In Cases D ($t = -0.5$) and E ($t = -1$), the particles change their orientation towards the invariant lines. Moreover, in Case E, when the APB energy is lower, misfit strain accommodation between the domains may lead to the formation of compact self accommodating patterns (triangles containing all six variants). The shape and orientation of a representative white variant in each case are shown in Fig. 5.27.

Volume fraction

In Cases A and B, the volume fractions occupied by the precipitates in the two alloys show a different behavior than in Cases C, D, and E. In the alloys with higher concentration, the volume fractions of the precipitates exceeds the alloy composition in the former cases, while in the latter cases, the volume fractions tend to approach the alloy composition, asymptotically. These behaviors can be rationalized using generalized Gibbs-Thomson effect, as described below.

When coherency strains are taken into account, the coherent equilibrium interfacial compositions of the precipitate phase and the matrix phase are schematically shown in Fig. 5.28 [3].

The deviation of composition on the matrix side of a curved interface, Δc^α , from its

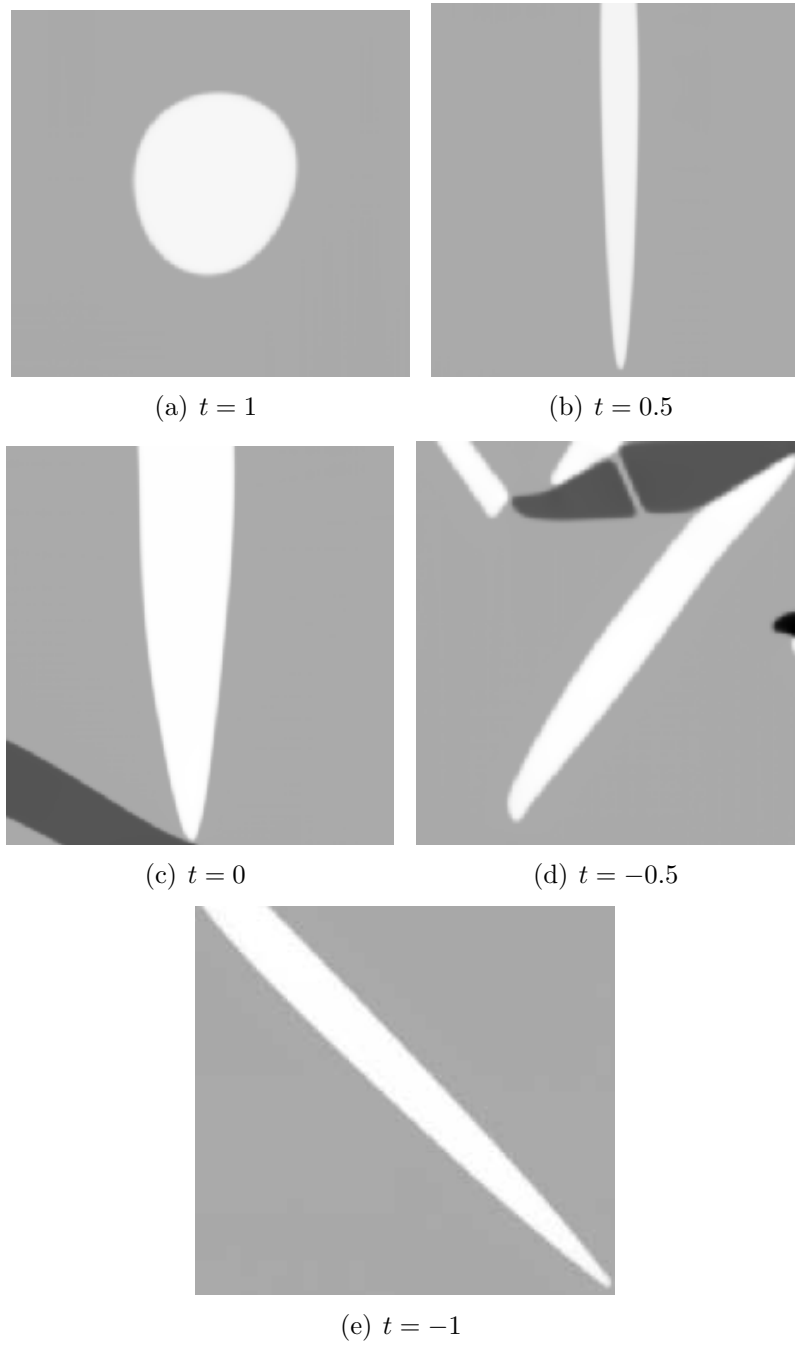


Figure 5.27: Shape and orientation of a white variant in the five systems with different values of misfit anisotropy parameter, t .

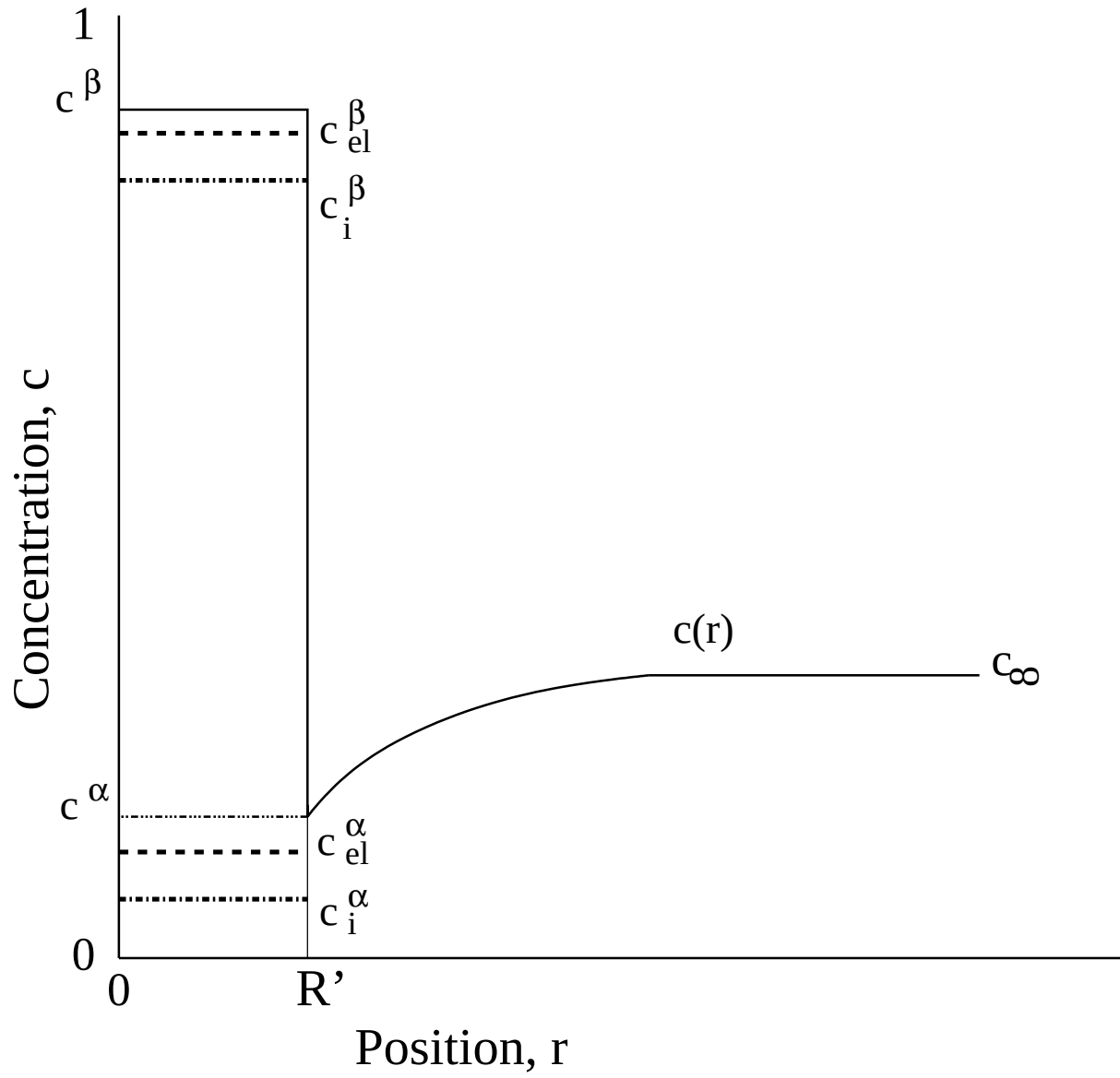


Figure 5.28: Schematic representation of the composition field when two phases are in coherent equilibrium. c^α and c^β are the local equilibrium interfacial concentrations. The coherent equilibrium compositions are denoted c_{el}^α and c_{el}^β . c_i^α and c_i^β are the stress-free equilibrium concentrations (after Laraia *et al.* [3])

equilibrium value (across a planar incoherent interface) is given by

$$\Delta c^\alpha = \Delta c_c^\alpha + \Delta c_{el}^\alpha, \quad (5.4)$$

where Δc_c^α is the contribution due to capillarity, and Δc_{el}^α is the contribution due to elastic stresses.

Δc_c^α is given by the Gibbs-Thomson equation [103]

$$\Delta c_c^\alpha = \frac{\chi^p \gamma}{(c_e^\beta - c_e^\alpha) \Psi^\alpha}, \quad (5.5)$$

where χ^β is the mean curvature of the interface of the precipitate phase, γ is the interfacial energy density, c_e^β and c_e^α are the equilibrium compositions of the matrix and precipitate phases, respectively, when they are separated by a planar interface, and

$$\Psi^\alpha = \left[\frac{\partial^2 f_0}{\partial c^2} \right]_{c_e^\alpha}$$

denotes the curvature of the bulk free energy evaluated at c_e^α .

Δc_{el}^α is given by [103]

$$\Delta c_{el}^\alpha = \frac{\sigma_{ij}^\alpha (\epsilon_{ij}^\alpha - \epsilon_{ij}^\beta) + [\sigma_{ij}^\beta (\epsilon_{ij}^\beta - \epsilon_{ij}^0) - \sigma_{ij}^\alpha \epsilon_{ij}^\alpha] / 2}{(c_e^\beta - c_e^\alpha) \Psi^\alpha}, \quad (5.6)$$

where $\boldsymbol{\sigma}^\alpha$, $\boldsymbol{\epsilon}^\alpha$, $\boldsymbol{\sigma}^\beta$ and $\boldsymbol{\epsilon}^\beta$ are the stress and strain fields on the matrix and precipitate sides of the interface, respectively, and ϵ^0 is the eigenstrain. For each case, $\boldsymbol{\sigma}^\alpha$, $\boldsymbol{\epsilon}^\alpha$, $\boldsymbol{\sigma}^\beta$, and $\boldsymbol{\epsilon}^\beta$ are calculated analytically using Eshelby's sharp interface theory [18, 62].

For Case A, Δc_{el}^α obtained from Eq. 5.6 is 0.193. If we neglect the contribution from capillarity, the composition on the precipitate side is given by $c_i^\beta = 1.145$, and on the matrix side, $c_i^\alpha = 0.193$.

However, in this case, the interfacial precipitate concentration, obtained from the com-

position profiles of an isolated precipitate in both alloys at $\tau = 1000$, is smaller (≈ 1.08) than that predicted by the Gibbs-Thomson effect (see Fig. 5.29). The corresponding value of order parameter, η , in both alloys is also smaller than the equilibrium value of order parameter ($\eta_e(c)$) obtained from Eq. 3.12 for the observed precipitate concentration. Hence, we may infer that the system prefers to reduce the elastic energy by lowering the value of η and c_i^β from the predicted equilibrium values, although these values correspond to an increase in the bulk chemical energy. Since the effective precipitate composition is lower, this leads to a larger volume fraction in both alloys than that predicted by the Gibbs-Thomson effect. Also we note that the reduction in precipitate concentration is more pronounced for the alloy with higher concentration which results in a volume fraction larger than the solute concentration.

Similar arguments apply to Case B, for which the one dimensional composition and order parameter profiles drawn through a relatively isolated particle for both alloys are shown in Fig. 5.30. The interfacial concentrations on the ‘precipitate’ side and the ‘matrix’ side of the particle in both alloys are lower than the equilibrium concentrations predicted using generalized Gibbs-Thomson effect (Eq. 5.6). Similar to the previous case, the reduction in precipitate composition is larger in the alloy with higher concentration.

Hence in both Cases A and B, a relatively large reduction in precipitate composition leads to a higher volume fraction of precipitates in the more concentrated alloy with $c_0 = 0.5$.

What about the other cases? Take Case C, for example. Fig. 5.31 shows the variation of composition and order parameter fields across a precipitate chosen from the microstructure of each alloy at $\tau = 1000$. The precipitate compositions as well as the corresponding order parameters in both alloys are closer to equilibrium (predicted from generalized Gibbs-Thomson effect). As a result, the volume fractions of the particles observed in this case are also closer to the predicted values. A similar trend is observed in Cases D and E.

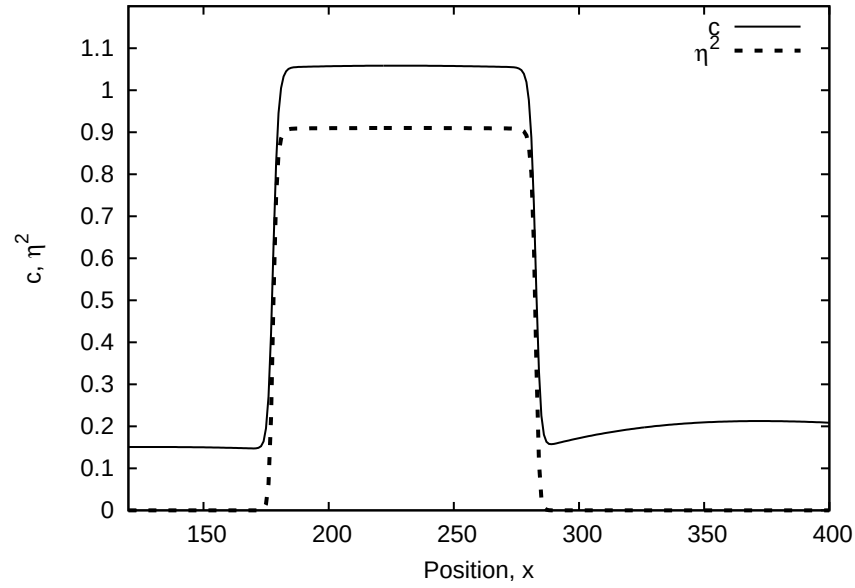
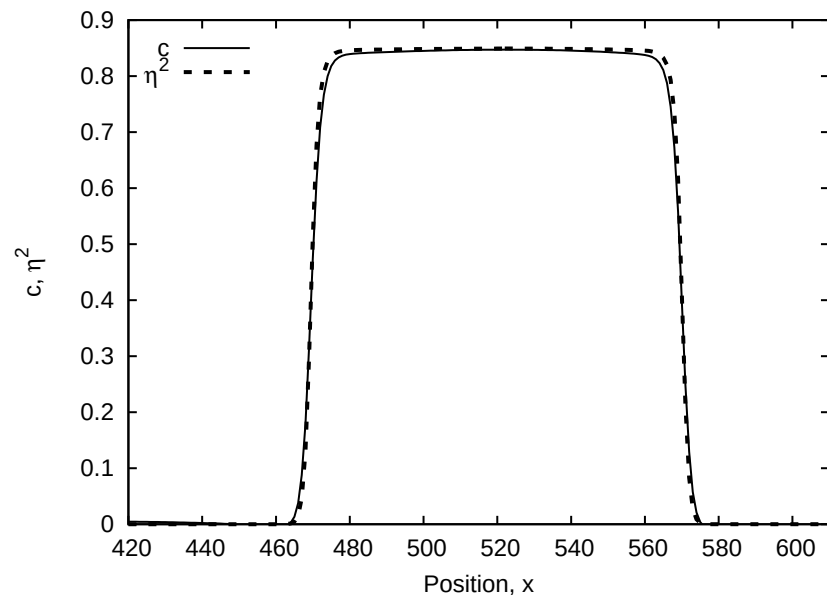
(a) $c_0 = 0.3$ (b) $c_0 = 0.5$

Figure 5.29: Composition and order parameter profiles (one dimensional section) for a precipitate in two alloys with concentrations (a) $c = 0.3$ and (b) $c = 0.5$ at $\tau = 1000$ in Case A. Note that the precipitate composition in (b) is lower than its bulk equilibrium value of 1.0.

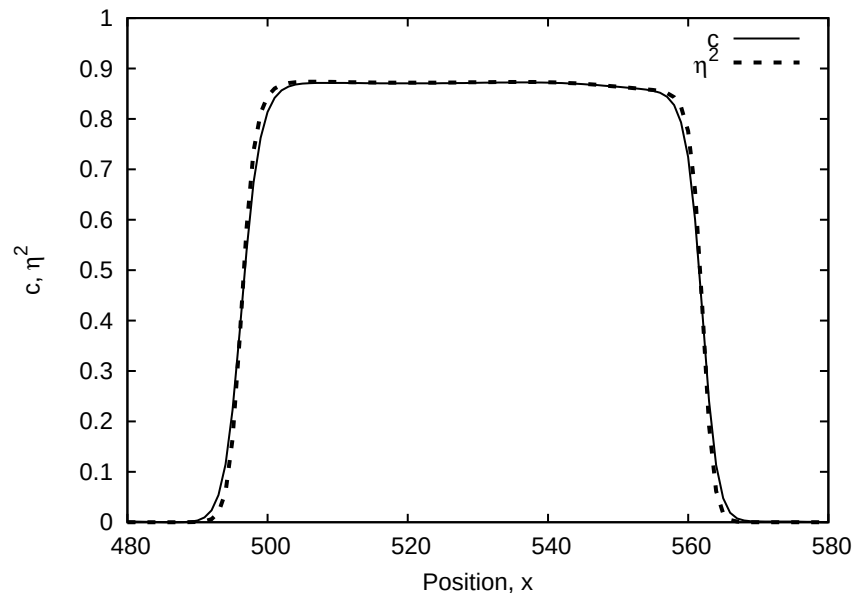
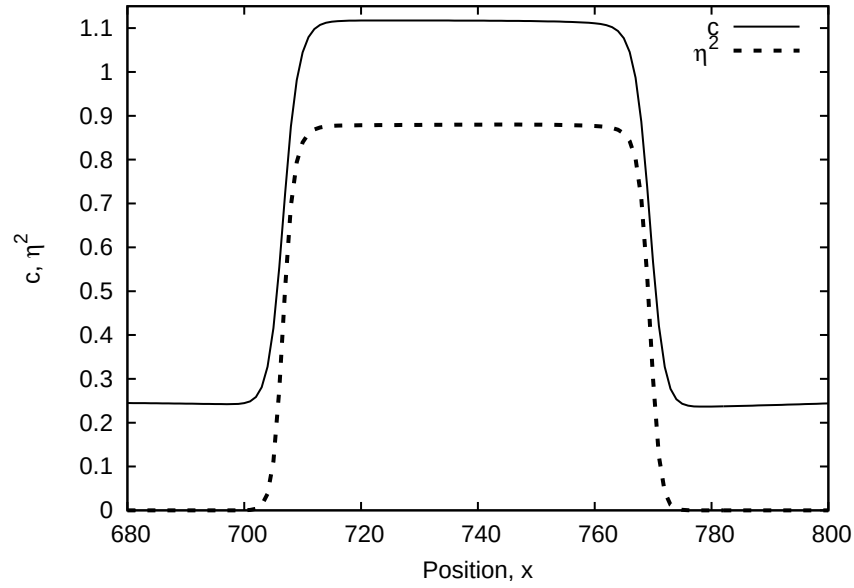


Figure 5.30: Composition and order parameter profiles (one dimensional section) for a precipitate in two alloys with concentrations (a) $c = 0.3$ and (b) $c = 0.5$ at $\tau = 1000$ in Case B. Note that the precipitate composition in (b) is lower than its bulk equilibrium value of 1.0.

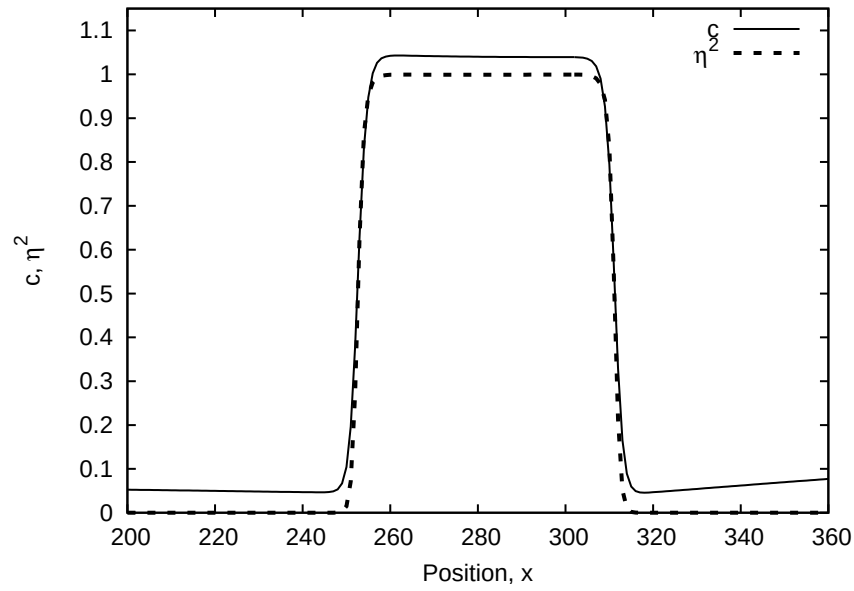
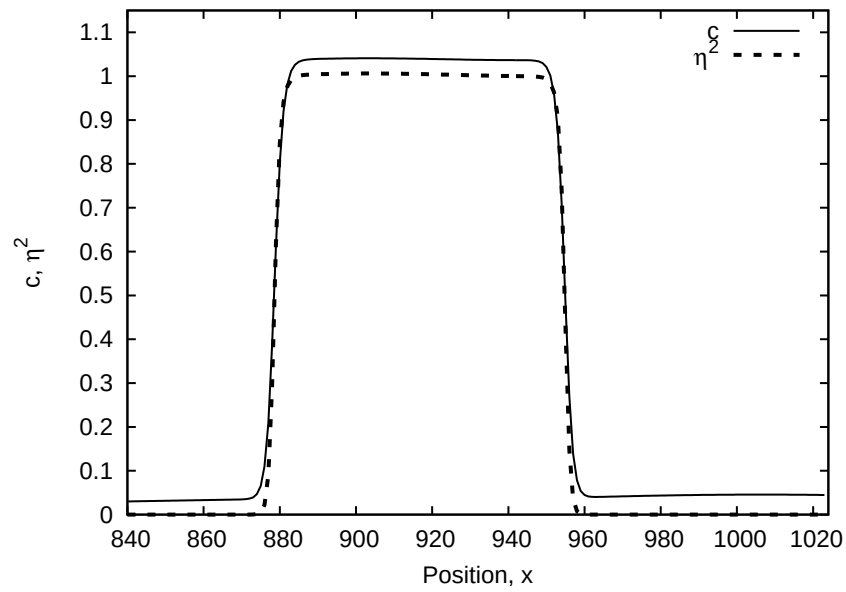
(a) $c_0 = 0.3$ (b) $c_0 = 0.5$

Figure 5.31: Composition and order parameter profiles (one dimensional section) for a precipitate in two alloys with concentrations (a) $c = 0.3$ and (b) $c = 0.5$ at $\tau = 1000$. ($t = 0$) Note that in both (a) and (b), the precipitate composition remains above 1.0.

Case E: special features

For each case, the anisotropic misfit strain ϵ_{ij}^0 associated with each variant can be resolved as a sum of hydrostatic and shear strains:

$$\epsilon_\eta \begin{pmatrix} 1 & 0 \\ 0 & t \end{pmatrix} = \epsilon_\eta \left\{ \epsilon_h \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \epsilon_s \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\}, \quad (5.7)$$

where ϵ_h and ϵ_s are the coefficients of the hydrostatic and shear components of misfit strain, respectively. The values of the coefficients, ϵ_h and ϵ_s , are listed in Table 5.3 for different values of t . Among all cases, the hydrostatic part of the misfit strain is zero

Table 5.3: Coefficients of the hydrostatic and shear components of misfit strain for different values of t ($\epsilon_\eta = 0.03$)

	t	ϵ_h	ϵ_s
Case A	1	1.0	0.0
Case B	0.5	0.75	0.25
Case C	0	0.5	0.5
Case D	-0.5	0.25	0.75
Case E	-1	0.0	1.0

in Case E. This may lead to self accommodating patterns, since the sum of the misfits associated with each variant is zero in this case. However, such patterns are possible only when antiphase domain boundaries have a low energy. Our results, in Section 5.4, show that this is indeed the case. A reduction in APB energy increases the number of edge-sharing domains which organize themselves into compact self accommodating patterns.

Retransformation of precipitate phase to matrix phase

Interactions between variants may lead to retransformation of precipitate phase into matrix phase at the region of overlap, as shown in Fig. 5.17. Retransformation may be caused when the local strain is unfavourable to the variant that dissolves. For example, in the figure, the principal form of local strain measured in the central region is $\begin{pmatrix} -0.0026 & 0 \\ 0 & 0.0013 \end{pmatrix}$; this strain does not favor the previously existing white variant whose misfit strain is $\begin{pmatrix} 0.03 & 0 \\ 0 & -0.03 \end{pmatrix}$. Thus, the white variant under such an unfavorable state of strain dissolves to yield a matrix phase.

Lateral movement of variants

Another interesting observation in Case E is a significant lateral movement of the variants shown in Fig. 5.19. The lateral migration may be caused by two mechanisms: (a) dissolution of the variant on one side and growth on the other side, and (b) solute transfer through the thickness of the particle, allowing one interface to advance and the other to recede.

The composition profiles of the variant at different times, shown in Fig. 5.32, suggests the second mechanism to be operational here.

Comparison with experiments

The microstructures reported in the experimental studies of Muraleedharan *et al.* [2] and Wen *et al.* [13] in TiAlNb alloy system have a close resemblance with our simulated microstructures for Case E. Several morphological features reported by them are also observed in the simulated microstructures for Case E. We compare some of these features

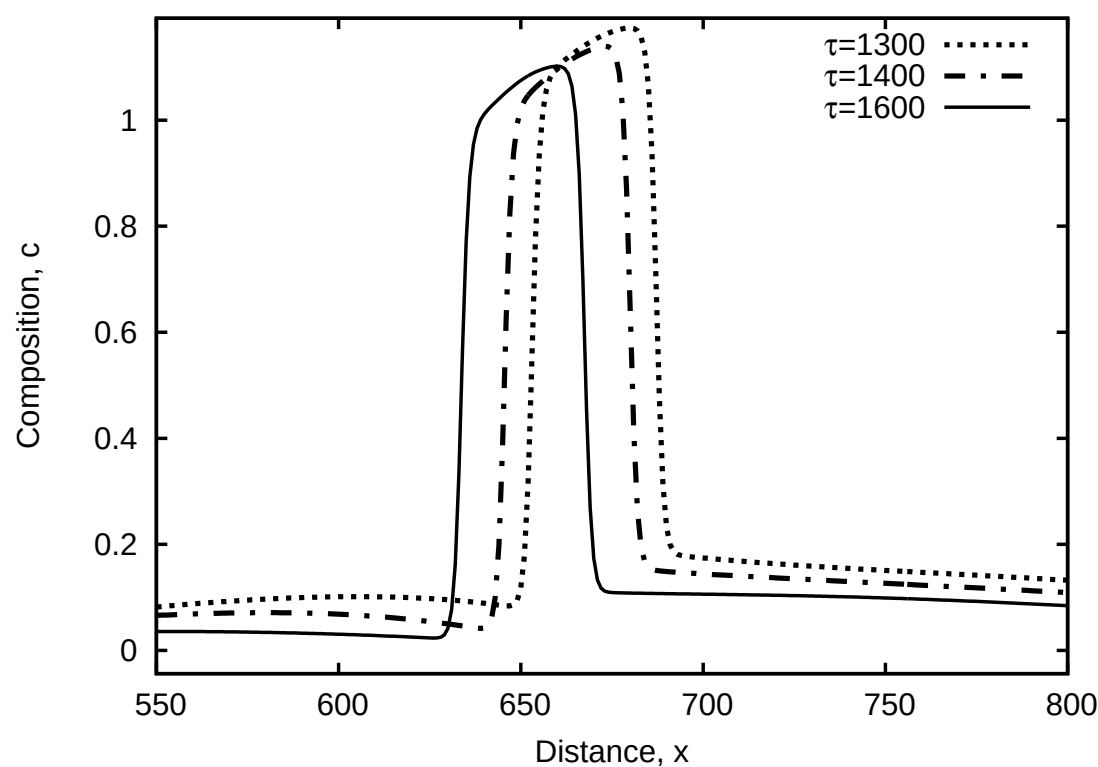


Figure 5.32: Composition profiles of a laterally migrating variant at different times. Composition is plotted against distance normal to the broad face of the black variant in Fig. 5.19.

in Fig. 5.33.

In Fig. 5.33a, our simulations show the three orientation variants meeting in the form of a three-sided star; the central region is occupied by the matrix phase. Such a star shaped three plate configuration with the central region being occupied by the matrix phase is also reported in the experiments (Fig. 5.33b). Muraleedharan *et al.* also reported that interaction between two variants can result in a retransformation of the precipitate phase into matrix phase at the intersection point (see Fig. 5.33d). Our simulations also show such a retransformation event giving rise to ‘zig-zag’ patterns (Fig. 5.33c). In the experiments, two types of wedge patterns have been reported: one in which growth of either variant, forming the arms of the wedge, stop at the point of intersection; another type of wedge is also possible where one of the intersecting variants continue to grow (see Fig. 5.33f). Our simulations show the occurrence of both types of wedges in Fig. 5.33e.

Secondary nucleation

Muraleedharan *et al.* have also reported what appears to be an occurrence of a fresh nucleation event when several variants intersect (see Fig 5.34 for detail). It is not clear whether the plate nucleates at the point of intersection or a preexisting plate dissolves at that point. We have not encountered such secondary nucleation events in our “three-variant” simulations. However, “two-variant” simulations in Case E with the addition of small sustained noise (to mimic the unsaturated fluctuations in the system) leads to a spontaneous nucleation of the third variant (see Fig. 5.35). At $\tau = 500$ and 540, the microstructures contain the white and gray variants elongating along respective invariant directions. Now, let us focus our attention to the right hand side of the snapshots. At $\tau = 540$, a gray variant hits a white variant followed by dissolution of the precipitate phase at the point of impingement. At $\tau = 600$, the narrower arm of the white variant dissolves and a black variant nucleates at the point of intersection. At later times, the black variant

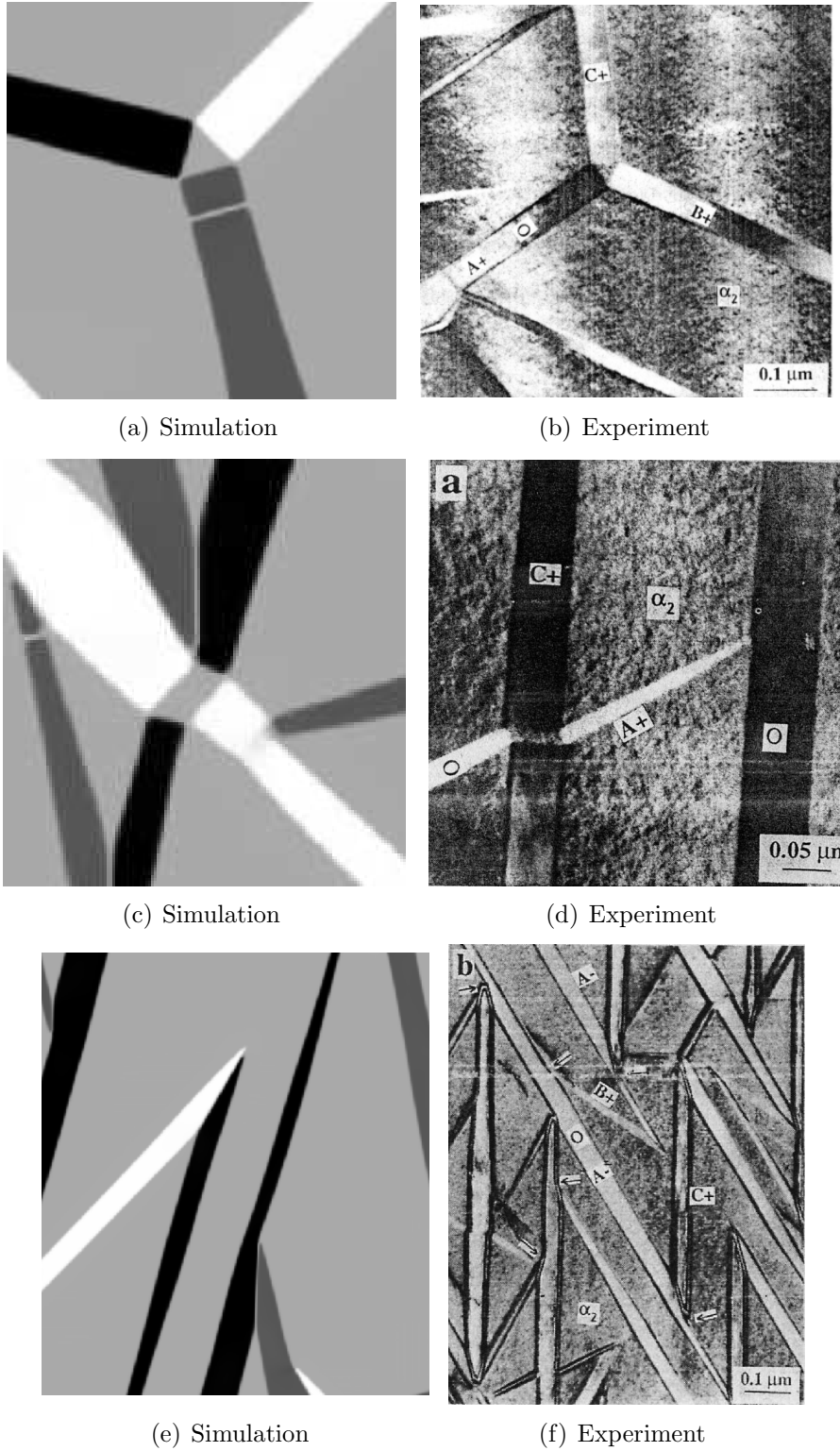


Figure 5.33: Comparison of the simulated morphological features with those obtained from experiments. The left column shows the results obtained from our simulations, the right column shows the experimental features from the work of Muraleedharan *et al.* [2].

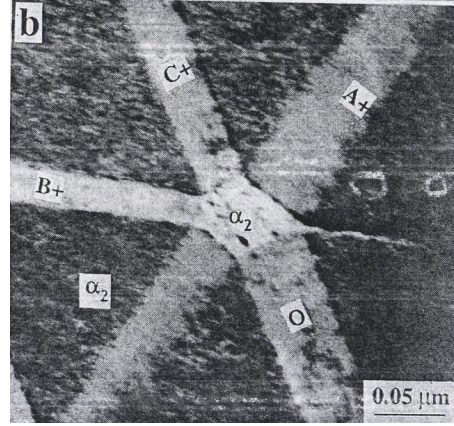


Figure 5.34: Fresh nucleation of additional plate at the point of intersection from the experimental work of Muraleedharan *et al.* [2].

grows and forms a spear-like pattern by association with the other variants. Such spear patterns formed by fresh nucleation events have also been reported by Muraleedharan *et al.* Another nucleation event occurs at $\tau = 1500$, as shown in the top left region of the microstructure in Fig. 5.35d.

Comparison with simulations

Wen *et al.* [13, 15] have used phase field simulations to study microstructural evolution during coherent hexagonal to orthorhombic transformation in the contexts of pure ordering and precipitation. However, they have considered only the case of pure shear misfit, which is Case E in our study. For this case, the morphological patterns reported by us are similar to those observed in their simulations (e.g. stars, triangles, wedges, checkerboard).

Although the APB energy and interfacial energy are not explicitly mentioned in their study, the microstructural features reported in their simulations more closely approximates those obtained for system E₃ in our study. Further, in all cases we track the change in volume fraction of the precipitate phase with time. Since their study does not report the evolution of volume fraction with time, a comparison of the same is not possible.

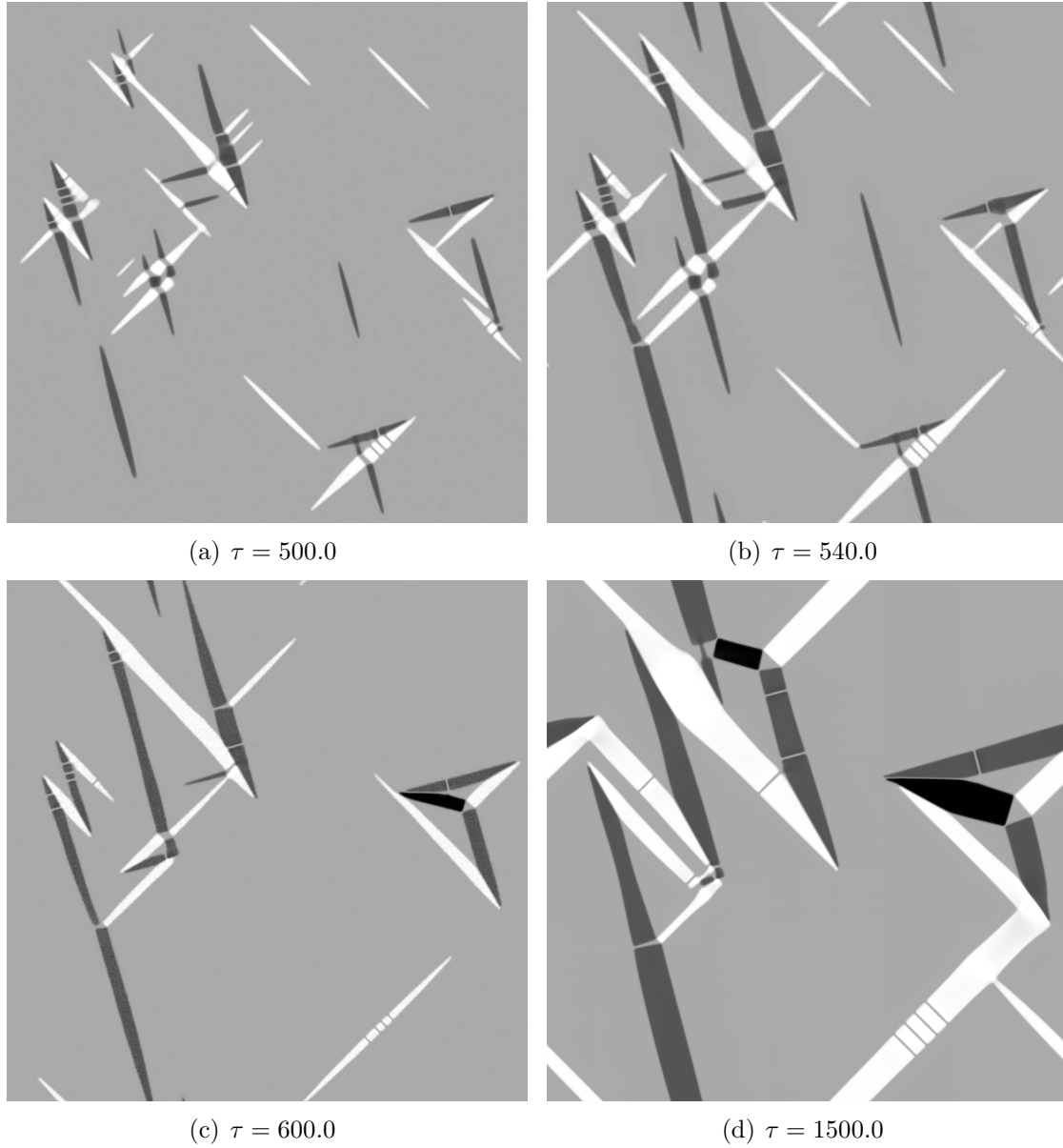


Figure 5.35: Spontaneous nucleation of third variant in a simulation in which only two variants were allowed to nucleate and grow. Alloy composition $c_0 = 0.3$. $G = 3000$; $t = -1$. The evolution time is indicated below each figure.

5.6 Summary

We have catalogued and rationalized the different types of microstructures arising in systems in which the precipitate phase has multiple variants and anisotropic misfit. The morphology of the precipitates is shown to be influenced by the level of anisotropy in misfit. In Case A, the variants are circular. In Cases B and C, the variants elongate along the directions of lowest absolute misfit. In Cases D and E ($t < 0$), the variants change their orientation along their respective invariant directions. Moreover, in Case E, where the misfit strains of the variants add up to zero, interaction between the variants lead to formation of various microstructural patterns such as stars, wedges, triangles and checkerboard. Further, when the APB energy is sufficiently low in this case, the variants appear in compact, self-accommodating clusters that retain their overall morphology well into the coarsening regime.

Chapter 6

Conclusions

We have used a phase field model to study the effect of anisotropic misfit on dynamical growth shapes and multivariant microstructures in two dimensions. The anisotropy in misfit is defined using a parameter $t = \epsilon_{yy}/\epsilon_{xx}$ where ϵ_{xx} and ϵ_{yy} are the principal components of the misfit strain tensor. The parameter t is systematically varied from +1 (pure dilatation) to -1 (pure shear). We report the results for microstructural evolution in systems with different values of t , representing different types of anisotropy in misfit:

- Case A: $t = 1$ (dilatational or isotropic misfit)
- Case B: $t = 0.5$ (principal misfit component along the y direction is less than that along the x direction, though they have the same sign)
- Case C: $t = 0$ (the principal misfit along the y direction is zero)
- Case D: $t = -0.5$ (principal misfit component along the y direction is less than that along the x direction, and they have the opposite sign))
- Case E: $t = -1$ (principal misfit components are equal, but with opposite signs; pure shear)

The key conclusions from our study of morphological evolution of isolated precipitate (single variant) and many interacting precipitates (multivariant) are listed below.

Dynamical growth shapes

- In the study of growth shapes of isolated precipitates, for each case, the shapes of the precipitate at different sizes are compared with the corresponding equilibrium shapes. For Case A, the precipitate remains circular at all sizes. For Cases B and C, the precipitate shape is elongated along the y direction (i.e. the direction of lower absolute misfit). In these cases, the symmetry of the precipitate shape remains unchanged at all times.
- In Cases D and E, where the misfit components have opposite sign, the symmetry of the precipitate shape changes during growth from one with higher symmetry to that with a lower symmetry. For Case D, the particle shape undergoes orthorhombic $\rightarrow$ monoclinic symmetry-breaking transition, while for Case E, it undergoes tetragonal $\rightarrow$ orthorhombic symmetry-breaking transition beyond a critical size.
- In both cases, the critical precipitate size at which the shape of the growing precipitate breaks its symmetry is larger than that for the equilibrium shapes. This critical size is larger when matrix supersaturation is higher and shear modulus is lower.

Multivariant microstructures

- In each case, the morphology of the precipitates depends crucially on the level of anisotropy in misfit. In Case A, the microstructure contains circular domains of the precipitate phase. In Cases B and C, the three orientational domains elongate

along the directions of lowest absolute misfit. In Cases D and E, each orientational domain is associated with an invariant line. Hence, in both cases, the domains re-orient themselves along the invariant line.

- In Case E, interaction between the domains result in the formation of complex microstructural patterns such as stars, wedges, triangles and checkerboard. The formation of these patterns often lead to re-transformation of matrix phase in the overlapping regions of the domains.
- We have examined the effect of antiphase boundary energy in Case E. In systems with high APB energy, isolated domains as well as clusters of domains appear during nucleation. With further growth, the domains interact to form a (often three sided) star pattern, observed experimentally in Ti-Al-Nb alloys. When the APB energy is sufficiently low, the variants appear in compact, self-accommodating clusters that retain their overall morphology well into the coarsening regime.

Appendix A

Derivation of elastic energy $\mathcal{F}_e$

To obtain the expression for $\mathcal{F}_e$ defined in Eq. 3.26 in reciprocal space, we start with Eq. 3.14

$$\mathcal{F}_e = \frac{1}{2} \int_V \sigma_{ij}^{el}(\mathbf{r}) \epsilon_{ij}^{el}(\mathbf{r}) dV, \quad (\text{A.1})$$

Using Eqns. 3.15, 3.19 and 3.16 in Eq. A.1, we obtain

$$\mathcal{F}_e = \frac{1}{2} \int_V C_{ijkl} (\varepsilon_{ij}(\mathbf{r}) + E_{ij} - \epsilon_{ij}^0(\mathbf{r})) (\varepsilon_{kl}(\mathbf{r}) + E_{kl} - \epsilon_{kl}^0(\mathbf{r})) dV. \quad (\text{A.2})$$

Since C_{ijkl} is symmetric C_{ijkl} ($C_{ijkl} = C_{klij}$), the above equation may be expressed as a summation of six integrals:

$$\begin{aligned} \mathcal{F}_e = & \frac{1}{2} \int_V C_{ijkl} \varepsilon_{ij}(\mathbf{r}) \varepsilon_{kl}(\mathbf{r}) dV + \int_V C_{ijkl} \varepsilon_{ij}(\mathbf{r}) E_{kl} dV \\ & - \int_V C_{ijkl} \varepsilon_{ij}(\mathbf{r}) \epsilon_{kl}^0 dV - \int_V C_{ijkl} E_{ij} \epsilon_{kl}^0(\mathbf{r}) dV \\ & + \frac{1}{2} \int_V C_{ijkl} E_{ij} E_{kl} dV + \frac{1}{2} \int_V C_{ijkl} \epsilon_{ij}^0(\mathbf{r}) \epsilon_{kl}^0(\mathbf{r}) dV \end{aligned} \quad (\text{A.3})$$

We denote these integrals as $I_1, I_2, \dots, I_6$:

$$\mathcal{F}_e = I_1 + I_2 + I_3 + I_4 + I_5 + I_6. \quad (\text{A.4})$$

We evaluate the integrals separately and sum them up to obtain $\mathcal{F}_e$.

Using Parseval's theorem ($\int_{-\infty}^{\infty} f(x)g(x)dx = \int_{-\infty}^{\infty} \tilde{f}(k)\tilde{g}^*(k)dk$), I_1 can be expressed in Fourier space as follows:

$$\begin{aligned} I_1 &= \frac{1}{2} \int_V C_{ijkl} \varepsilon_{ij}(\mathbf{r}) \varepsilon_{kl}(\mathbf{r}) dV \\ &= \frac{1}{2} \int_V C_{ijkl} \tilde{\varepsilon}_{ij}(\mathbf{k}) \tilde{\varepsilon}_{kl}^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \end{aligned} \quad (\text{A.5})$$

Using Eq. 3.17 in the above equation, we get

$$\begin{aligned} I_1 &= \frac{1}{2} \int_V C_{ijkl} \tilde{u}_{i,j}(\mathbf{k}) \tilde{u}_{k,l}^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \quad (\because C_{ijkl} = C_{jikl}) \\ &= \frac{1}{2} \int_V C_{ijkl} [n_j n_n \Omega_{mi}(\mathbf{n}) \sum_p \hat{\sigma}_{mn}^{(p)} \Delta\theta_p(\mathbf{k})] [n_l n_s \Omega_{rk}(\mathbf{n}) \sum_q \hat{\sigma}_{rs}^{(q)} \Delta\theta_q^*(\mathbf{k})] \frac{d^3 \mathbf{k}}{(2\pi)^3} \\ &\quad (\because \tilde{u}_{k,l}(\mathbf{k}) = J|\mathbf{k}| n_l \tilde{u}_k(\mathbf{k}) = n_j n_l \Omega_{ik}(\mathbf{n}) \sum_p \hat{\sigma}_{ij}^{(p)} \Delta\theta_p(\mathbf{k})) \\ &= \frac{1}{2} \int_V [C_{ijkl} n_j n_l] \Omega_{mi}(\mathbf{n}) \Omega_{rk}(\mathbf{n}) n_n n_s \sum_{p,q} \hat{\sigma}_{mn}^{(p)} \hat{\sigma}_{rs}^{(q)} \Delta\theta_p(\mathbf{k}) \Delta\theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \\ &= \frac{1}{2} \int_V \Omega_{ik}^{-1}(\mathbf{n}) \Omega_{mi}(\mathbf{n}) \Omega_{rk}(\mathbf{n}) n_n n_s \sum_{p,q} \hat{\sigma}_{mn}^{(p)} \hat{\sigma}_{rs}^{(q)} \Delta\theta_p(\mathbf{k}) \Delta\theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \\ &= \frac{1}{2} \int_V \delta_{km} \Omega_{rk}(\mathbf{n}) n_n n_s \sum_{p,q} \hat{\sigma}_{mn}^{(p)} \hat{\sigma}_{rs}^{(q)} \Delta\theta_p(\mathbf{k}) \Delta\theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \quad (\because \Omega_{ik}^{-1} \Omega_{mi} = \delta_{km}) \\ &= \frac{1}{2} \sum_{p,q} \int_V n_n \hat{\sigma}_{kn}^{(p)} \Omega_{rk}(\mathbf{n}) \hat{\sigma}_{rs}^{(q)} n_s \Delta\theta_p(\mathbf{k}) \Delta\theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \quad (\because \delta_{km} \hat{\sigma}_{mn}^{(p)} = \hat{\sigma}_{kn}^{(p)}) \\ &= \frac{1}{2} \sum_{p,q} \int_V n_j \hat{\sigma}_{ij}^{(p)} \Omega_{ki}(\mathbf{n}) \hat{\sigma}_{kl}^{(q)} n_l \Delta\theta_p(\mathbf{k}) \Delta\theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3}. \end{aligned}$$

(Changing dummy indices $n \rightarrow j, k \rightarrow i, r \rightarrow k, s \rightarrow l$) (A.6)

Since $\int_V \varepsilon_{ij}(\mathbf{r}) dV = 0$, I_2 is evaluated as:

$$\begin{aligned} I_2 &= \frac{1}{2} \int_V \varepsilon_{ij} E_{kl} dV \\ &= 0. \end{aligned} \tag{A.7}$$

Using the definition $\epsilon_{kl}^0(\mathbf{r}) dV = \sum_p \epsilon_{kl}^{(p)} \theta_p(\mathbf{r}) dV$ and Parseval's theorem, I_3 is obtained as follows:

$$\begin{aligned} I_3 &= - \int_V C_{ijkl} \varepsilon_{ij}(\mathbf{r}) \epsilon_{kl}^0(\mathbf{r}) dV \\ &= - \sum_q \int_V C_{ijkl} \varepsilon_{ij}(\mathbf{r}) \epsilon_{kl}^{(q)} \theta_q(\mathbf{r}) dV \\ &= - \sum_q \int_V C_{ijkl} \varepsilon_{ij}(\mathbf{r}) \epsilon_{kl}^{(q)} (\Delta \theta_q(\mathbf{r}) + \bar{\theta}_q) dV \quad (\because \Delta \theta_q(\mathbf{r}) = \theta_q(\mathbf{r}) + \bar{\theta}_q) \\ &= - \sum_q \int_V C_{ijkl} \epsilon_{kl}^{(q)} \varepsilon_{ij}(\mathbf{r}) \Delta \theta_q(\mathbf{r}) dV + \int_V C_{ijkl} \epsilon_{kl}^{(q)} \varepsilon_{ij}(\mathbf{r}) w_q dV \\ &= - \sum_q \int_V C_{ijkl} \epsilon_{kl}^{(q)} \varepsilon_{ij}(\mathbf{r}) \Delta \theta_q(\mathbf{r}) dV \quad (\because \int_V \varepsilon_{ij}(\mathbf{r}) dV = 0) \\ &= - \sum_q \int_V C_{ijkl} \epsilon_{kl}^{(q)} \varepsilon_{ij}(\mathbf{k}) \Delta \theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \\ &= - \sum_q \int_V C_{ijkl} \epsilon_{kl}^{(q)} \tilde{u}_{i,j}(\mathbf{k}) \Delta \theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \\ &= - \sum_{p,q} \int_V C_{ijkl} \epsilon_{kl}^{(q)} [n_j \hat{\sigma}_{mn}^{(p)} n_n \Omega_{mi}(\mathbf{n}) \Delta \theta_p(\mathbf{k})] \Delta \theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \\ &= - \sum_{p,q} \int_V \hat{\sigma}_{ij}^{(q)} n_j \sigma_{mn}^{(p)} n_n \Omega_{mi}(\mathbf{n}) \Delta \theta_p(\mathbf{k}) \Delta \theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \\ &= - \sum_{p,q} \int_V n_j \hat{\sigma}_{ij}^{(q)} \Omega_{ki}(\mathbf{n}) \sigma_{kl}^{(p)} n_l \Delta \theta_p(\mathbf{k}) \Delta \theta_q^*(\mathbf{k}) \frac{d^3 \mathbf{k}}{(2\pi)^3} \end{aligned} \tag{A.8}$$

I_4 is obtained using the definitions of homogeneous strain (see Eq. 3.18) and total eigen-

strain (see Eq. 3.5):

$$\begin{aligned}
I_4 &= - \int_V C_{ijkl} E_{ij} \epsilon_{kl}^0(\mathbf{r}) dV \\
&= - \int_V C_{ijkl} \sum_p \epsilon_{kl}^{(p)} \theta_p(\mathbf{r}) E_{ij} dV \\
&= - \int_V \sum_p \hat{\sigma}_{ij}^{(p)} \theta_p(\mathbf{r}) E_{ij} dV \\
&= - \sum_{p,q} \hat{\sigma}_{ij}^{(p)} V \bar{\theta}_p \epsilon_{ij}^{(q)} \bar{\theta}_q \quad (\because \int_V \theta_p(\mathbf{r}) dV = V \bar{\theta}_p) \\
&= -V \sum_{p,q} C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} \bar{\theta}_p \bar{\theta}_q
\end{aligned} \tag{A.9}$$

I_5 is expressed as follows:

$$\begin{aligned}
I_5 &= \frac{1}{2} \int_V C_{ijkl} E_{ij} E_{kl} dV \\
&= \frac{1}{2} \int_V C_{ijkl} \sum_p \epsilon_{ij}^{(p)} \bar{\theta}_p \sum_q \epsilon_{kl}^{(q)} \bar{\theta}_q dV \\
&= \frac{V}{2} \sum_{p,q} C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} \bar{\theta}_p \bar{\theta}_q
\end{aligned} \tag{A.10}$$

I_6 is obtained using the definition of eigenstrain and substituting $\theta_p(\mathbf{r}) = \Delta\theta_p(\mathbf{r}) + \bar{\theta}_p$ in Eq. 3.5:

$$\begin{aligned}
I_6 &= \frac{1}{2} \int_V C_{ijkl} \epsilon_{ij}^0(\mathbf{r}) \epsilon_{kl}^0(\mathbf{r}) dV \\
&= \frac{1}{2} \sum_{p,q} \int_V C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} \theta_p(\mathbf{r}) \theta_q(\mathbf{r}) dV \\
&= \frac{1}{2} \sum_{p,q} \int_V C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} (\Delta\theta_p(\mathbf{r}) + \bar{\theta}_p) (\Delta\theta_q(\mathbf{r}) + \bar{\theta}_q) dV
\end{aligned}$$

$$\begin{aligned}
I_6 &= \frac{1}{2} \sum_{p,q} \int_V C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} (\Delta\theta_p(\mathbf{r}) \Delta\theta_q(\mathbf{r})) dV + \sum_{p,q} \int_V C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} (\Delta\theta_p(\mathbf{r}) \bar{\theta}_q) dV \\
&\quad + \frac{1}{2} \sum_{p,q} \int_V C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} (\bar{\theta}_p \bar{\theta}_q) dV \\
&= \frac{1}{2} \sum_{p,q} \int_V C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} (\Delta\theta_p(\mathbf{k}) \Delta\theta_q^*(\mathbf{k})) \frac{d^3\mathbf{k}}{(2\pi)^3} + 0 + \frac{V}{2} C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} \bar{\theta}_p \bar{\theta}_q \quad (\text{A.11}) \\
&\quad (\because \int_V \Delta\theta_p(\mathbf{r}) dV = \int_V (\theta_p(\mathbf{r}) - \bar{\theta}_p) dV = V\bar{\theta}_p - V\bar{\theta}_p = 0)
\end{aligned}$$

Substituting the final expressions of $I_1, I_2, \dots, I_6$ (obtained in Eqs. A.6–A.11) in Eq. A.4 and rearranging, we get $\mathcal{F}_e$:

$$\begin{aligned}
\mathcal{F}_e &= \frac{1}{2} \sum_{p,q} \int_V \left\{ C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} - n_j \hat{\sigma}_{ij}^{(p)} \Omega_{ki}(\mathbf{n}) \hat{\sigma}_{kl}^{(q)} n_l \right\} \Delta\theta_p(\mathbf{k}) \Delta\theta_q^*(\mathbf{k}) \frac{d^3\mathbf{k}}{(2\pi)^3} \\
&= \frac{1}{2} \sum_{p,q} \int_V B_{pq}(\mathbf{n}) \Delta\theta_p(\mathbf{k}) \Delta\theta_q^*(\mathbf{k}) \frac{d^3\mathbf{k}}{(2\pi)^3} \\
&= \frac{1}{2} \sum_{p,q} \int_{V'} B_{pq}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) \tilde{\theta}_q^*(\mathbf{k}) \frac{d^3\mathbf{k}}{(2\pi)^3}, \quad (\text{A.12}) \\
&\quad \left(\because \Delta\theta_p(\mathbf{k}) = \begin{cases} \tilde{\theta}_p(\mathbf{k}) & \text{if } \mathbf{k} \neq 0 \\ 0 & \text{otherwise} \end{cases} \right)
\end{aligned}$$

where $B_{pq} = C_{ijkl} \epsilon_{ij}^{(p)} \epsilon_{kl}^{(q)} - n_j \hat{\sigma}_{ij}^{(p)} \Omega_{ki}(\mathbf{n}) \hat{\sigma}_{kl}^{(q)} n_l$. $\int_{V'}$ signifies that a volume of $(2\pi)^3/V$ about $\mathbf{k} = 0$ is excluded from the integration.

Appendix B

Evaluation of variational derivative

$$(\delta\mathcal{F}_e/\delta\theta_i)$$

We begin with the expression for elastic energy in Fourier space (Eq. 3.26):

$$\mathcal{F}_e = \frac{1}{2} \sum_{p,q=0}^3 \int \frac{d^3\mathbf{k}}{(2\pi)^3} B_{pq}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) \tilde{\theta}_q^*(\mathbf{k}), \quad (\text{B.1})$$

The variation $\delta\mathcal{F}_e$ corresponding to the variation $\delta\theta_i(\mathbf{r})$ is given by

$$\delta\mathcal{F}_e = \frac{1}{2} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \sum_{p,i=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) \delta\tilde{\theta}_i^*(\mathbf{k}) + \frac{1}{2} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \sum_{i,q=0}^3 B_{iq}(\mathbf{n}) \delta\tilde{\theta}_i(\mathbf{k}) \tilde{\theta}_q^*(\mathbf{k}), \quad (\text{B.2})$$

where $\delta\tilde{\theta}_i(\mathbf{k}) (= \int \delta\theta_i(\mathbf{r}) e^{-J\mathbf{k}\cdot\mathbf{r}} d^3\mathbf{r})$ is the Fourier transform of $\delta\theta_i(\mathbf{r})$.

Labeling the two terms on the right hand side of Eq. B.2 as δI_1 and δI_2 , we have

$$\delta\mathcal{F}_e = \delta I_1 + \delta I_2. \quad (\text{B.3})$$

Considering δI_1 first, we proceed as follows:

$$\begin{aligned}
\delta I_1 &= \frac{1}{2} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \sum_{p,i=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) \delta\tilde{\theta}_i^*(\mathbf{k}) \\
&= \frac{1}{2} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \sum_{p,i=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) \int d^3\mathbf{r} \delta\theta_i(\mathbf{r}) e^{J\mathbf{k}\cdot\mathbf{r}} \\
&\quad (\because \delta\theta_i(\mathbf{r}) \text{ is real, } \delta\tilde{\theta}_i^*(\mathbf{k}) = \delta\tilde{\theta}_i(-\mathbf{k})) \\
&= \frac{1}{2} \int d^3\mathbf{r} \delta\theta_i(\mathbf{r}) \int \frac{d^3\mathbf{k}}{(2\pi)^3} \sum_{p,i=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) e^{J\mathbf{k}\cdot\mathbf{r}}
\end{aligned} \tag{B.4}$$

So the variational derivative of δI_1 with respect to θ_i is given by

$$\frac{\delta I_1}{\delta\theta_i} = \frac{1}{2} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \sum_{p,i=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) e^{J\mathbf{k}\cdot\mathbf{r}} \tag{B.5}$$

The right hand side of the above equation is written in the form of an inverse Fourier transform. Hence, we may rewrite this equation as follows:

$$\frac{\delta I_1}{\delta\theta_i} = \left[\frac{1}{2} \sum_{p,i=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}) \right]_{\mathbf{r}}, \tag{B.6}$$

where the notations $[\dots]_{\mathbf{r}}$ and $[\dots]_{\mathbf{k}}$ stand for inverse Fourier transform and Fourier transform, respectively. Therefore, the Fourier transform of $\frac{\delta I_1}{\delta\theta_i}$ is

$$\left[\frac{\delta I_1}{\delta\theta_i} \right]_{\mathbf{k}} = \frac{1}{2} \sum_{p,i=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}). \tag{B.7}$$

Similarly, for δI_2 we have

$$\delta I_2 = \frac{1}{2} \int d^3\mathbf{r} \delta\theta_i(\mathbf{r}) \int \frac{d^3\mathbf{k}}{(2\pi)^3} \sum_{i,q=0}^3 B_{iq}(\mathbf{n}) \tilde{\theta}_q^*(\mathbf{k}) e^{-J\mathbf{k}\cdot\mathbf{r}} \tag{B.8}$$

Substituting $\mathbf{k} = -\mathbf{k}'$ in Eq. B.8, we get

$$\frac{\delta I_2}{\delta\theta_i} = \frac{1}{2} \int \frac{d^3\mathbf{k}'}{(2\pi)^3} \sum_{i,q=0}^3 B_{iq}(\mathbf{n}) \tilde{\theta}_q^*(-\mathbf{k}') e^{J\mathbf{k}' \cdot \mathbf{r}} \quad (\text{B.9})$$

Since $\tilde{\theta}_q^*(-\mathbf{k}') = \tilde{\theta}_q(\mathbf{k}')$, we may rewrite the above equation as

$$\frac{\delta I_2}{\delta\theta_i} = \frac{1}{2} \int \frac{d^3\mathbf{k}'}{(2\pi)^3} \sum_{i,q=0}^3 B_{iq}(\mathbf{n}) \tilde{\theta}_q(\mathbf{k}') e^{J\mathbf{k}' \cdot \mathbf{r}} \quad (\text{B.10})$$

Again, the right hand side of the above equation is in the form of an inverse Fourier transform; thus, the Fourier transform of $\frac{\delta I_2}{\delta\theta_i}$ can be expressed as:

$$\left[\frac{\delta I_2}{\delta\theta_i} \right]_{\mathbf{k}} = \frac{1}{2} \sum_{i,q=0}^3 B_{iq}(\mathbf{n}) \tilde{\theta}_q(\mathbf{k}). \quad (\text{B.11})$$

Since $B_{pi}(\mathbf{n}) = B_{iq}(\mathbf{n})$, combining Eqns. B.7 and B.11 we get the variational derivative of $\mathcal{F}_e$ with respect to θ_i in the Fourier space as

$$\left[\frac{\delta \mathcal{F}_e}{\delta\theta_i} \right]_{\mathbf{k}} = \sum_{p=0}^3 B_{pi}(\mathbf{n}) \tilde{\theta}_p(\mathbf{k}). \quad (\text{B.12})$$

Appendix C

Non-dimensionalisation

All the equations in our formulation presented in Chapter 3 are in their non-dimensional form. As noted previously in section 3.6, we use a characteristic energy E_n , time T_n , and length L_n to render the equations non-dimensional. In this section, we describe the procedure to non-dimensionalise the quantities in the equations. In what follows, the primed quantities denote dimensional quantities.

Let us begin with the Cahn-Hilliard free energy functional in the dimensional form:

$$\begin{aligned}
 \mathcal{F}' &= N'_v \int_V \{f'_0 + \kappa'_c(\nabla' c')^2 + \kappa'_\eta(\nabla' \eta_i)^2\} dV' + \frac{1}{2} \int_V \{C'_{ijkl} \epsilon_{ij}^{\text{el}} \epsilon_{kl}^{\text{el}}\} dV', \\
 &= N'_v \int_V \{f'_0 + \kappa'_c(\nabla' c')^2 + \kappa'_\eta(\nabla' \eta_i)^2\} dV' + \frac{1}{2} N'_v \int_V \left\{ \frac{C'_{ijkl}}{N'_V} \epsilon_{ij}^{\text{el}} \epsilon_{kl}^{\text{el}} \right\} dV', \\
 &= N'_v \int_V \{f'_0 + \kappa'_c(\nabla' c')^2 + \kappa'_\eta(\nabla' \eta_i)^2\} dV' + \frac{1}{2} N'_v \int_V \{C'^{S'}_{ijkl} \epsilon_{ij}^{\text{el}} \epsilon_{kl}^{\text{el}}\} dV', \\
 &= N'_v \int_V \{f'\} dV'
 \end{aligned} \tag{C.1}$$

where $C'^{S'}_{ijkl} = C'_{ijkl}/N'_V$ and f' denotes the total free energy per atom.

N'_v has units of atoms per unit volume. Since the total free energy $\mathcal{F}'$ has the units of energy, f'_0 has the units of energy per atom. Also, the scaled stiffness tensor $C'^{S'}_{ijkl}$ has the units of energy per atom.

The dimensional composition variable c' is rendered non-dimensional using the relation

$$c = \frac{c' - c'_\alpha}{c'_\beta - c'_\alpha}, \quad (\text{C.2})$$

where c'_α and c'_β denote the non-dimensional compositions of the matrix and precipitate phases, respectively. The non-dimensional counterparts of N'_v , f'_0 , κ'_c , κ'_η , and $C'^{S'}_{ijkl}$ are obtained as

$$N_v = N'_v L_n^3, \quad (\text{C.3})$$

$$f_0 = \frac{f'_0}{E_n}, \quad (\text{C.4})$$

$$\kappa_c = \frac{\kappa'_c}{E_n L_n^2}, \quad (\text{C.5})$$

$$\kappa_\eta = \frac{\kappa'_\eta}{E_n L_n^2}, \quad (\text{C.6})$$

$$C^S_{ijkl} = \frac{C'^{S'}_{ijkl}}{E_n}. \quad (\text{C.7})$$

The final expression of Eq. C.1 can be written in the non-dimensional form as

$$\begin{aligned} \mathcal{F} &= \frac{\mathcal{F}'}{E_n} \\ &= \frac{N_v}{E_n L_n^3} \int_V \{f(c, \eta_i, \nabla c, \nabla \eta_i, \dots) E_n\} dV L_n^3 \\ &= N_v \int_V \{f(c, \eta_i, \nabla c, \nabla \eta_i, \dots)\} dV. \end{aligned} \quad (\text{C.8})$$

By definition, the effective chemical potential μ'_c is given by

$$\begin{aligned} \mu'_c &= \frac{1}{N'_v} \frac{\delta \mathcal{F}'}{\delta c'} \\ &= \frac{1}{N'_v} \frac{\delta \mathcal{F}'}{\delta c} \frac{\partial c}{\partial c'} \\ &= \frac{1}{N'_v (c'_\beta - c'_\alpha)} \frac{\delta \mathcal{F}'}{\delta c} \quad (\because \frac{\partial c}{\partial c'} = \frac{1}{(c'_\beta - c'_\alpha)} \text{ from Eq. C.2}). \end{aligned} \quad (\text{C.9})$$

Since $\delta\mathcal{F}'/\delta c'$ has the units of energy per unit volume, $\mu'_c = (1/N'_v)(\delta\mathcal{F}'/\delta c')$ has the units of energy per atom. Hence the non-dimensional potential μ_c is obtained as

$$\begin{aligned}\mu_c &= \frac{\mu'_c}{E_n} \\ &= \frac{L_n^3}{N_v E_n} \frac{1}{(c'_\beta - c'_\alpha)} \frac{\delta\mathcal{F}}{\delta c} \frac{E_n}{L_n^3} \\ &= \frac{1}{N_v (c'_\beta - c'_\alpha)} \frac{\delta\mathcal{F}}{\delta c}.\end{aligned}\tag{C.10}$$

Now let us consider the Cahn-Hilliard equation in dimensional form:

$$\frac{\partial c'}{\partial \tau'} = M' \nabla'^2 \mu'_c.\tag{C.11}$$

The left hand side of Eq.. C.11 can be expressed as

$$\begin{aligned}\frac{\partial c'}{\partial \tau'} &= \frac{\partial c'}{\partial c} \frac{\partial c}{\partial \tau'} \\ &= \frac{(c'_\beta - c'_\alpha)}{T_n} \frac{\partial c}{\partial \tau'}.\end{aligned}\tag{C.12}$$

Using Eqns. C.10 and C.12, non-dimensional form of Eqn. C.11 is obtained as follows:

$$\begin{aligned}
\frac{\partial c}{\partial \tau} &= \frac{M'T_n}{(c'_\beta - c'_\alpha)} \nabla'^2 \mu' \\
&= \frac{M'T_n}{(c'_\beta - c'_\alpha)L_n^2} \nabla^2 \mu' \\
&= \frac{M'T_n}{(c'_\beta - c'_\alpha)N_v L_n^2} \nabla^2 \left[\frac{\delta \mathcal{F}'}{\delta c'} \right] \\
&= \frac{M'T_n L_n^3}{(c'_\beta - c'_\alpha)N_v L_n^2} \nabla^2 \left[\frac{\delta \mathcal{F}}{\delta c'} \frac{E_n}{L_n^3} \right] \\
&= \frac{M'T_n E_n}{(c'_\beta - c'_\alpha)L_n^2} \nabla^2 \left[\frac{\delta(\mathcal{F}/N_v)}{\delta c'} \right] \\
&= \frac{M'T_n E_n}{(c'_\beta - c'_\alpha)L_n^2} \nabla^2 \left[\frac{\delta(\mathcal{F}/N_v)}{\delta c'} \right] \\
&= \frac{M'T_n E_n}{(c'_\beta - c'_\alpha)L_n^2} \nabla^2 \left[\frac{\delta(\mathcal{F}/N_v)}{\delta c} \frac{\delta c}{\delta c'} \right] \\
&= \frac{M'T_n E_n}{(c'_\beta - c'_\alpha)^2 L_n^2} \nabla^2 \left[\frac{\delta(\mathcal{F}/N_v)}{\delta c} \right]
\end{aligned} \tag{C.13}$$

The dimensionless mobility M may be identified as

$$M = \frac{M'T_n E_n}{(c'_\beta - c'_\alpha)^2 L_n^2}, \tag{C.14}$$

Therefore, we obtain

$$\frac{\partial c}{\partial \tau} = M \nabla^2 \left[\frac{\delta(\mathcal{F}/N_v)}{\delta c} \right]. \tag{C.15}$$

Since the non-dimensional mobility M has the value of unity, the characteristic time T_n is obtained as follows:

$$T_n = \frac{(c'_\beta - c'_\alpha)^2 L_n^2}{M'E_n} \tag{C.16}$$

Further, $A'_1 = E_n$ renders the nondimensional value of A_1 as unity.

We consider an unit cell with a lattice parameter of 4 Å. Then the non-dimensional number of atoms per unit volume, N_v , is given by $4L_n^3/[(4 \times 10^{-10})^3]$. In the simulations,

the dimensionless interfacial width, $W_i = 4 \times \Delta x = 2$. Taking the typical interfacial width in dimensional units, W_d , to be 1.6 nm, the characteristic length, L_n , is obtained as

$$L_n = \frac{W_d}{W_i} = 8 \times 10^{-10} \text{ m}. \quad (\text{C.17})$$

Hence dimensionless N_v is calculated to be 32.

The scaled non-dimensional interfacial energy, σ_i/N_v , is found to be 0.64 from our simulations. Hence, $\sigma_i = 32 \times 0.64 = 20.48$. We assume the typical interfacial energy to be 0.1 J/m^2 . Since the interfacial energy is scaled by E_n/L_n^2 , the characteristic energy E_n is obtained as follows:

$$E_n = L_n^2/(\sigma_i) = 3.125 \times 10^{-21} \text{ Joules}. \quad (\text{C.18})$$

Therefore, $A'_1 = E_n = 3.125 \times 10^{-21} \text{ Joules}$.

The dimensional shear modulus G' is obtained as follows:

$$G' = \frac{GE_n}{L_n^3}. \quad (\text{C.19})$$

Hence a scaled dimensionless value of $G = 1500$ corresponds to $G' \approx 293 \text{ GPa}$.

The relation between dimensional mobility M' and dimensional diffusivity D' is given by [104]:

$$D' = \frac{\partial^2 f'_0}{\partial c'^2} M'. \quad (\text{C.20})$$

The non-dimensional bulk free energy density, f'_0 is given as

$$\begin{aligned} f'_0 = & A'_1 \left[\frac{c' - c'_\alpha}{c'_\beta - c'_\alpha} \right]^2 + A'_2 \left[1 - \frac{c' - c'_\alpha}{c'_\beta - c'_\alpha} \right] (\eta_1^2 + \eta_2^2 + \eta_3^2) \\ & + A'_{41}(\eta_1^4 + \eta_2^4 + \eta_3^4) + A'_{42}(\eta_1^2 \eta_2^2 + \eta_2^2 \eta_3^2 + \eta_1^2 \eta_3^2) \\ & + A'_{61}(\eta_1^6 + \eta_2^6 + \eta_3^6) + A'_{62}(\eta_1^2 \eta_2^2 \eta_3^2) \end{aligned} \quad (\text{C.21})$$

Differentiating f'_0 twice with respect to c' , we obtain

$$\frac{\partial^2 f'_0}{\partial c'^2} = \frac{2A'_1}{(c'_\beta - c'_\alpha)^2} \quad (\text{C.22})$$

Therefore,

$$M' = \frac{(c'_\beta - c'_\alpha)^2}{2A'_1} D'. \quad (\text{C.23})$$

If we take a typical value of 0.1 for $(c'_\beta - c'_\alpha)$ and a dimensional diffusivity value of $10^{-18} \text{ m}^2/\text{sec}$, using Eq. C.16 we obtain the characteristic time $T_n = 1.28 \text{ s}$.

Appendix D

Estimation of critical nucleus size

For all cases of anisotropic misfit, we have measured the size of the critical nucleus for alloy compositions, $c_0 = 0.3$ and $c_0 = 0.5$.

In our 2D model system, the total free energy ($\Delta\mathcal{G}$) of a circular precipitate of radius r is given by:

$$\Delta\mathcal{G} = \pi r^2[\Delta\mathcal{G}_v + \Delta\mathcal{G}_e] + 2\pi r\gamma_i, \quad (\text{D.1})$$

where $\Delta\mathcal{G}_v$ is the free energy change per unit volume (area in 2D) of the precipitate formed, $\Delta\mathcal{G}_e$ is the elastic strain energy associated with the circular precipitate of unit area, and γ_i is the interfacial energy per unit length.

The critical radius of the nucleus, r^* , is the one corresponding to the maximum in $\Delta\mathcal{G}$ (in Eq. D.1). The value of r^* is given by

$$r^* = \frac{\gamma_i}{\Delta\mathcal{G}_v + \Delta\mathcal{G}_e}. \quad (\text{D.2})$$

$\Delta\mathcal{G}_v$ is given by

$$\Delta\mathcal{G}_v = N_v\Delta\mathcal{G}_m, \quad (\text{D.3})$$

where $\Delta\mathcal{G}_m$ is the free energy per molecule driving the nucleation process, and N_v is the

number of molecules per unit volume.

For any alloy composition c_0 , $\Delta\mathcal{G}_m$ is obtained from free energy versus composition curve for the matrix (α) phase (in Fig. 3.5):

$$\Delta\mathcal{G}(c_0) = - \left[f^\alpha(c_0) + (1 - c_0) \left(\frac{\partial f^\alpha}{\partial c} \right)_{c=c_0} \right] \quad (\text{D.4})$$

where, $f^\alpha = A_1 c^2$ is given in Eq. 3.9.

Hence, substituting the expression for f^α in Eq. D.4 we obtain $\Delta\mathcal{G}_v$ as follows:

$$\Delta\mathcal{G}_v = -2N_v A_1 c_0 \left(1 - \frac{c_0}{2} \right). \quad (\text{D.5})$$

Following Mura [18], $\Delta\mathcal{G}_e$ is obtained for a circular precipitate as:

$$\Delta\mathcal{G}_e = \frac{G}{2(1-\nu)} \left[\frac{3}{4}(\epsilon_{11}^0)^2 + \frac{3}{4}(\epsilon_{22}^0)^2 + \frac{1}{2}\epsilon_{11}^0\epsilon_{22}^0 \right], \quad (\text{D.6})$$

where G is the shear modulus, ν is the Poisson's ratio, and ϵ_{ij}^0 is the eigenstrain.

Eq. D.2 is derived with the assumption that the nucleus composition is the same as the equilibrium precipitate composition. Taking $\gamma_i = 0.64$, and $G = 500$, the values of r^* for the two different alloys in Cases A–E are listed in Table D. As shown in the table, when $c_0 = 0.3$, the particle will never grow in Case A.

Table D.1: Radius of critical nucleus, r^* , for two alloys with compositions $c_0 = 0.3$ and 0.5 , for the five values of misfit anisotropy parameter t ($G = 500$)

	c_0	G	t	$\Delta\mathcal{G}_v$	$\mathcal{G}_e$	r^*
Case A	0.3	500	1	-0.51	0.643	-
Case B			0.5	-0.51	0.357	4.2
Case C			0	-0.51	0.24	2.37
Case D			-0.5	-0.51	0.22	2.2
Case E			-1	-0.51	0.32	3.4
Case A	0.5	500	1	-0.75	0.643	6.4
Case B			0.5	-0.75	0.357	1.6
Case C			0	-0.75	0.24	1.25
Case D			-0.5	-0.75	0.22	1.2
Case E			-1	-0.75	0.32	1.5

Appendix E

Effect of elastic anisotropy on growth shapes

In this thesis, we have considered only elastically isotropic systems ($A_Z = 1$) for studying morphological evolution of precipitates. However, cubic anisotropy in elastic constants can be easily incorporated in our model through the anisotropy parameter A_Z . In the single particle setting, we have examined the effect of elastic anisotropy on the symmetry-breaking transitions of growth shapes in Cases D and E.

When cubic elastic anisotropy is included in these cases, three competing factors decide the shape of the growing precipitate, namely, (a) the relative magnitudes of the misfit strains along the x and y directions, (b) the direction of the invariant line, and (c) the elastically soft (hard) directions associated with elastic anisotropy.

For both cases, we have used $G = 750$ and $c_\infty = 0.3$. We choose three different values of the anisotropy parameter A_Z :

1. $A_Z = 1/3$ where the $\langle 11 \rangle$ directions are the elastically soft directions,
2. $A_Z = 1$ for an elastically isotropic system, and

3. $A_Z = 3$ where the $\langle 10 \rangle$ directions are the elastically soft directions.

The results for Cases D and E are shown in Figs. E.1 and E.2, respectively. In the left column of these figures, we present the growth shapes of the precipitates at various times for three different values of A_Z ; in the right column, we show the corresponding bifurcation diagrams. For each case, the critical size, R_c , is obtained from the corresponding bifurcation diagrams for different values of A_Z . Figs. E.3 and E.4 show the variation of critical size, R_c , with increasing A_Z for Cases D and E, respectively. In both cases, R_c increases as A_Z changes from $1/3$ to 3 ;

The effect of A_Z on R_c can be explained in terms of the driving force for symmetry-breaking transition. When elastic energy alone is taken into consideration, the driving force for symmetry-breaking transition in Case D is given by the difference in elastic energy of a plate oriented along the invariant line and that oriented along the y axis. (In Case E, the driving force is given by the difference in elastic energy of a plate extending along the invariant line and that of a square plate with its diagonals along x and y axes.)

Since the energy of the plate oriented along the invariant direction is zero in both cases, the driving force for transition is the energy of the plate elongated along the y axis (in Case D). (For Case E, the driving force for transition is given by the energy of the square plate with its diagonals extending along x and y axes.)

For a given G and c_∞ , the driving force is higher for $A_Z = 1/3$ (since the x and y axes are elastically hard) than for $A_Z = 1$ and 3 . Hence in both cases, the critical size of the particle for symmetry-breaking transition is lower for $A_Z < 1$.

Although our results are consistent with those for the equilibrium shapes [1], the critical sizes for symmetry-breaking transition for growth shapes are larger than the corresponding equilibrium shapes.

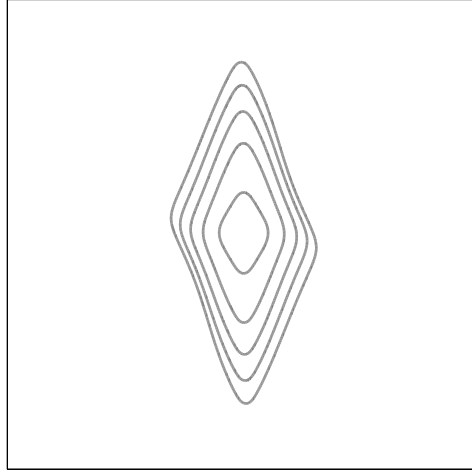
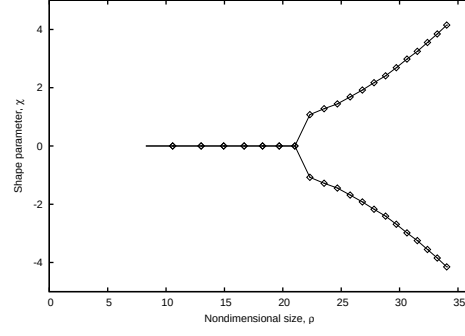
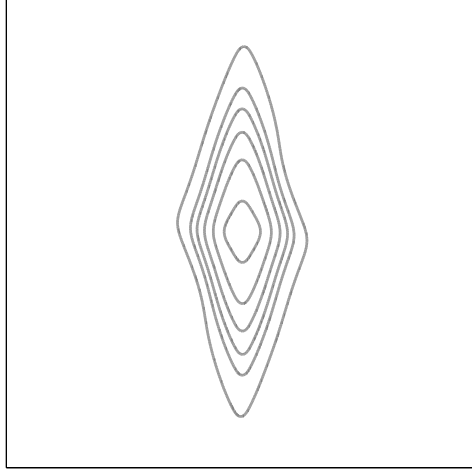
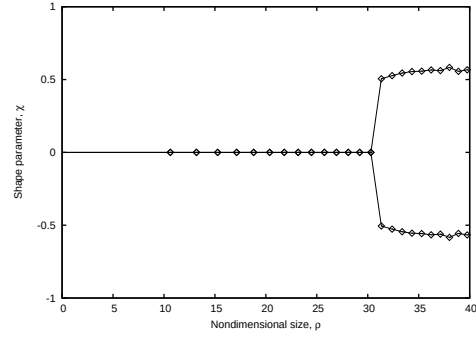
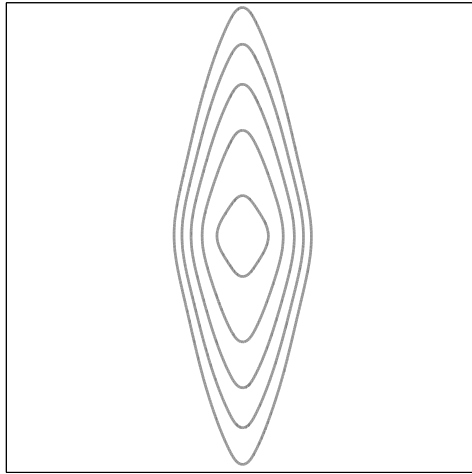
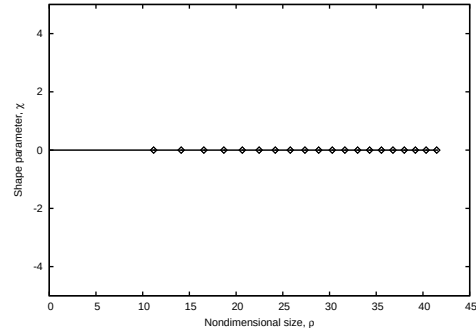
(a) $A_Z = 1/3$ (b) $A_Z = 1/3$ (c) $A_Z = 1$ (d) $A_Z = 1$ (e) $A_Z = 3$ (f) $A_Z = 3$

Figure E.1: Dynamic growth shapes of precipitates and the corresponding bifurcation diagrams in systems with $A_Z = 1/3, 1, 3$ for Case D ($t = -0.5$). In each system, $G = 750$ and $c_\infty = 0.3$. The value of A_Z is indicated in each figure.

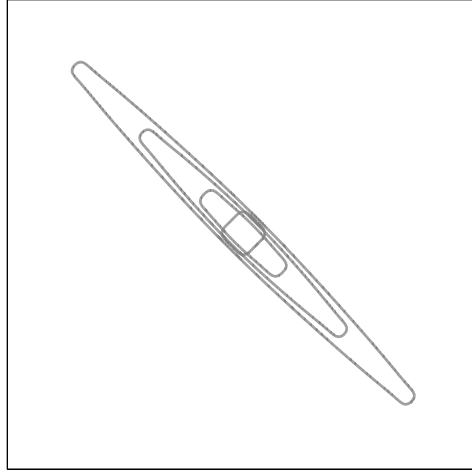
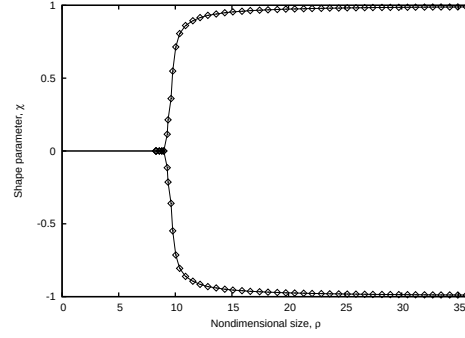
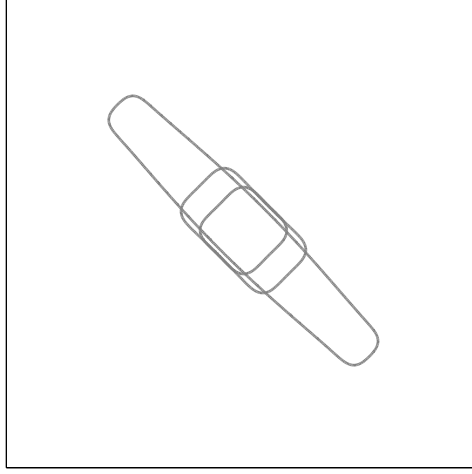
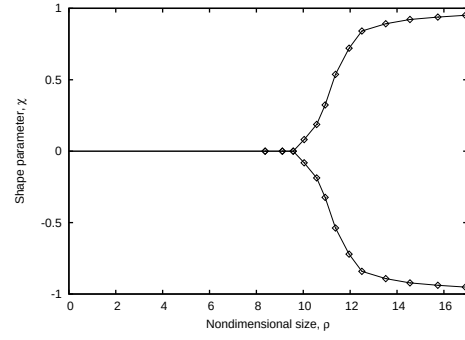
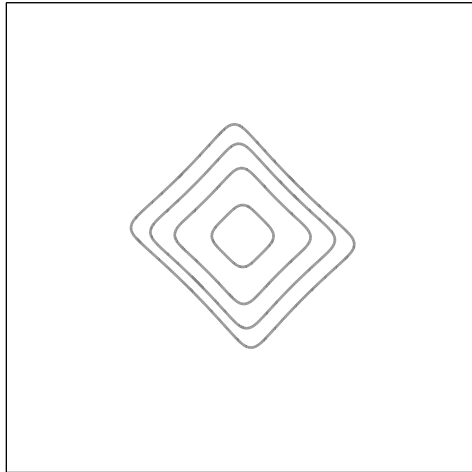
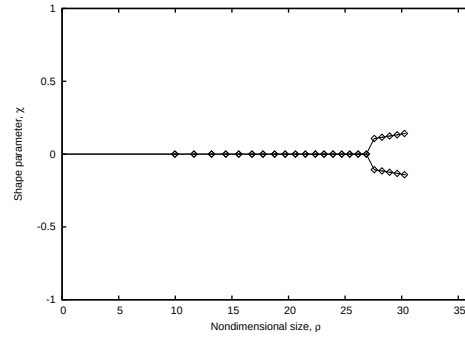
(a) $A_Z = 1/3$ (b) $A_Z = 1/3$ (c) $A_Z = 1$ (d) $A_Z = 1$ (e) $A_Z = 3$ (f) $A_Z = 3$

Figure E.2: Dynamic growth shapes of precipitates and the corresponding bifurcation diagrams in systems with $A_Z = 1/3, 1, 3$ for Case E ($t = -1$). In each system, $G = 750$ and $c_\infty = 0.3$. The value of A_Z is indicated in each figure.

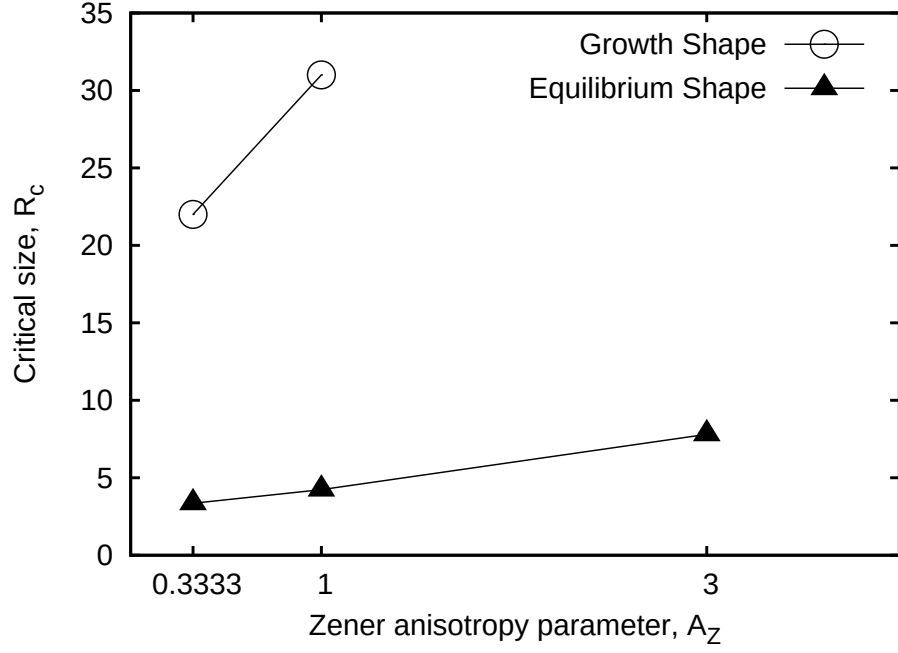


Figure E.3: Variation of critical size R_c with anisotropy parameter A_Z for Case D ($t = -0.5$). $G = 750$, $c_\infty = 0.3$.

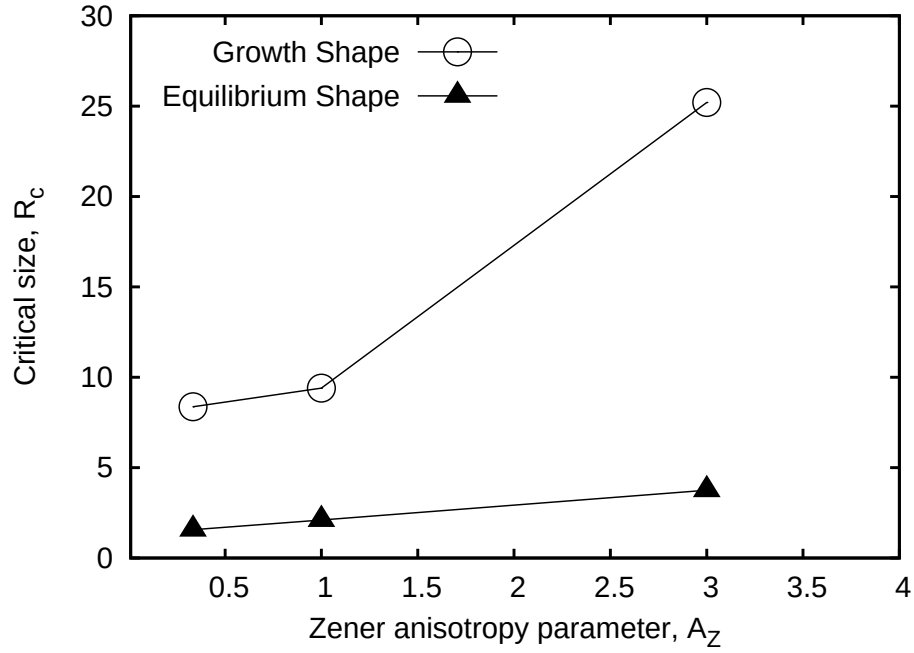


Figure E.4: Variation of critical size R_c with anisotropy parameter A_Z for Case E ($t = -1$). $G = 750$, $c_\infty = 0.3$.

References

- [1] R. Sankarasubramanian, C. S. Jog, and T. A. Abinandanan, *Symmetry-breaking transitions in equilibrium shapes of coherent precipitates: Effect of elastic anisotropy and inhomogeneity*, Metallurgical and Materials Transactions A, **33**(4), 1083–1090 (2002).
- [2] K. Muraleedharan, D. Banerjee, S. Banerjee, and S. Lele, *The α_2 -to-O transformation in Ti-Al-Nb alloys*, Philosophical Magazine A, **71**(5), 1011–1036 (1995).
- [3] V. J. Laraia, W. C. Johnson, and P. W. Voorhees, *Growth of a coherent precipitate from a supersaturated solution*, Journal of Materials Research, **3**(2), 257–266 (1988).
- [4] A. G. Khachaturyan, *Theory of Structural Transformations in Solids*, John Wiley and Sons, New York (1983).
- [5] K. Muraleedharan and D. Banerjee, *Phase transformations involving the α_2 and O phases in Ti-Al-Nb alloys*, Scripta Metallurgica et Materialia, **29**(4), 527–532 (1993).
- [6] L. A. Bendersky, A. Roytburd, and W. J. Boettinger, *Phase transformations in the (Ti, Al)₃ Nb section of the Ti-Al-Nb system-I. Microstructural predictions based on a subgroup relation between phases*, Acta Metallurgica et Materialia, **42**(7), 2323–2335 (1994).

-
- [7] L. A. Bendersky and W. J. Boettinger, *Phase transformations in the $(\text{Ti}, \text{Nb})_3 \text{Al}$ section of the Ti-Al-Nb system. II. Experimental TEM study of microstructures*, *Acta Metallurgica et Materialia*, **42**(7), 2337–2352 (1994).
- [8] C. Leroux, A. Loiseau, D. Broddin, and G. Vantendeloo, *Electron microscopy study of the coherent two-phase mixtures $L1_0+L1_2$ in CoPt alloys*, *Philosophical Magazine B*, **64**(1), 57–82 (1991).
- [9] A. H. Heuer, *Transformation Toughening in ZrO_2 -Containing Ceramics*, *Journal of the American Ceramic Society*, **70**(10), 689–698 (1987).
- [10] M. Doi, *Elasticity effects on the microstructure of alloys containing coherent precipitates*, *Progress in Materials Science*, **40**(2), 79–180 (1996).
- [11] P. Fratzl, O. Penrose, and J. L. Lebowitz, *Modeling of Phase Separation in Alloys with Coherent Elastic Misfit*, *Journal of Statistical Physics*, **95**(5), 1429–1503 (1999).
- [12] Y. Le Bouar, A. Loiseau, and A. G. Khachaturyan, *Origin of chessboard-like structures in decomposing alloys. Theoretical model and computer simulation*, *Acta Materialia*, **46**(8), 2777–2788 (1998).
- [13] Y. H. Wen, Y. Wang, L. A. Bendersky, and L. Q. Chen, *Microstructural evolution during the $\alpha_2 \rightarrow \alpha_2 + O$ transformation in Ti-Al-Nb alloys: phase-field simulation and experimental validation*, *Acta Materialia*, **48**(16), 4125–4135 (2000).
- [14] Y. Wang, H.-Y. Wang, L.-Q. Chen, and A. G. Khachaturyan, *Microstructural development of coherent tetragonal precipitates in magnesium-partially-stabilized zirconia: a computer simulation*, *Journal of the American Ceramic Society*, **78**(3), 657–661 (1995).
- [15] Y. H. Wen, Y. Wang, and L. Q. Chen, *Effect of elastic interaction on the formation of a complex multi-domain microstructural pattern during a coherent hexagonal to orthorhombic transformation*, *Acta Materialia*, **47**(17), 4375–4386 (1999).

-
- [16] Y. H. Wen, Y. Wang, and L. Q. Chen, *Phase-field simulation of domain structure evolution during a coherent hexagonal-to-orthorhombic transformation*, Philosophical Magazine A, **80**(9), 1967–1982 (2000).
- [17] C. S. Jog, R. Sankarasubramanian, and T. A. Abinandanan, *Symmetry-breaking transitions in equilibrium shapes of coherent precipitates*, Journal of the Mechanics and Physics of Solids, **48**(11), 2363–2389 (2000).
- [18] T. Mura, *Micromechanics of Defects in Solids*, Kluwer Academic Publishers (1982).
- [19] P. W. Voorhees and W. C. Johnson, *Thermodynamics of elastically stressed crystals*, in H. Ehrenreich and F. Spaepen, editors, *Solid state physics: advances in research and applications*, volume 59, pp. 1–201, Elsevier academic press (2004).
- [20] J. D. Eshelby, *Elastic inclusions and inhomogeneities*, Progress in Solid Mechanics, **2**(3), 87–140 (1961).
- [21] A. J. Ardell and R. B. Nicholson, *On the modulated structure of aged Ni-Al alloys : with an appendix On the elastic interaction between inclusions by J. D. Eshelby*, Acta Metallurgica, **14**(10), 1295–1309 (1966).
- [22] A. Maheshwari and A. J. Ardell, *Morphological evolution of coherent misfitting precipitates in anisotropic elastic media*, Physical Review Letters, **70**(15), 2305–2308 (1993).
- [23] T. Miyazaki, H. Imamura, and T. Kozakai, *The formation of "γ' precipitate doublets" in Ni-Al alloys and their energetic stability*, Materials Science and Engineering, **54**(1), 9–15 (1982).
- [24] N. V. Starostina, S. V. Prikhodko, A. J. Ardell, and S. Prasad, *Coarsening of Ni₃Ge precipitates in Ni-Ge alloys aged under uniaxial compression*, Materials Science and Engineering A, **397**(1-2), 264–270 (2005).

-
- [25] S. Yeom, D. Yoon, and M. Henry, *The morphological changes of γ' precipitates in a Ni-8 Al(st.%) alloy during their coarsening*, Metallurgical Transactions A, **24**(9), 1975–1981 (1993).
- [26] A. D. Sequeira, H. A. Calderon, G. Kostorz, and J. S. Pedersen, *Bimodal size distributions of γ' precipitates in Ni-Al-Mo-II. Transmission electron microscopy*, Acta Metallurgica et Materialia, **43**(9), 3441–3451 (1995).
- [27] M. Fahrman, P. Fratzl, O. Paris, E. Fahrman, and W. C. Johnson, *Influence of coherency stress on microstructural evolution in model Ni-Al-Mo alloys*, Acta Metallurgica et Materialia, **43**(3), 1007–1022 (1995).
- [28] Cho and A. J. Ardell, *Coarsening of Ni_3Si precipitates at volume fractions from 0.03 to 0.30*, Acta Materialia, **46**(16), 5907–5916 (1998).
- [29] J. K. Tien and S. M. Copley, *Effect of uniaxial stress on the periodic morphology of coherent γ' precipitates in nickel- base superalloy crystals*, Metallurgical Transactions, **2**(1), 215–219 (1971).
- [30] J. Chang and S. Allen, *Elastic energy changes accompanying γ'' rafting in nickel-base superalloys*, Journal of Materials Research, **6**(9), 1843–1855 (1991).
- [31] M. Doi, T. Miyazaki, and T. Wakatsuki, *The effect of elastic interaction energy on the morphology of γ' precipitates in nickel-based alloys*, Materials Science and Engineering, **67**(2), 247–253 (1984).
- [32] M. Kaufman, P. Voorhees, W. Johnson, and F. Biancaniello, *An elastically induced morphological instability of a misfitting precipitate*, Metallurgical and Materials Transactions A, **20**(10), 2171–2175 (1989).
- [33] Y. Yoo, D. Yoon, and M. Henry, *The effect of elastic misfit strain on the morphological evolution of γ -precipitates in a model Ni-base superalloy*, Metals and Materials, **1**, 47 (1995).

-
- [34] H. A. Calderon, C. Kisielowski, and T. Mori, *Statistical analysis of γ' quartet split patterns in γ - γ' Ni alloys revealed by high resolution electron microscopy*, Philosophical Magazine Letters, **87**(1), 33–40 (2007).
- [35] Y. Y. Qiu, *The splitting behavior of γ' particles in Ni-based alloys*, Journal of Alloys and Compounds, **270**(1-2), 145–153 (1998).
- [36] G. K. Bansal and A. H. Heuer, *Precipitation in partially stabilized zirconia*, Journal of the American Ceramic Society, **58**(5-6) (1975).
- [37] A. H. Heuer, R. Chaim, and V. Lanteri, *The displacive cubic $\rightarrow$ tetragonal transformation in ZrO_2 alloys*, Acta Metallurgica, **35**(3), 661–666 (1987).
- [38] V. Lanteri, R. Chaim, and A. H. Heuer, *On the Microstructures Resulting from the Diffusionless Cubic $\rightarrow$ Tetragonal Transformation in ZrO_2 - Y_2O_3 Alloys*, Journal of the American Ceramic Society, **69**(10), C258–261 (1986).
- [39] T. Sakuma, Y. Yoshizawa, and H. Suto, *The modulated structure formed by isothermal ageing in ZrO_2 -5.2 mol% Y_2O_3 alloy*, Journal of Materials Science, **20**(3), 1085–1092 (1985).
- [40] T. Sakuma, Y. Yoshizawa, and H. Suto, *The metastable two-phase region in the zirconia-rich part of the ZrO_2 - Y_2O_3 alloy*, Journal of Materials Science, **21**, 1436 (1986).
- [41] A. H. Heuer, R. Chaim, and V. Lanteri, *Review: Phase transformations and microstructural characterisation of alloys in the system Y_2O_3 - ZrO_2* , in S. Somiya, N. Yamamoto, and H. Yanagida, editors, *Science and technology of Zirconia III*, volume 24, pp. 3–20, American Ceramic Society, Westerville, OH (1988).
- [42] M. Doi and T. Miyazaki, *On the spinodal decomposition in zirconia-yttria (ZrO_2 - Y_2O_3) alloys*, Philosophical Magazine B, **68**(3), 305–315 (1993).

-
- [43] C. A. Bateman and M. R. Notis, *Coherency effects during precipitate coarsening in partially stabilized zirconias*, *Acta Metallurgica et Materialia*, **40**(9), 2413–2421 (1992).
- [44] J. A. Manriquez and L. Rabenberg, *Extended domains of oriented γ'' particles in Ni–Fe–Cr base alloys*, *Scripta Metallurgica et Materialia*, **28**(5), 581–586 (1993).
- [45] W. H. Tian and M. Nemoto, *Morphology of $L1_0$ -TiAl(Ag) precipitation in Ag-modified $L1_2$ -Al₃Ti*, *Intermetallics*, **8**(4), 345–352 (2000).
- [46] K.-I. Udoh, El, Y. Tanaka, K. Hisatsune, K. Yasuda, G. Van Tendeloo, and J. Van Landuyt, *Structural aspects of AuCu I or AuCu II and a cuboidal block configuration of f.c.c. disordered phase in AuCu–Pt and AuCu–Ag pseudobinary alloys*, *Materials Science and Engineering A*, **203**(1-2), 154–164 (1995).
- [47] W. H. Tian and M. Nemoto, *Crystallography and morphology of $D0_{23}$ -Al₁₁Ti₅ precipitation in Ag-modified $L1_2$ -Al₃Ti*, *Intermetallics*, **8**(8), 835–843 (2000).
- [48] J. F. Nie and B. C. Muddle, *High temperature precipitation hardening in a rapidly quenched AlTiNi alloy II. Precipitate characterisation*, *Materials Science and Engineering A*, **221**(1-2), 22–32 (1996).
- [49] K. Muraleedharan, T. Nandy, D. Banerjee, and S. Lele, *Phase stability and ordering behaviour of the O phase in Ti–Al–Nb alloys*, *Intermetallics*, **3**(3), 187–199 (1995).
- [50] C. Boehlert, B. Majumdar, V. Seetharaman, and D. Miracle, *Part I. The microstructural evolution in Ti–Al–Nb O+Bcc orthorhombic alloys*, *Metallurgical and Materials Transactions A*, **30**(9), 2305–2323 (1999).
- [51] R. Sinclair and J. Dutkiewicz, *Lattice imaging of the B19 ordering transformation and interfacial structures in Mg₃Cd.*, *Acta Metallurgica*, **25**(3), 235–249 (1977).

-
- [52] J. Vicens and P. Delavignette, *Particular domain configuration observed in a new phase of the Ta-N system*, Physica Status Solidi (A) Applied Research, **33**(2), 497–509 (1976).
- [53] C. Manolikas and S. Amelinckx, *Phase transitions in ferroelastic lead orthovanadate as observed by means of electron microscopy and electron diffraction. I. Static observations*, Physica Status Solidi (A), **60**(2), 607–617 (1980).
- [54] C. Manolikas and S. Amelinckx, *Phase transitions in ferroelastic lead orthovanadate as observed by means of electron microscopy and electron diffraction. II. Dynamic Observations*, Physica Status Solidi (A), **61**(1), 179–188 (1980).
- [55] T. Saburi and C. M. Wayman, *Crystallographic similarities in shape memory martensites*, Acta Metallurgica, **27**(6), 979–995 (1979).
- [56] K. Bhattacharya, *Wedge-like microstructure in martensites*, Acta Metallurgica et Materialia, **39**(10), 2431–2444 (1991).
- [57] C. M. Wayman, *Shear Transformations and Microstructure*, Materials Characterization, **39**(2-5), 235–260 (1997).
- [58] M. Umemoto and C. M. Wayman, *Crystallography and morphology studies of Fe-Pt martensites: Lenticular to thin plate transition and thin plate morphologies*, Acta Metallurgica, **26**(10), 1529–1549 (1978).
- [59] T. A. Schroeder and C. M. Wayman, *The formation of martensite and the mechanism of the shape memory effect in single crystals of Cu-Zn alloys*, Acta Metallurgica, **25**(12), 1375–1381 (1977).
- [60] K. Madangopal and S. Banerjee, *The self-accommodating morphology of NiTi shape memory alloy martensite*, Scripta Metallurgica et Materialia, **24**(12), 2291–2296 (1990).

-
- [61] D. Srivastava, K. Madangopal, S. Banerjee, and S. Ranganathan, *Self accomodation morphology of martensite variants in Zr-2.5wt% Nb alloy*, Acta Metallurgica et Materialia, **41**(12), 3445–3454 (1993).
- [62] J. D. Eshelby, *The Determination of the Elastic Field of an Ellipsoidal Inclusion, and Related Problems*, Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences (1934-1990), **241**(1226), 376–396 (1957).
- [63] F. K. Legoues, H. I. Aaronson, and Y. W. Lee, *Influence of crystallography upon critical nucleus shapes and kinetics of homogeneous f.c.c.-f.c.c. nucleation-III. The influence of elastic strain energy*, Acta Metallurgica, **32**(10), 1845–1853 (1984).
- [64] I. M. Kaganova and A. L. Roitburd, *Equilibrium between elastically-interacting phases*, Soviet Physics - JETP, **67**(6), 1173–1183 (1988).
- [65] C. Herring, *The use of classical macroscopic concepts in surface-energy problems*, in R. Gomer and C. S. Smith, editors, *Structure and Properties of Solid Surfaces*, pp. 5–81, The University of Chicago Press, Chicago, Illinois (1953).
- [66] A. G. Khachaturyan and G. A. Shatalov, *Elastic-interaction potential of defects in a crystal*, Soviet Physics Solid State, **11**, 118–123 (1969).
- [67] E. Kostlan and J. W. Morris, *The preferred habit of a coherent thin-plate inclusion in an anisotropic elastic solid*, Acta Metallurgica, **35**(8), 2167–2175 (1987).
- [68] W. C. Johnson and J. W. Cahn, *Elastically induced shape bifurcations of inclusions*, Acta Metallurgica, **32**(11), 1925–1933 (1984).
- [69] I. Schmidt and D. Gross, *The equilibrium shape of an elastically inhomogeneous inclusion*, Journal of the Mechanics and Physics of Solids, **45**(9), 1521–1549 (1997).
- [70] R. Sankarasubramanian, *Symmetry-breaking shape transitions in equilibrium shapes of coherent precipitates*, Ph.D. thesis, Indian Institute of Science (2000).

-
- [71] J. K. Lee, *Morphology of coherent precipitates via a discrete atom method*, Materials Science and Engineering A, **238**(1), 1–12 (1997).
- [72] J. K. Lee, *Coherency strain analyses via a discrete atom method*, Scripta Metallurgica et Materialia, **32**(4), 559–564 (1995).
- [73] J. K. Lee, *A study on coherency strain and precipitate morphology via a discrete atom method*, Metallurgical and Materials Transactions A, **27**(6), 1449–1459 (1996).
- [74] P. Fratzl and O. Penrose, *Ising model for phase separation in alloys with anisotropic elastic interaction—I. Theory*, Acta Metallurgica et Materialia, **43**(8), 2921–2930 (1995).
- [75] P. Fratzl and O. Penrose, *Ising model for phase separation in alloys with anisotropic elastic interaction—II. A computer experiment*, Acta Materialia, **44**(8), 3227–3239 (1996).
- [76] J. W. Cahn and J. E. Hilliard, *Free Energy of a Nonuniform System. I. Interfacial Free Energy*, The Journal of Chemical Physics, **28**(2), 258–267 (1958).
- [77] J. W. Cahn, *On spinodal decomposition*, Acta Metallurgica, **9**(9), 795–801 (1961).
- [78] S. M. Allen and J. W. Cahn, *A microscopic theory for antiphase boundary motion and its application to antiphase domain coarsening*, Acta Metallurgica, **27**(6), 1085–1095 (1979).
- [79] L. Chen, *Phase-field models for microstructure evolution*, Annual Review of Materials Research, **32**(1), 113–140 (2002).
- [80] W. Boettinger, J. Warren, C. Beckermann, and A. Karma, *Phase-field simulation of solidification*, Annual Review of Materials Research, **32**(1), 163–194 (2002).
- [81] L. Q. Chen and Y. Wang, *The continuum field approach to modeling microstructural evolution*, JOM, **48**(12), 13–18 (1996).

-
- [82] Y. Wang, D. Banerjee, C. C. Su, and A. G. Khachaturyan, *Field kinetic model and computer simulation of precipitation of $L1_2$ ordered intermetallics from f.c.c. solid solution*, *Acta Materialia*, **46**(9), 2983–3001 (1998).
- [83] Y. Wang, H. Wang, L.-Q. Chen, and A. G. Khachaturyan, *Shape evolution of a coherent tetragonal precipitate in partially stabilized cubic ZrO_2 : A computer simulation*, *J Am Ceram Soc*, **76**(12), 3029–3033 (1993).
- [84] D. Fan and L. Q. Chen, *Computer Simulation of Twin Formation during the Displacive $c \rightarrow t'$ Phase Transformation in the Zirconia-Yttria System*, *Journal of the American Ceramic Society*, **78**(3), 769–773 (1995).
- [85] Y. W. Cui, T. Koyama, I. Ohnuma, K. Oikawa, R. Kainuma, and K. Ishida, *Simulation of hexagonal-orthorhombic phase transformation in polycrystals*, *Acta Mater*, **55**(1), 233–241 (2007).
- [86] T. A. Abinandanan and F. Haider, *An extended Cahn-Hilliard model for interfaces with cubic anisotropy*, *Philosophical Magazine A*, **81**(10), 2457–2479 (2001).
- [87] R. Kobayashi, *Modeling and numerical simulations of dendritic crystal growth*, *Physica D*, **63**(3-4), 410–423 (1993).
- [88] S. Y. Hu and L. Q. Chen, *A phase-field model for evolving microstructures with strong elastic inhomogeneity*, *Acta Mater*, **49**(11), 1879–1890 (2001).
- [89] M. P. Gururajan and T. A. Abinandanan, *A phase field study of microstructural evolution in elastically inhomogeneous systems*, in J. M. Howe, D. E. Laughlin, J. K. Lee, U. Dahmen, and W. A. Soffa, editors, *Proceedings of the International Conference on Solid-Solid Phase Transformations in Inorganic Materials*, volume 2 of *International Conference on Solid-Solid Phase Transformations in Inorganic Materials*, pp. 747–756 (2005).

-
- [90] M. P. Gururajan and T. A. Abinandanan, *Phase field study of precipitate rafting under a uniaxial stress*, Acta Materialia, **55**(15), 5015–5026 (2007).
- [91] H. E. Cook, *Brownian motion in spinodal decomposition*, Acta Metallurgica, **18**(3), 297–306 (1970).
- [92] L. Q. Chen and A. G. Khachaturyan, *Computer simulation of structural transformations during precipitation of an ordered intermetallic phase*, Acta Metallurgica et Materialia, **39**(11), 2533–2551 (1991).
- [93] J. P. Simmons, C. Shen, and Y. Wang, *Phase field modeling of simultaneous nucleation and growth by explicitly incorporating nucleation events*, Scripta Materialia, **43**(10), 935–942 (2000).
- [94] P. C. Hohenberg and B. I. Halperin, *Theory of dynamic critical phenomena*, Reviews of Modern Physics, **49**(3), 435–479 (1977).
- [95] C. Shen, J. P. Simmons, and Y. Wang, *Effect of elastic interaction on nucleation: II. Implementation of strain energy of nucleus formation in the phase field method*, Acta Materialia, **55**(4), 1457–1466 (2007).
- [96] L. Q. Chen and J. Shen, *Applications of semi-implicit Fourier-spectral method to phase field equations*, Computer Physics Communications, **108**(2-3), 147–158 (1998).
- [97] J. Zhu, L. Q. Chen, J. Shen, and V. Tikare, *Coarsening kinetics from a variable-mobility Cahn-Hilliard equation: Application of a semi-implicit Fourier spectral method*, Physical Review E, **60**(4), 3564–3572 (1999).
- [98] M. Frigo and S. G. Johnson, *FFTW: an adaptive software architecture for the FFT* (<http://www.fftw.org>), in *Proceedings of the 1998 IEEE International Conference on Acoustics, Speech and Signal Processing, 1998.*, volume 3, pp. 1381–1384 (1998).

-
- [99] M. E. Thompson, C. S. Su, and P. W. Voorhees, *Equilibrium shape of a misfitting precipitate*, Acta Metallurgica et Materialia, **42**, 2107–2122 (1994).
- [100] R. Mueller and D. Gross, *3D simulation of equilibrium morphologies of precipitates*, Computational Materials Science, **11**(1), 35–44 (1998).
- [101] J.-H. Choy and J. K. Lee, *On the shape evolution of a two-dimensional coherent precipitate with a general misfit strain*, Materials Science and Engineering A, **285**(1-2), 195–206 (2000).
- [102] Y. Wang and A. G. Khachaturyan, *Shape instability during precipitate growth in coherent solids*, Acta Metallurgica et Materialia, **43**(5), 1837–1857 (1995).
- [103] W. C. Johnson, *Precipitate shape evolution under applied stress - thermodynamics and kinetics.*, Metallurgical Transactions A, **18**(2), 233–247 (1987).
- [104] J. W. Cahn, *Spinodal Decomposition*, Transactions of the Metallurgical Society of AIME, **242**, 166–180 (1968).